Development of Web-Based Anonymous Feedback System for CCIS Students Using Sentiment Analysis and Algorithmic Concern Ranking

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I thank the Lord for giving me the capability to do this research project. I also thank Mr. Noel S. Rafanan, my professor in CMPSC 146 Software Engineering for the assistance he rendered in the development of the project and its documentation.

The following are the development background and evaluation of ResBack for the College of Computing and Information Sciences of Mariano Marcos State University. There were twenty-five students who already answered a pre-questionnaire using Google Forms. The stakeholder validation and evaluation of the system will be done once the necessary permission is given.

CHAPTER 1 INTRODUCTION

1.1 Background of the Project

Student feedback allows the institution to gauge their experiences of the teaching process, the academic facilities and other services provided, as well as the facilities themselves. Feedback in written form will reveal issues which may be hard to identify through scoring. Richardson [1] explains the importance of the feedback tool in evaluating teaching and the effectiveness of the program. Shah et al. [2] note the need for institutions to use feedback obtained.

For the College of Computing and Information Sciences (CCIS), Mariano Marcos State University, ResBack's objective is to arrange the comments for authorized examination without compromising the anonymity of the submitter. Initially, twenty-five participants were surveyed using a Google Forms survey from September 24 to 29, 2026. All twenty-three respondents agreed or strongly agreed that they would feel more comfortable giving feedback if they could remain anonymous, and all twenty-five of them linked anonymity to truthfulness [3].

ResBack requires an account that privately links submission information with its owners for personal history and abuse protection purposes. The dashboard and feedback report for faculty lack any information about the identity of the submitter. As such, anonymity in this project is synonymous with identity suppression in the review process. However, the link through accounts stays in the database, while comments could reveal identities.

The sentiment analysis categorizes the sentiments of the comments as positive, neutral, and negative sentiments.

The reviews of the studies by Kastrati et al. [4], Dalipi et al. [5], and Shaik et al. [6] reveal that the sentiment analysis has been applied to the field of education and it is faced with challenges like context interpretation, negation detection, opinion contradiction, and language diversity. In the Philippines, sentiment analysis has been conducted through the analysis of the English and Filipino comments [7], academic service analytics [8], and multilingual faculty evaluation [9].

ResBack is an amalgamation of submitting concerns via a web-browser and pre-trained Gemma inference via the Google Gemini API. The system checks for the validity of the results generated by the model, language normalization and concerns categories, as well as the provision of summaries and report downloads. The concerns are ranked based on a programmed formula which takes into consideration the negative volume, negative ratio, certainty of the model's output, and the recency of the concerns. This project does not use model training on any language dataset.

The project combines identity protection, multi-language analysis, topic classification, concern prioritization, and role-based reporting for CCIS. It helps authorized users analyze the feedback received. Investigation, disciplinary action, and concern resolution are still the responsibility of the organization.

1.2 Statement of the Problem and Project Gap

The key issue that arises is how to gather CCIS feedback, preserve anonymity in the submission interface for the submitter, and assist the designated person in finding recurring problems. For this, there needs to be consistent labeling of sentiments and languages used, as well as summaries of topics.

Previous works have shown analytics for student feedback [7][8]. ResBack fills a gap in local integration through the use of English, Tagalog, Ilocano, and bilingual processing together with submission of data using private accounts and a ranked rule. This will be evaluated for CCIS.

The study addresses the following questions.

1. How is a web-based system able to create a feedback mechanism for authenticated CCIS users but exclude the identities of the submitters on the faculty dashboards and feedback reports?
2. How is a pretrained language model able to classify feedback into positive, neutral, or negative and provide analysis results that are valid and actionable?
3. How is the system able to identify and categorize English, Tagalog, Ilocano, and mixed language feedback types?
4. How can submissions be categorized in terms of concern topics so that repetitive concerns can be analyzed together?
5. How can a ranking rule be developed based on negative volume, negative ratio, confidence, and recency but without making the score seem like urgency?
6. How can filtered dashboards, Excel files, and PDF reports help facilitate review by faculty and administration?
7. What evaluation criteria will be used in order to ensure functionality, analysis quality, usability, privacy and ranking behavior?

1.3 Objectives of the Project

1.3.1 Main Objective

To design and evaluate ResBack, a web-based feedback application on CCIS that ensures the anonymity of the sender while including pre-trained sentiment analysis, multilingual categorization, topic clustering, and explicit concerns prioritization in the interface for evaluation.

1.3.2 Specific Objectives

1. Describe the CCIS implementation's feedback, access, privacy, and reporting standards and verify them with the appropriate stakeholders.
2. Offer registration, login, private feedback submission, and feedback history by user with access controls from the server side.
3. Include pretrained Gemma inference and verify its results for sentiment, confidence, keywords, language, and topic.
4. Standardize the English, Tagalog, Ilocano, Taglish, Iloclish, Taglocano, Trilingual, and Other Language categories.
5. Categorize the feedback by standardized concern names and compute the weighted ranking with the number of supporting evidence samples and selected examples.
6. Include dashboards with filtering options based on date and language and generate Excel and PDF reports without identifying the submitter.
7. Evaluate the application through functional and failure-path testing, independent feedback annotations, ranking verification, and stakeholder usability study.

8. Iteratively refine the application through Agile Data Science 2.0 and develop the respective design documents, test cases, and operating instructions.

1.4 Scope and Limitations of the Project

This web application is used by CCIS students, faculty, administrative staff members, and a privileged super administrator. The users can provide their feedback and check their history. The faculty members and administrative users can use the analytics and reports facility, while the administrative users handle the user accounts and remove feedback. The mobile application is out of the scope of this project.

The feedback is allowed up to 10–2,000 characters only in the active CCIS category. The controls available are duplicate detection over a 24-hour period, a maximum of five submission per account in ten minutes, and a rejectable dictionary. The rejected submission stays saved, is exempted from model evaluation, and does not appear in faculty analytics/reporting. This does not validate any complaint.

Analysis covers overall sentiment, model-reported confidence, up to five keywords, concern topics, detected languages, and language confidence. Topics include internet access, classrooms, facilities, cleanliness, teaching, schedules, enrollment, the library, safety, equipment, administration, and other concerns. A submission may receive several topics, so topic counts can exceed submission counts.

The languages to be covered by the system include English, Tagalog, Ilocano, and any combination of these languages. Accuracy will depend on the language, sarcasm, regional variations, spelling, and sentiment. Every individual comment gets a single sentiment score regardless of the number of topics being discussed. Pretrained Gemma will be used.

Development relies on Laravel, PHP, XAMPP Database Server for the local environment, browsers and Google APIs. The model needs to have internet connection for analysis as well as access to API. Delay may happen during synchronous analysis when the API is slow. Failure of the analysis does not give any sentiment label.

Concern Ranking reflect the filtered information but not the severity of the issue itself. It is necessary to examine the weighting system along with the thirty-day recency half-life. Emergency response, automation of problem-solving process, and the appeals process are out of scope for now.

1.5 Significance of the Project

For students, using ResBack include having a common medium through which feedback can be provided, having their identity protected in the reviewing interface, and access to their submission history. Local language and multilingual support is intended to facilitate participation.

For faculty and administrators, Sentiment summaries, topic groupings, filters, and example content may help identify patterns. Reports can be downloaded for review and discussion, while comment access will ensure that the context is retained.

For CCIS and the university, the project provides documented access rules, system diagrams, and evaluation procedures that can inform future expansion. Any improvement in institutional responsiveness will depend on how personnel use and act on the feedback.

Researchers can build on the documented model integration, safeguards, and ranking through language-specific evaluation and sensitivity analysis.

CHAPTER 2 REVIEW OF RELATED LITERATURE

2.1 Review of Related Literature and Systems

This chapter reviews student feedback, privacy, sentiment analysis, multilingual processing, concern ranking, and dashboards. It relates findings from these areas to ResBack's design and compares similar educational systems.

2.1.1 Student Feedback and Institutional Improvement

Feedback tools for the purposes of teaching, satisfaction with the program, and quality perception are discussed by Richardson [1], along with constraints of interpretation and application. The feedback approaches in Australia and Scotland are examined by Shah et al. [2].

ResBack supports this process through grouped topics and reports. The current application does not record a complete case-resolution dialogue or notify students of every administrative action. Closing the feedback loop remains an institutional responsibility and a possible system extension.

2.1.2 Identity Protection Privacy and Anonymity

The concepts of consent, de-identification, surveillance, and interpretation of student data are addressed by Slade and Prinsloo [10]. Privacy as a factor is linked to transparency, trust, and accountability by Pardo and Siemens [11]. Such principles would be conducive to implementing separate access controls on accounts and feedback reports.

In their study of university peer assessment, Fraile et al. [12] state that 73.6% of students favored anonymity, although this preference did not always enhance perception

measurement and identifiable discussion was still useful. The above finding can help to address the issue of user preferences; it cannot be automatically applied to CCIS feedback.

ResBack keeps identities hidden from faculty views and exports but maintains private user connections. It is important for users to be aware of this situation and have access to data at the back end. As per the implementation rules of the Philippine Data Privacy Act, transparency, legitimate purpose, and proportionality are highlighted [13].

2.1.3 Sentiment Analysis in Educational Feedback

Pang, Lee, and Vaithyanathan [14] provide an example of sentiment classification based on movie reviews, forming a base for sentiment-based text organization. The application of the technique in multilingual educational feedback analysis needs to be evaluated specifically.

Kastrati et al. [4] survey 92 papers on NLP and deep learning in student feedback. The paper of Dalipi et al. [5] focuses on sentiment analysis in MOOCs, while Shaik et al. [6] talk about different analytical levels, negation, and polarity. In their turn, Grimalt-Álvaro and Usart [15] pinpoint some limitations in the use of sentiment analysis tools in education.

In their analysis of lexicon-based and machine learning systems for student feedback Dervenis, Kanakis and Fitsilis [16], offer evidence that comparisons should be conducted on a similar task. The ResBack method will be evaluated against independently generated labels, and an alternative model/lexicon used as a benchmark.

2.1.4 Philippine Applications and Multilingual Feedback

Balahadia et al. [7] analyze English and Filipino teacher-evaluation comments and display sentiment results graphically. Justo [8] applies sentiment analytics to academic services

among computer studies students at Laguna State Polytechnic University. These applications support ResBack's coverage of teaching and broader CCIS services.

Bonifacio et al. [9] introduce ClasSentiments, which uses English, Tagalog, and Bisaya texts for evaluating teaching quality at a Philippine State University. The labeling of the comments, visualization of the data and human oversight make for an extremely similar functional comparison. Their models and evaluation results cannot be compared to those of ResBack's Gemma.

Bautista [17] mentions Tagalog-English code-switching as a linguistic tool, while Herrera, Aich, and Parde [18] present TweetTaglish for analyzing this language context. The two references mentioned above accept multilingual input, though neither gives a CCIS sentiment corpus or classifies Ilocano.

Samanta et al. [19] study sentiment detection in code-mixed text, whereas Conneau et al. [20] show multilingual representation learning using XLM-R. These studies lead us to perform experiments in different languages and mixed languages. ResBack will test language identification and sentiment classification independently.

2.1.5 Pretrained Language Models and Structured Analysis

The Gemma [21] and Gemma 4 [22] technical reports describe the pretrained model families and their benchmark performance. They establish model provenance but do not measure accuracy on CCIS feedback.

Kuzmanova et al. [23] use Gemini in feedback provided by students at the university level, which involves linguistic variability, slang, and sentiment. The work done by Nguyen et al. [24] includes EduPulse, which is used for feedback in Vietnam that involves sentiment, categorization, suggestion detection, and summarization.

Requests made by ResBack require structured JSON and validation of field values prior to storage. This is done to ensure usable data format but human readable labels are necessary for meaning assessment.

2.1.6 Concern Topics Confidence and Prioritization

Pontiki et al. [25], there is a difference between identifying aspects and finding sentiments towards the aspects. ResBack provides topic labels to multi-topics reviews but only provides one single sentiment label, which makes it less effective than aspect based sentiment analysis.

Guo et al. [26], neural-network confidence does not necessarily equate to the accuracy of the predictions made by neural networks. Calibration results provided by the researchers cannot be used as proof of the validity of Gemma's calculated confidence figures.

ResBack's weighting system for its ranking (40 percent negative volume, 25 percent negative ratio, 20 percent confidence, and 15 percent recency) is a project heuristic. Their appropriateness will be evaluated by means of controlled cases, faculty evaluations, and sensitivity testing.

2.1.7 Dashboards Reporting and Human Review

Schwendimann et al. [27] discuss educational dashboards and information visualization. Kaliisa et al. [28] cover the design of teacher-oriented dashboards, stakeholder participation, privacy, and evaluation. Masiello et al. [29] summarize the existing body of work on dashboards and their limitations.

These studies deal mostly with information about learning activities, while ResBack deals with the feedback that has been submitted. The design principle which is applicable here is

helping users to answer real-life review questions. This principle is implemented by ResBack through date filtering, language filtering, comments, and export.

2.1.8 Comparison of Related Systems and Synthesis

Table 1 compares the reported features, methods, and language coverage of related systems. Unreported features are not assumed to be absent.

Table 1 Comparison of related feedback systems

Related system	Setting and capability	Connection and distinction
Balahadia et al. [7]	Comments on English and Filipino teacher evaluations; opinion extraction.	Facilitates feedback categorization. ResBack incorporates identity protection based on accounts and ranking of topics.
Justo [8]	Academic services of LSPU; Sentiment analysis and Data Analytics.	Service feedback analysis support. CCIS language capabilities and ranking need local assessment.
Bonifacio et al. [9]	Evaluation process of multilingual faculty using transformer model.	Close Philippine comparison. The published work cannot be directly applied to ResBack.
Kuzmanova et al. [23]	Analysis of LLM-based feedback by students through Gemini.	Facilitates LLM Analysis Studies. ResBack enables validation of structured results and deals with failed queries.
Nguyen et al. [24]	Vietnamese educational feedback; LLM-assisted opinion mining.	Facilitates practical integration. Specific language and deployment environments require individual testing.

Related system	Setting and capability	Connection and distinction
Dervenis et al. [16]	Comparison of student feedback sentiment analysis techniques.	Supports measuring model suitability rather than assuming one method performs best.
ResBack project	Feedback from CCIS; Gemma's conclusion, topic summary, report, and weighted ranking.	Initial questionnaire completed; models and task-oriented systems yet to be evaluated.

The literature supports a central feedback channel, privacy controls, automated analysis evaluation, multilingual topic categorization, and clear reports. The ResBack system incorporates all these elements for CCIS and suggests evaluation of these for this environment.

2.2 Conceptual Framework

The theoretical framework takes the input-process-output format but is human-aided. The inputs consist of comments, timestamps, accounts, and access policies. The process entails validation, screening, pretrained inference, label standardization, aggregation, ranking, and reporting. The outputs consist of classified feedback, topics summary, ranked concerns, dashboards, and downloaded reports.

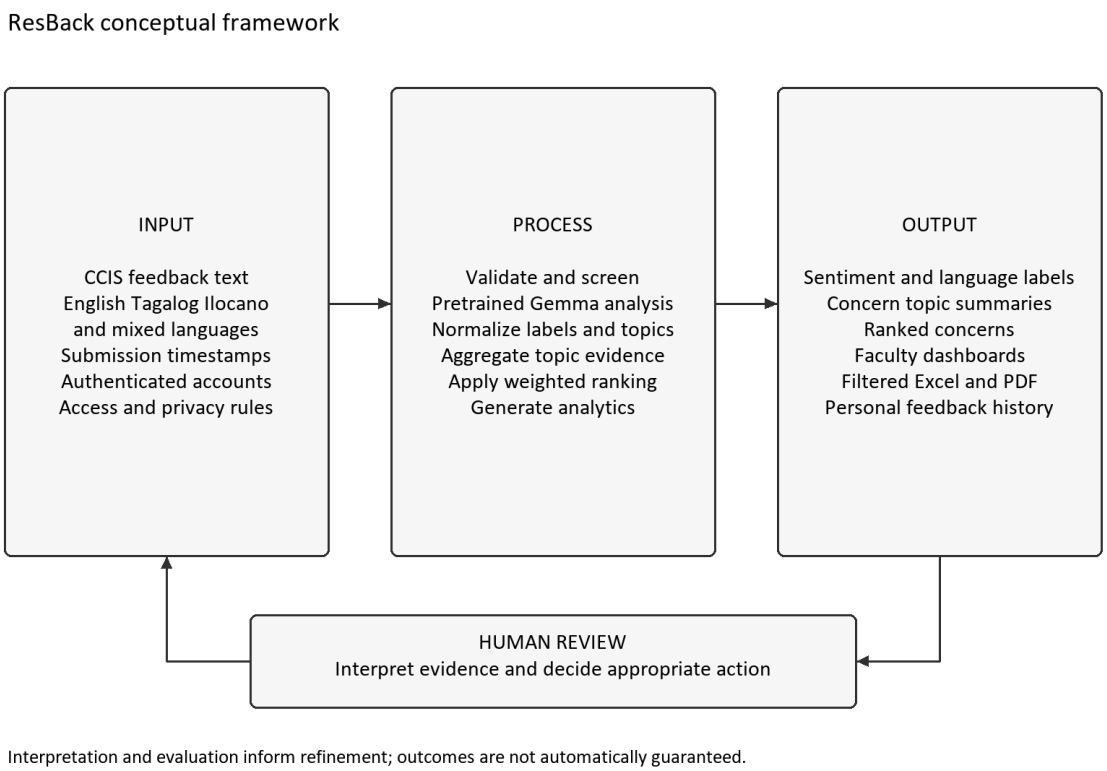


Figure 1 ResBack conceptual framework

Figure 1 shows that comments provide analytical content, timestamps provide recency scores, topic tags allow aggregation, and access policies control usage. The screening comes before inference and validation of the model's responses. There are privacy controls across the entire process.

Authorizations are required for reviewing the summaries along with comments. The feedback received from evaluation is used for changing interfaces, prompts, safety measures, and ranking criteria. The return arrow stands for human intervention along with system improvement and not for concern resolution through automation.

CHAPTER 3 METHODOLOGY

3.1 Project and Research Design

The project is following the software engineering development methodology approach to develop and test the Resback for CCIS. The development process includes requirement gathering, implementation, functional testing, and evaluation of analysis quality and usability. Requirement gathering from students through the initial questionnaire has been done. Model evaluation and usability testing by stakeholders have been suggested, and these activities have been defined in this chapter.

Agile Data Science 2.0 as outlined by Jurney [30] involves the process of iterations using Records, Charts, Reports, Recommendations, and Actions. In the case of Resback, these phases are connected to data gathering, visualization, report downloads, prioritizing concerns and human approval. Findings from the testing and review phase may call for modifications in previous phases. The process has been customized according to the current Laravel application without the need to use all the technologies that have been utilized in the book.

Resback incorporates a trained machine learning model using the Google Gemini API and implements programmed rules for input filtering, language and topic normalization, and concern prioritization. The research team does not train or fine-tune Gemma. Evaluation will thus include testing the trained model against CCIS feedback, checking that the calculation is correct, and watching participants perform tasks with the system. Questionnaire ratings of anticipated utility do not guarantee the validity or usability of the model.

3.2 Method and Approach

3.2.1 Setting Stakeholders and Preliminary Requirements Questionnaire

This research project will take place in CCIS in Mariano Marcos State University. The students can give feedbacks as well as check their submission records. The faculty members have access to give feedbacks and access to see their authorized reports and analytics. The Admin and Super Admin can view the analytics, manage the account based on their authority and delete the feedback.

The first Google Forms survey consists of consent questions, profile questions, 25 statements on a five-point Likert Scale of agreement, and five open-ended questions [3]. The dataset collected data for 25 volunteer students between September 24th and 29th, 2026: 17 from the Computer Science department and 8 from Information Technology. They all chose the age group of 18 to 25 years old. There are no faculty or administration in the dataset.

Table 2 links the project requirements to implementation areas and verification activities. These links will support traceability when features, prompts, or analytical rules are revised.

Table 2 Requirements and implementation traceability

ID	Requirement	Implementation or verification focus
R1	Authenticated private submission and history.	Account controls, FeedbackController, ownership checks.

ID	Requirement	Implementation or verification focus
R2	Suppress submitter identity in review outputs.	Dashboard and export field review; privacy task questions.
R3	Accept only active CCIS feedback categories.	StoreFeedbackRequest; category access tests.
R4	Screen abuse and limit repeated submissions.	FeedbackSubmissionGuard and FeedbackAutoRejectService.
R5	Validate sentiment and multilingual analysis.	SentimentService and LanguageCategoryService; annotated corpus.
R6	Normalize topics and calculate concern ranking.	ConcernTopicService and ConcernRankingService; arithmetic and sensitivity tests.
R7	Provide filtered dashboards and matching reports.	DashboardController, Excel and PDF controllers; record reconciliation.
R8	Restrict administrative and account actions.	Role middleware and account authorization; hierarchy tests.
R9	Expose analysis status and retain failed submissions.	Pending, analyzed, failed, and rejected lifecycle tests.
R10	Support repeatable evaluation and maintenance.	Test records, diagrams, modular services, and documented configuration.

The 25 students provided 625 ratings across the 25 agreement statements. Each item mean is the sum of its ratings divided by 25. Table 3 reports the unweighted mean of the

five item means within each questionnaire area. These descriptive results summarize this preliminary sample and do not constitute a usability or model-performance score.

Table 3 Preliminary questionnaire descriptive areas

Descriptive area	Items	Mean rating from 1 to 5
Perceived value of feedback	1–5	4.39
Anonymity and expression	6–10	4.62
Sentiment categorization	11–15	4.23
Concern ranking	16–20	4.18
Willingness and perceived usefulness	21–25	4.27

Anonymity has the highest area mean, at 4.62 on the five-point scale. The source summary records 23 of the 25 respondents, or 92 percent, as more comfortable providing feedback through this approach; the corresponding item mean is 4.68. The item about anonymity supporting honest expression has a mean of 4.64. An item mean alone does not establish that every respondent selected an agreement response.

The open-ended answers obtained from the questionnaire were thematically analyzed in relation to safety in expression, institution-related aspects, seriousness of the issue, students involved, accessibility, and progress updates. One participant was concerned with irrelevant submission and jokes. Blank answers and those with no substantial information were ignored in the thematic analysis. This is based on observations made from the questionnaires and not from interviews.

These results indicate that there is an identity protection system, personal feedback experience, summary information, and submission security measures. There is a need for resolution tracking, which highlights the limitation since the application tracks the submission and analysis but it doesn't track the resolution process entirely. The survey questionnaire lacks measurement of classification accuracy, validation of the ranking weights, and number of affected students.

Faculty and administrator interviews and walkthroughs that are planned will focus on access needs, language needs, understanding of the dashboard, and review processes. Recruitment, permission, and consent will be tracked. The preliminary survey sample will be noted as separate from the usability sample, even though some individuals participate in both efforts.

3.2.2 Agile Data Science 2.0 Development Process

The Records phase sets up accounts, feedback, categories, analysis findings, and ranks with explicit ownership, validation, status, and timestamps. The deliverables will be the data model, migration scripts, access control details, and sample records. Feedback and its analysis are stored separately such that a submission may still be retrievable even if external analysis fails.

Sentiment distributions, language classifications, trends, and concern summaries will be displayed at the Charts stage based on the data available in the records. Verification will involve matching the data totals in the display against the controlled data, as well as the manner in which the pending, failed, and missing analysis are indicated.

The Reports phase entails combining feedback tables, Excel export, and PDF analysis report through the use of identical supported filters. Tests will be done to see how the chosen records, statuses, and totals correlate with the controlled data. The review output should not include any rejected submissions and identification accounts, but only the feedback information.

Recommendations is a phase in which the feedback is organized according to concerns and in which the weighted ranking process described in Section 3.2.6 takes place. Count data, negative ratios, reported confidence by the models, recency of the complaint, and examples are made explicit in the justification for the ranking. The weights and low evidence markers will be evaluated using controlled studies, faculty reviews, and sensitivity analyses.

Under the Actions stage, the task allows authorized persons to investigate issues and determine what actions should be taken by the institution. The research team will take into consideration evaluation results when improving the documents, questions, interfaces, and ranking criteria. The refinement process will not involve automated issue solving. A new version of the Gantt chart, which includes all the same dates and development tasks, is in Appendix B.

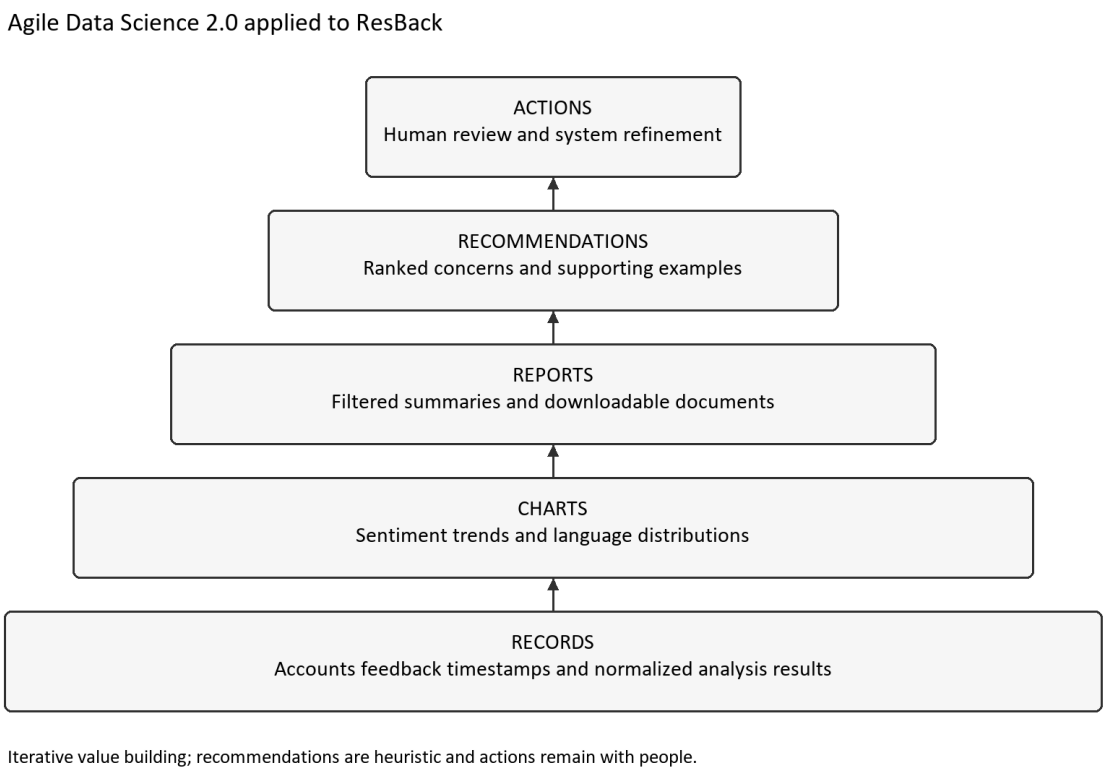


Figure 2 Agile Data Science 2.0 stages adapted to ResBack

3.2.3 Application Architecture and Role Permissions

The ResBack framework relies on Laravel routing, controllers, services, Eloquent model, and Blade views. Authentication, validation, and authorization define access to restricted operations. The controllers manage the submission and reports, whereas the services perform the screening, analysis, normalization, and ranking. Chart.js is used for charts, PhpSpreadsheet for generating Excel files, and Dompdf for PDF reports.

The figure below illustrates the architecture of the local development that is being done using PHP with Laravel framework, MySQL or MariaDB from XAMPP, local application files and the Google Gemini API over HTTPS protocol. The second figure below (figure 31 in appendix I) is the proposed design of the online deployment to be done using Docker on Render with Supabase PostgreSQL and Supabase Storage for profile pictures. This system still needs database compatibility tests and integration to be called deployment.

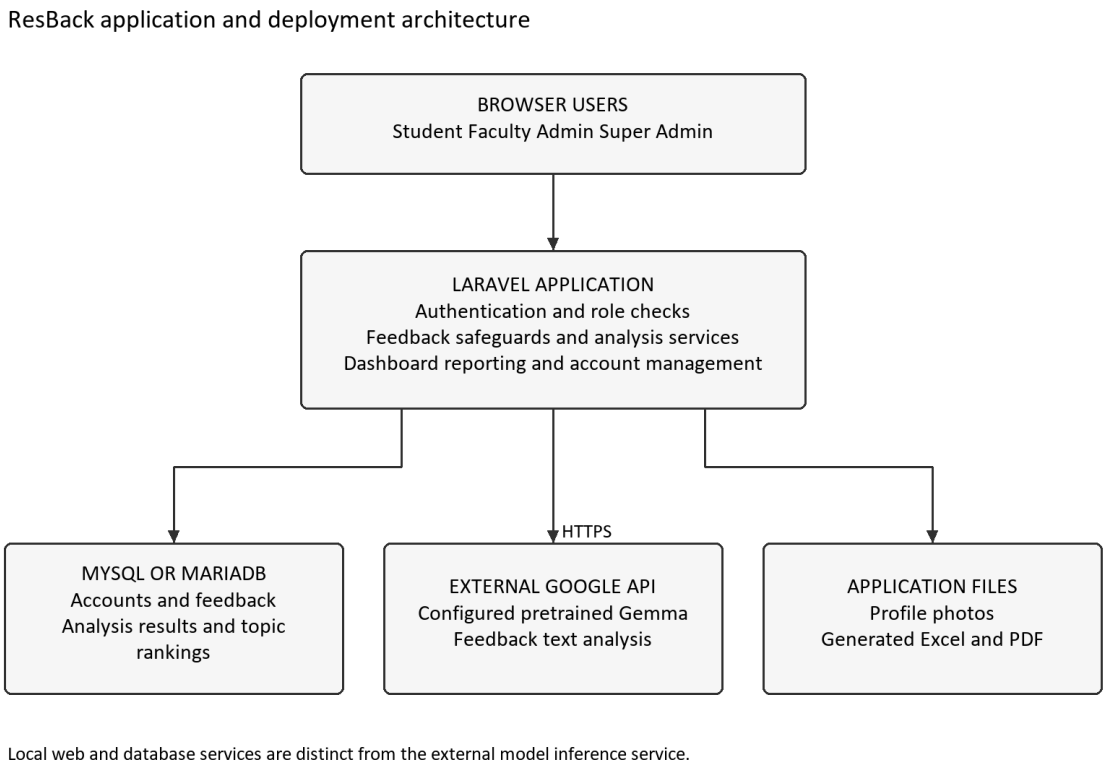


Figure 3 Local development architecture

Figure 28, which is located in Appendix G, has segmented the application into categories such as Web Presentation, Identity and Accounts, Feedback Management, Sentiment Analysis, Concern Ranking, and Dashboard & Reporting. All these are segments that logically fall into one Laravel application.

Component Diagram utilizes provided and required interfaces for service contracts and delegation connectors to connect application boundaries to inner components. External inference is provided by the Google Gemini API. Dompdf and PhpSpreadsheet work inside the application, while the profile-photo implementation currently works using the Laravel public filesystem. Application records and the database-backed session and cache records are kept in relational storage. Accepted feedback is processed synchronously, while rejected feedback is not used for external inference.

Student users are allowed to provide feedback, see only their feedback history, and make changes to their profile. The faculty users are able to submit feedback, as well as access dashboards and reports. The admins and super-admins can view reports and analytics, manage accounts in accordance with their role hierarchy, and delete feedback, but are not able to submit feedback.

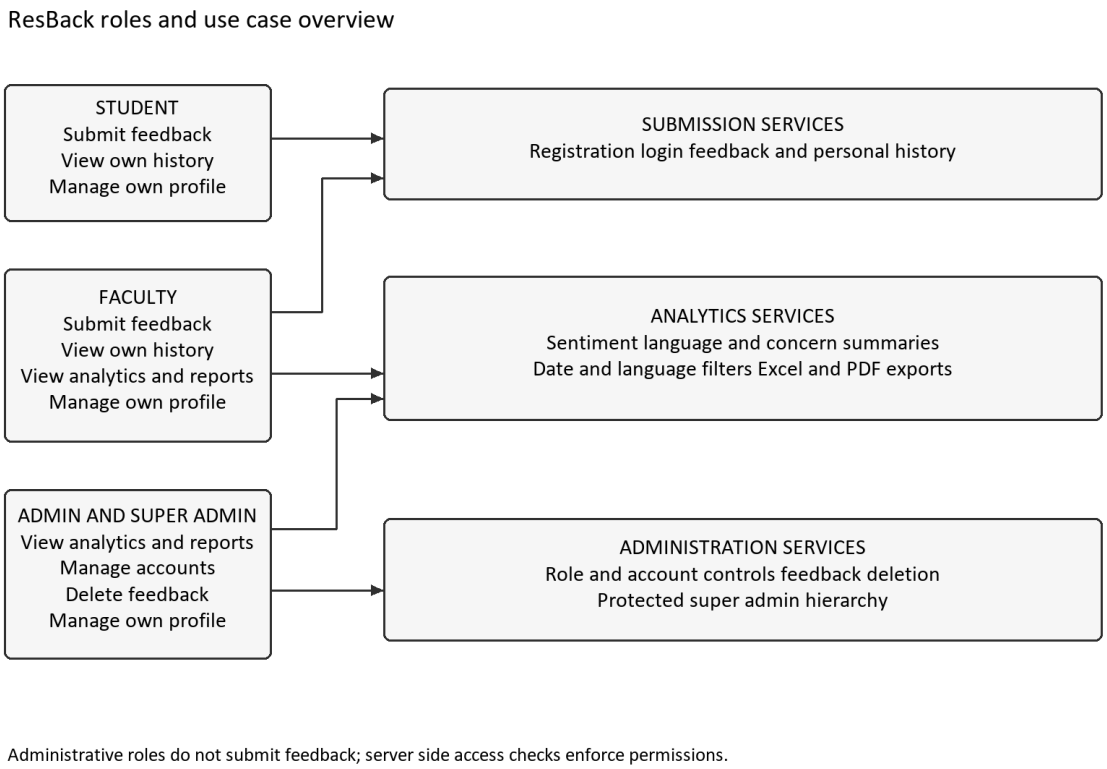


Figure 4 Roles and use case overview

These appendices have been presented in the prescribed order. Appendix A: Fishbone Diagram, Appendix B: Gantt Chart, Appendix C: Use Case Diagram, Appendix D: Use Case Description, Appendix E: Activity Diagrams, Appendix F: Class Diagrams, Appendix G: Component Diagram, Appendix H: Package Diagram, and Appendix I: Deployment Diagrams (Local & Online). The use case diagram consists of 17 use cases, along with their 17 descriptions. There are 12 activity diagrams and 6 class diagrams describing the application.

3.2.4 Data Model and Feedback Lifecycle

The fundamental entities are the following: User, Category, Feedback, Sentiment Result, and Concern Ranking. A user can be associated with more than one instance of feedback; moreover, each instance of feedback is linked with a category. The successful analysis leads to the creation or updating of the associated sentiment result. A category can have many instances of topic rankings.

Feedback log contains the original content, private owner, hashed network address, processing status, and time stamps. Analysis log contains sentiment, confidence of model, keywords, normalized topic names, language details, and the response of provider. Ranking log contains numbers and scores for all categories and topics. In the current ranking model, keyword field contains the normalized topic name.

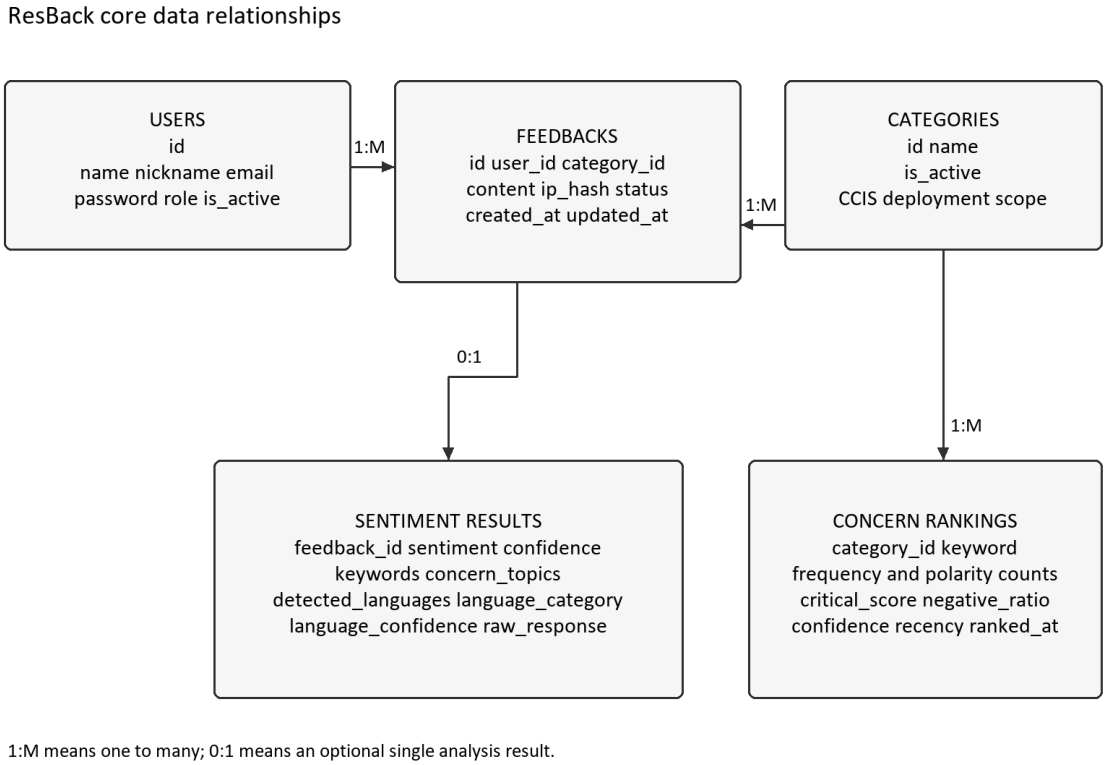


Figure 5 Core data relationships

There are four different statuses for the feedback life cycle. Pending refers to the saved submission that requires analysis. Analyzed refers to the submission that has a stored usable result. Failed refers to the saved submission whose analysis is not capable of producing a usable result. Rejected refers to the submission that is filtered based on the configured screening rules, without any inference or inclusion in analytics.

3.2.5 Submission Safeguards and Analysis Procedure

The server will accept only those comments which belong to the active CCIS category and contain 10-2000 characters. The duplicate check will be based on the same user and the same category for the last 24 hours, ignoring case and whitespace differences. There is a different rate-limited restriction for the user: saving 5 comments in 10 minutes.

The configured prohibited content dictionary filters validated feedback prior to inference. If there is a match, the submission will be stored as a rejected submission; if not, the submission will be stored as a pending submission. The text of the feedback remains intact, while a hash of the network address is maintained. Both accepted and rejected submissions are included in the submission limits.

The feedback sent by SentimentService includes the following elements – sentiment, confidence, keywords, topics, languages, and language_confidence – and is provided in the JSON format. The prompt asks the model to treat the feedback data as content rather than instructions to be executed. The response parser skips the thought segments, parses the JSON received, verifies the usable sentiment, confidence, keywords, and language information, and ensures that the confidence numbers are within the range between zero and one.

LanguagesCategoryService translates detected languages into English, Tagalog, Ilocano, Taglish, Iloclish, Taglocano, Trilingual, or Other Language. ConcernsTopicService translates concern titles into topics that are supported and resorts to using keywords and feedback body as evidence when required. Usable translation is recorded, feedback is analyzed, and category ranking is updated. Failure to provide configuration, loss of connection, API failure, or unprocessable analysis results makes the saved feedback failed.

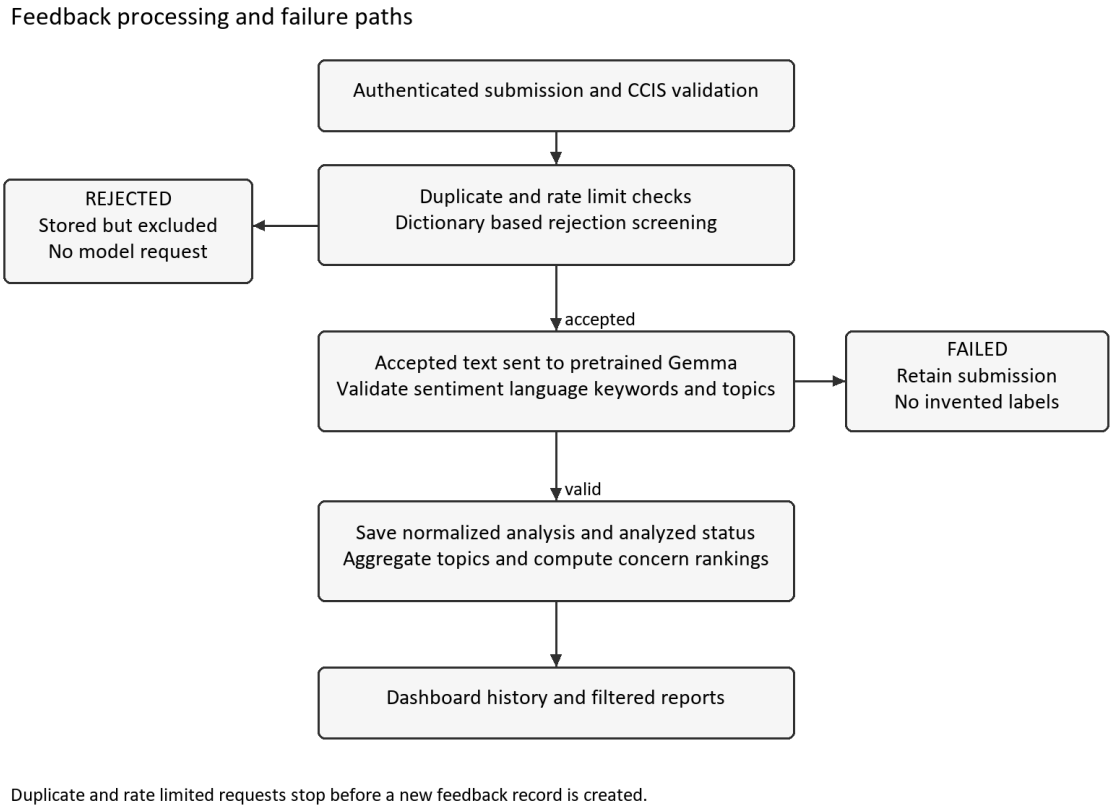


Figure 6 Feedback processing and failure paths

The default identifier for the repository default model is gemma-4-26b-a4b-it. The specific deployment can override the default identifier. The parameters currently in use for the request include a temperature of 0.1 and a maximum output length of 256 tokens. For the evaluation process, the following details will be logged: the actual model

that was used, version of prompt, request parameters, date of evaluation, and identified failures.

3.2.6 Concern Ranking Procedure

ConcernRankingService groups stored analysis results by normalized topic within the selected category, date, and language scope. It counts sentiments and calculates four factors for each topic. A comment assigned to several topics contributes its single overall sentiment to every assigned topic. This is topic grouping with comment-level sentiment; it is not a separate sentiment prediction for each aspect.

Let n be a topic's negative count, t its total analyzed count, and nmax the largest negative count among the selected topics, with a minimum of one. The volume factor is $V = \ln(1 + n) / \ln(1 + nmax)$, with $V = 0$ when $n = 0$. The negative ratio is $R = n / t$. The confidence factor C is the mean model-reported confidence of negative comments. The recency factor T is their mean $\exp(-\ln(2) \times \text{age in days} / 30)$, using non-negative ages relative to the calculation time. If there are no negative comments, C and T are zero.

Concern score $S = 100 \times (0.40V + 0.25R + 0.20C + 0.15T)$, to one decimal place. Normalization of logarithmic volume lessens the impact of very high values, and the ratio represents the proportion of negative comments. Recency factor of each comment decreases by half within 30 days. Weights and half-life are heuristic assumptions of this project and need to be evaluated. Confidence in model is not a probability of right classification [26].

For instance, scores for $V = 1.00$, $R = 0.75$, $C = 0.80$, and $T = 0.50$ yield 82.3. This is a demonstrative computation. The score for a topic having no negative comments equals zero. The existing low-evidence flagging is established when the number of comments on

a topic is less than three irrespective of the polarity of these comments. Scores are ranked in descending order, and in case of a tie, negative comment count and topic name break the tie.

Volume normalization depends on the largest negative count in the selected dataset. A topic score may therefore change when filters or other topic counts change, even if that topic's own records are unchanged. Verification will fix the calculation time and selected scope for reproducibility. Sensitivity analysis will compare alternative weights and recency half-lives, including a calculation without the confidence factor.

3.2.7 Proposed Evaluation Participants and Data

This is the usability sample that has been developed using purposive sampling which involves 30 CCIS students, five faculty members, and two administrators provided there is consent and availability. The student selection process shall involve variation among the year of study, browser usage, and language used. The reviewers' task shall involve some responsibilities associated with the tasks being tested.

This proposed corpus of frozen evaluations comprises 210 comments without any identifiers, wherein there are ten comments for each of the three sentiment classes in each of the seven language categories. The language categories considered are English, Tagalog, Ilocano, Taglish, Iloclish, Taglocano, and Trilingual. The other cases will be analyzed with respect to Other Language, ambiguous contents, screening matches, and service failures.

Comments may be contributed with consent or authored as explicitly identified test scenarios by contributors competent in the relevant languages. Source type will be recorded, duplicates removed, and unnecessary identifiers removed before external

processing. Natural and authored cases will be reported separately where sample size permits. Prompt-development examples will be separate from the 210 evaluation comments; the evaluation set will be frozen before the final prompt is assessed.

The annotation will be carried out independently by two annotators who possess the required language skills. Guidelines will include definitions for positive sentiment, neutral sentiment, negative sentiment, mixed sentiment, loanwords, and multiple topics. The agreement will be noted before any disagreements are resolved to create the reference labels. The guidelines and resolution process will make the evaluation method replicable.

3.2.8 Functional Testing and Quality Evaluation

Function tests will include authentication, ownership, role-based authorization, input validations, submission protections, analysis pass/fail status, normalization, filters, account management, deletion, and export tests. The current unit tests using PHPUnit will serve as the basis of the implementation. Each evaluation performed will document the environment, expected and actual results, failures, fixes, and re-tests. Successful test cases will be documented only if the run was conducted.

The chosen product quality factors will be structured according to ISO/IEC 25010:2023 [31]. Functional suitability, performance efficiency, reliability, security, compatibility, and maintainability will be studied. Interoperability will be studied through compatibility due to API and exportability. Maintainability testing will test modular service architecture, traceability, and reproducible testing. Usability will be analyzed using role-based task analysis and SUS. This selection of factors does not represent certification or a comprehensive evaluation of all factors of the standard.

End-to-end testing will make use of test-only accounts and test database. Calls to external models will be mocked for deterministic tests on the application and tested separately using inference requests. Failure modes can be in the absence of configuration, timeout on connections, failure response, malformed JSON, and incorrect labels. Testing for export will validate chosen records and their total amounts without the identifying fields of the accounts and rejected feedback.

Performance statistics will include the submission latency, the dashboard latency, the export time, failure percentage of analysis, size of the dataset, environment and API condition. Latency distribution will be computed through repeated submissions, and not just through one measurement. The external inference time will be tracked separately from local processing where possible, as synchronous requests influence overall submission time.

Table 4 Proposed evaluation measures and records

Area	Proposed measure	Evidence to record
Functional behavior	Expected outcomes for permissions, safeguards, filters, and exports.	Case results, failures, fixes, and retests.
Sentiment analysis	Accuracy, class precision and recall, macro-F1.	Confusion matrix; labels; ambiguous and failed attempts.
Language and topics	Language accuracy; topic micro- and macro-F1.	Category counts, missed or extra labels, annotator agreement.

Area	Proposed measure	Evidence to record
Concern ranking	Arithmetic checks, rank agreement, leading-concern agreement, sensitivity.	Controlled data; independent faculty rankings and comments.
Usability	Task completion, errors, assistance, time, and SUS.	Role-specific observations; scores and achieved sample.
Performance and reliability	Latency distributions, export duration, failure rate.	Environment, dataset size, API conditions, repeated runs.
Security and privacy	Ownership, role, export identity, and data handling checks.	Access test results; disclosures; approved data procedures.

3.2.9 Proposed Usability and Analysis Measures

Usability tasks will be carried out according to role permissions. Students will submit feedback, check result status, and find their history. Faculty will filter analytics, review support cases, and create reports. Administrators will conduct permissible user and feedback tasks using test data. Results of task performance, error rates, help needed, and task duration will be recorded before the completion of the ten-question System Usability Scale (SUS) [32].

SUS uses ten items rated on a five-point response scale. For odd-numbered items, the contribution is the response minus one. For even-numbered items, the contribution is five minus the response. The ten contributions are summed and multiplied by 2.5 to produce a score from zero to 100. This score is a usability index, not a percentage of completed tasks.

Evaluation of sentiment classification performance will rely on comparison between predictions and human-adjudicated reference labels via accuracy, class-based precision and recall, macro-F1, and confusion matrix analysis. Accuracy of language category classification will be computed both globally and for each category. Topic assignment performance will rely on micro-F1 and macro-F1 as well as specific instances of misses and false positives. Non-responsive submissions and failed request cases will be tracked separately along with valid response coverage.

The controlled record will validate the mathematics involved in ranking, tiebreakers, dates, filtering, and low evidence indicators. The faculty reviewers will consider the identical anonymous evidence independently prior to seeing the system ranking. The comparison will focus on the agreement in rank and leading problems, with both the rank statistic and tiebreaker method decided prior to evaluation. Disagreement will not be considered evidence that either ranking is right.

The acceptance criteria shall be determined in advance of the evaluation through discussion with the stakeholders. Ownership and role checks should succeed, exports should correspond to the chosen scope, and processing errors need to stay visible. Error reports, sample limitations, and coverage attained shall be part of model and usability reports. A benchmark appropriate for the subset of the language under test can be evaluated using the same set of reference data; its language coverage and configuration shall be described before comparison.

3.2.10 Ethical Procedures and Data Management

Participation is voluntary and dependent on institutional approval. Information sheets will describe the aim of the research, tasks to be performed, information collection and usage,

withdrawal rules, and model processing by third parties. Participation or non-participation will not influence grades or regular access to accounts. Proposed research procedures and preliminary questionnaire will be clearly marked in research documentation.

Research testing shall make use of test accounts, limited identifiers, limited research file access, and approved retention/deletion periods. Sensitive allegations will not be used as testing material. Identity masking on the dashboard is not able to protect identity from being exposed through free-text input nor is it able to remove the ownership of the application. Data privacy shall be assessed by the institution based on the Data Privacy Act Principles [13].

3.3 Machine Learning Integration and Evaluation

The following diagram depicts the machine learning pipeline linking the Gemma integration implementation to the processes of data collection, data pre-processing, feature extraction, model evaluation, and model usage. Further details about safety measures, data requirements, and evaluation metrics can be found in Sections 3.2.5-3.2.9. The project leverages an external pre-trained model [21][22] whose experiments will not train any model parameters.

3.3.1 Data Acquisition

The input for the operational use case comes in the form of feedback provided by authentic Student and Faculty users along with details such as category and timestamp. The application maintains a private ownership link, but does not include identification fields related to account in the review output. The research data will include the

individually documented consented/authored evaluation comments mentioned in Section 3.2.7.

3.3.2 Preprocessing and Data Cleaning

The application preprocessing ensures that the category and text size are validated, performs duplicate checks on normalized text, limits submissions, and checks configured prohibited keywords. The original text that is accepted goes through the model without undergoing stemming, stop-word removal, and TF-IDF processing. Preparation for the research will remove any unnecessary identifiers and duplicates, as well as language and references tags. Sentiment in wording will be maintained even during the process of cleaning.

3.3.3 Feature Extraction Using the Pretrained Model

Gemma utilizes representations obtained during pre-training. ResBack does not use any internal embeddings or manually create a feature vector. It receives sentiment, self-reported confidence, keywords, concern topics, and language data. Validation and normalization processes turn them into analytic features that will be filtered, aggregated, and ranked. The ranking criteria are defined by the programmer using the previously saved results and timestamps.

3.3.4 Model Configuration and Performance Evaluation

The same parameters will be used for each experiment, including the model, prompt, parameters, and evaluation samples. Two annotators will determine reference labels individually. Metrics from Section 3.2.9 will include sentiment accuracy, precision, recall, and macro-F1; language accuracy; and topic micro-F1 and macro-F1. Output will include

counts of samples, coverage, failure, and examples. The separate prompt development dataset will prevent evaluation leakage. Loss and epochs will not be part of the implementation.

3.3.5 Utilization of the Model

Validated feedback is subjected to synchronous inference, verification, normalization, result persistence, and ranking restoration. Dashboard and reporting tools show the chosen records for review by designated reviewers. The rejected records will not go through inference while the unsuccessful analysis will maintain their visibility. Reviewers study the actual comments and the related evidences before taking action in the institution.

REFERENCES

[1] J. T. E. Richardson, “Instruments for Obtaining Student Feedback: A Review of the Literature,” Assessment & Evaluation in Higher Education, vol. 30, no. 4, pp. 387–415, 2005. [Online]. Available: https://www.tandfonline.com/doi/abs/10.1080/02602930500099193

[2] M. Shah, M. Cheng, and R. Fitzgerald, “Closing the Loop on Student Feedback: The Case of Australian and Scottish Universities,” Higher Education, vol. 74, pp. 115–129, 2017. [Online]. Available: https://shura.shu.ac.uk/30828/

[3] C. J. G. Queypo, “ResBack preliminary requirements questionnaire responses,” Unpublished Google Forms response dataset, 25 student records, September 24–29, 2026. [Online]. Available: https://docs.google.com/spreadsheets/d/148lKjRJ4cuUbL5wjAZgBJGqB2NjcOWQOqkyIDnTaSZw/edit

[4] Z. Kastrati, F. Dalipi, A. S. Imran, K. P. Nuci, and M. A. Wani, “Sentiment Analysis of Students’ Feedback with NLP and Deep Learning: A Systematic Mapping Study,” Applied Sciences, vol. 11, no. 9, article 3986, 2021. [Online]. Available: https://www.mdpi.com/2076-3417/11/9/3986

[5] F. Dalipi, K. Zdravkova, and F. Ahlgren, “Sentiment Analysis of Students’ Feedback in MOOCs: A Systematic Literature Review,” Frontiers in Artificial Intelligence, vol. 4, article 728708, 2021. [Online]. Available: https://www.frontiersin.org/journals/artificial-intelligence/articles/10.3389/frai.2021.728708/full

[6] T. Shaik, X. Tao, C. Dann, H. Xie, Y. Li, and L. Galligan, “Sentiment Analysis and Opinion Mining on Educational Data: A Survey,” Natural Language Processing Journal, vol. 2, article 100003, 2023. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2949719122000036

[7] F. F. Balahadia, M. C. G. Fernando, and I. C. Juanatas, “Teacher’s Performance Evaluation Tool Using Opinion Mining with Sentiment Analysis,” 2016 IEEE Region 10 Symposium (TENSYMP), pp. 95–98, 2016. [Online]. Available: https://paraverse.feutech.edu.ph/repository/article/10.1109/TENCONSpring.2016.7519384

[8] R. C. Justo, “Academic-Related Services of Laguna State Polytechnic University with Sentiment Analysis and Data Analytics Using Multinomial Naïve Bayes Algorithm,” UE Research Journal, vol. 26, no. 1, 2024. [Online]. Available: https://www.ejournals.ph/article.php?id=33695

[9] K. L. Bonifacio, J. Estrada, M. M. Talandron-Felipe, G. Ayunar, F. Bascones Jr., N. J. Casildo, and J. Marcelo, “Automated Multilingual Sentiment Analysis of Student Comments in Faculty Evaluations using Transformer-Based AI,” Proceedings of the 33rd International Conference on Computers in Education, 2025. [Online]. Available: https://library.apsce.net/index.php/ICCE/article/view/6051

[10] S. Slade, and P. Prinsloo, “Learning Analytics: Ethical Issues and Dilemmas,” American Behavioral Scientist, vol. 57, no. 10, pp. 1510–1529, 2013. [Online]. Available: https://journals.sagepub.com/doi/10.1177/0002764213479366

[11] A. Pardo, and G. Siemens, “Ethical and Privacy Principles for Learning Analytics,” British Journal of Educational Technology, vol. 45, no. 3, pp. 438–450, 2014. [Online]. Available: https://bera-journals.onlinelibrary.wiley.com/doi/abs/10.1111/bjet.12152

[12] J. Fraile, M. Monguillot, C. González-Arévalo, P. Lehane, and E. Panadero, “Investigating Anonymity in Formative and Summative Peer Assessment: Effects on University Students’ Social-Affective Factors, Perceptions and Preference,” Studies in Educational Evaluation, vol. 83, article 101410, 2024. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0191491X24000890

[13] National Privacy Commission, “Implementing Rules and Regulations of Republic Act No 10173 known as the Data Privacy Act of 2012,” Philippines, 2016. [Online]. Available: https://privacy.gov.ph/implementing-rules-regulations-data-privacy-act-2012/

[14] B. Pang, L. Lee, and S. Vaithyanathan, “Thumbs Up? Sentiment Classification Using Machine Learning Techniques,” Proceedings of EMNLP 2002, pp. 79–86, 2002. [Online]. Available: https://aclanthology.org/W02-1011/

[15] C. Grimalt-Álvaro, and M. Usart, “Sentiment Analysis for Formative Assessment in Higher Education: A Systematic Literature Review,” Journal of Computing in Higher Education, vol. 36, pp. 647–682, 2024. [Online]. Available: https://link.springer.com/article/10.1007/s12528-023-09370-5

[16] C. Dervenis, G. Kanakis, and P. Fitsilis, “Sentiment analysis of student feedback: A comparative study employing lexicon and machine learning techniques,” Studies in Educational Evaluation, vol. 83, article 101406, 2024. [Online]. Available: https://doi.org/10.1016/j.stueduc.2024.101406

[17] M. L. S. Bautista, “Tagalog-English Code Switching as a Mode of Discourse,” Asia Pacific Education Review, vol. 5, pp. 226–233, 2004. [Online]. Available: https://link.springer.com/article/10.1007/BF03024960

[18] M. Herrera, A. Aich, and N. Parde, “TweetTaglish: A Dataset for Investigating Tagalog-English Code-Switching,” Proceedings of LREC 2022, pp. 2090–2097, 2022. [Online]. Available: https://aclanthology.org/2022.lrec-1.225/

[19] B. Samanta, N. Ganguly, and S. Chakrabarti, “Improved Sentiment Detection via Label Transfer from Monolingual to Synthetic Code-Switched Text,” Proceedings of ACL 2019, pp. 3528–3537, 2019. [Online]. Available: https://aclanthology.org/P19-1343/

[20] A. Conneau, K. Khandelwal, N. Goyal, V. Chaudhary, G. Wenzek, F. Guzmán, et al., "Unsupervised Cross-Lingual Representation Learning at Scale," Proceedings of ACL 2020, pp. 8440–8451, 2020. [Online]. Available: https://aclanthology.org/2020.acl-main.747/

[21] Gemma Team et al., "Gemma: Open Models Based on Gemini Research and Technology," Technical report, arXiv:2403.08295, 2024. [Online]. Available: https://arxiv.org/abs/2403.08295

[22] Gemma Team et al., "Gemma 4 Technical Report," Technical report, arXiv:2607.02770, 2026. [Online]. Available: https://arxiv.org/abs/2607.02770

[23] J. Kuzmanova, K. Zdravkova, and I. Chorbev, "Integrating Large Language Models for Comprehensive Study and Sentiment Analysis of Student Feedback," Proceedings of RANLP 2025, pp. 604–613, 2025. [Online]. Available: https://aclanthology.org/2025.ranlp-1.71/

[24] X. P. Nguyen, X. P. Nguyen, V. T. Nguyen, V. T. Dang, and N. L.-T. Nguyen, "EduPulse: A Practical LLM-Enhanced Opinion Mining System for Vietnamese Student Feedback in Educational Platforms," Proceedings of EACL 2026 Industry Track, pp. 338–365, 2026. [Online]. Available: https://aclanthology.org/2026.eacl-industry.25/

[25] M. Pontiki, D. Galanis, J. Pavlopoulos, H. Papageorgiou, I. Androutsopoulos, and S. Manandhar, "SemEval-2014 Task 4: Aspect Based Sentiment Analysis," Proceedings of SemEval 2014, pp. 27–35, 2014. [Online]. Available: https://aclanthology.org/S14-2004/

[26] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On Calibration of Modern Neural Networks," Proceedings of ICML 2017, PMLR, vol. 70, pp. 1321–1330, 2017. [Online]. Available: https://proceedings.mlr.press/v70/guo17a.html

[27] B. A. Schwendimann, M. J. Rodriguez-Triana, A. Vozniuk, L. P. Prieto, M. S. Boroujeni, A. Holzer, et al., "Perceiving Learning at a Glance: A Systematic Literature Review of Learning Dashboard Research," IEEE Transactions on Learning Technologies, vol. 10, no. 1, pp. 30–41, 2017. [Online]. Available: https://eric.ed.gov/?id=EJ1141028

[28] R. Kaliisa, I. Jivet, and P. Prinsloo, “A Checklist to Guide the Planning, Designing, Implementation, and Evaluation of Learning Analytics Dashboards,” International Journal of Educational Technology in Higher Education, vol. 20, article 28, 2023. [Online]. Available: https://link.springer.com/article/10.1186/s41239-023-00394-6

[29] I. Masiello, Z. A. Mohseni, F. Palma, S. Nordmark, H. Augustsson, and R. Rundquist, “A Current Overview of the Use of Learning Analytics Dashboards,” Education Sciences, vol. 14, no. 1, article 82, 2024. [Online]. Available: https://www.mdpi.com/2227-7102/14/1/82

[30] R. Jurney, “Agile Data Science 2.0 Building Full Stack Data Analytics Applications with Spark,” O’Reilly Media, 2017. [Online]. Available: https://www.oreilly.com/library/view/agile-data-science/9781491960103/

[31] International Organization for Standardization, “ISO/IEC 25010:2023 Systems and software engineering Systems and software Quality Requirements and Evaluation SQuaRE Product quality model,” International standard, 2023. [Online]. Available: https://www.iso.org/standard/78176.html

[32] J. Brooke, “SUS A quick and dirty usability scale,” In Usability Evaluation in Industry, P. W. Jordan, B. Thomas, B. A. Weerdmeester, and I. L. McClelland, Eds., Taylor & Francis, 1996. [Online]. Available: https://hci-studies.org/methods-and-measures/downloads/SUS_Brooke1996.pdf

APPENDIX A FISHBONE DIAGRAM

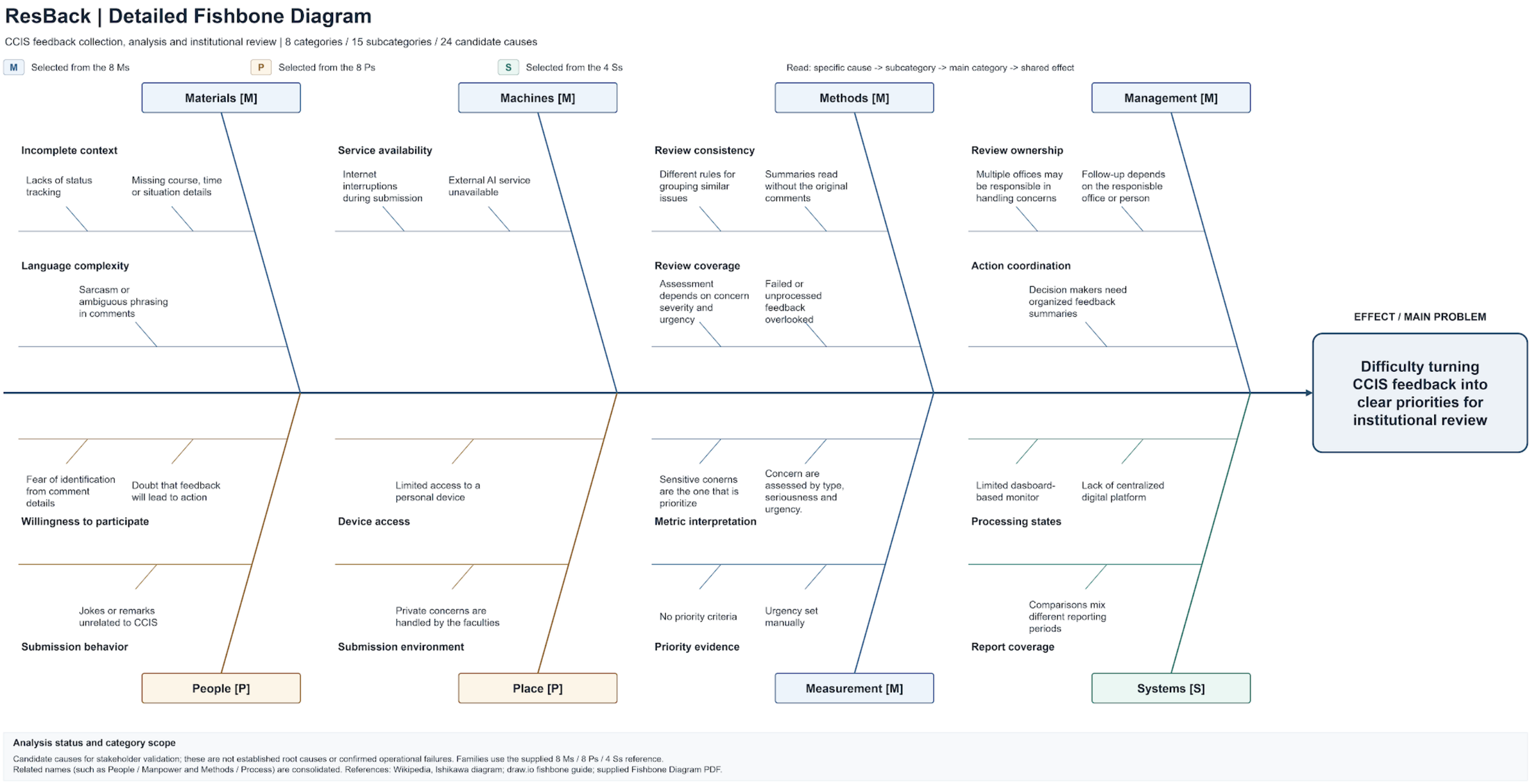


Figure 7 ResBack fishbone diagram

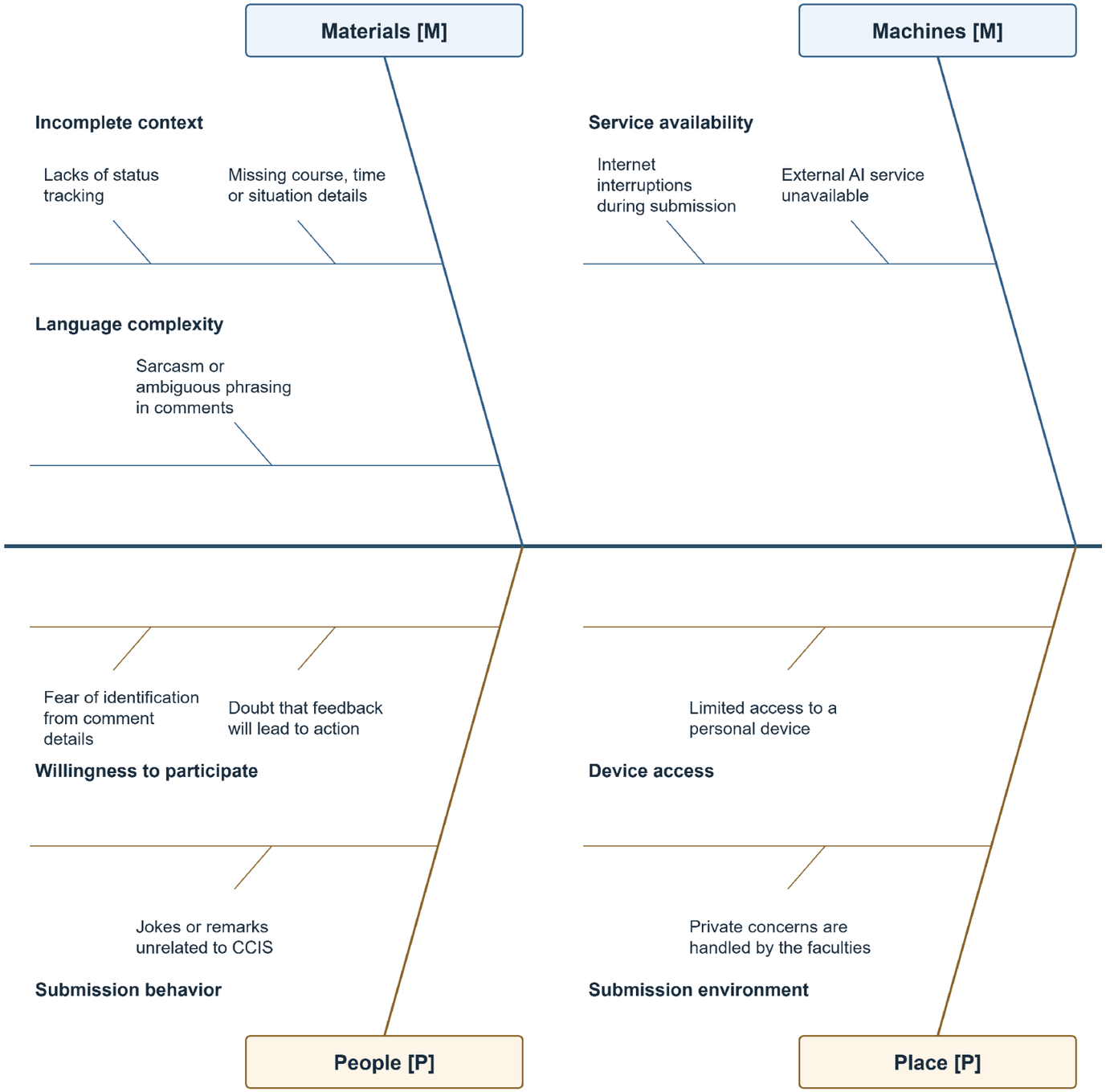


Figure 7 ResBack fishbone diagram enlarged continuation panel 1 of 2

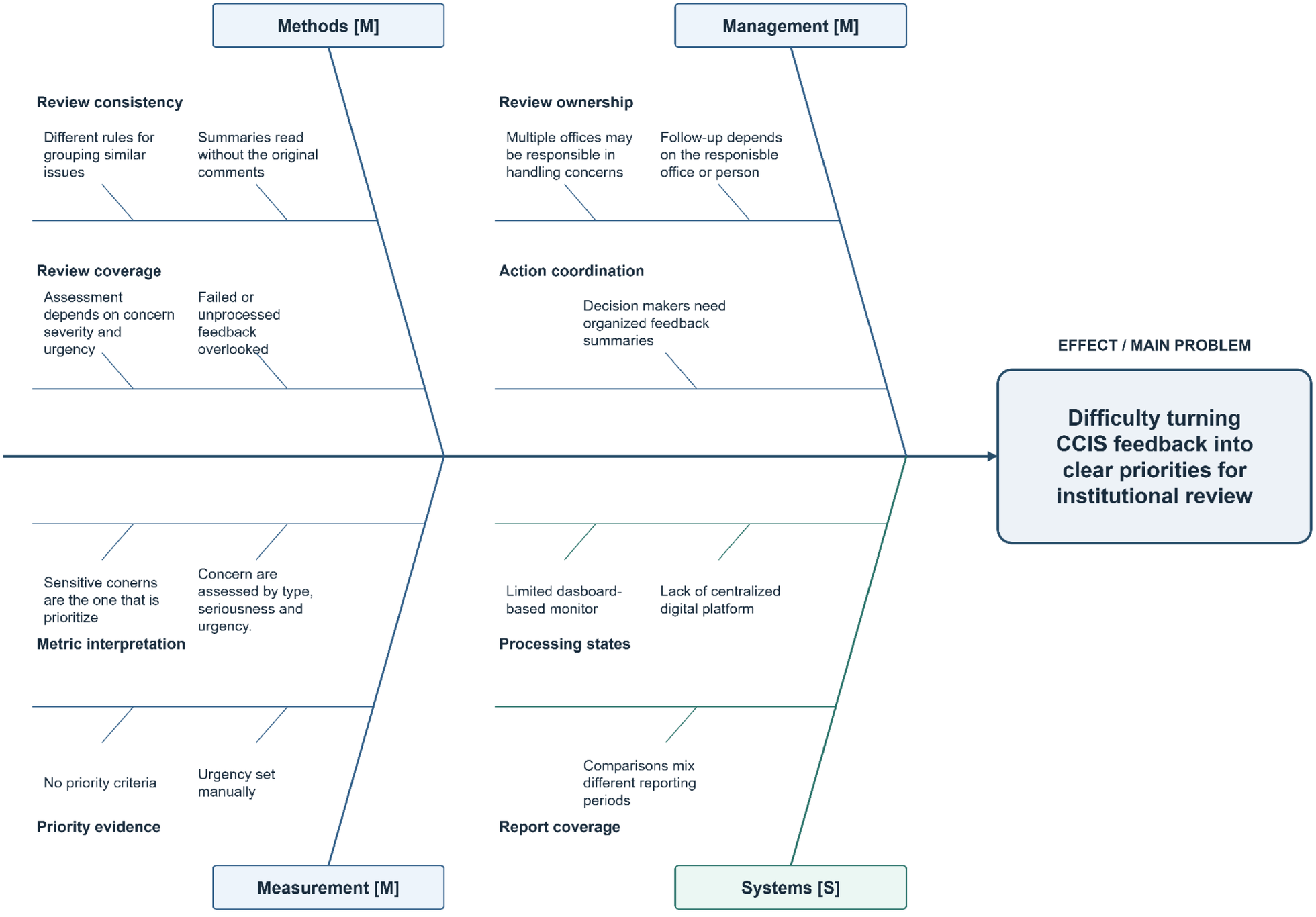


Figure 7 ResBack fishbone diagram enlarged continuation panel 2 of 2

APPENDIX B GANTT CHART

Figure 8 below shows the ResBack project schedule modified for use in Agile Data Science 2.0 framework. This project schedule contains all the task dates and the assigned team members from August 3 to October 31, 2026. The preparation stage assists in the completion of the following five stages of data value: Records, Charts, Reports, Recommendations, and Actions.

All task progress notations are made For confirmation. Bars for scheduling show planned task times and not actual completion. The chart uses the normal formatting of the template. Expanded continuation boxes list the tasks so that task descriptions can be aligned with their timeline and deliverables.

Week 1 is from August 3-9. All subsequent weeks start on a Monday; Week 5 starts on August 31, Week 9 starts on September 28, and Week 13 runs from October 26-31. The final partial week is included so that the end of all tasks can be shown until October 31.

GANTT CHART FOR THE PROJECT

Project:	Development of a Web-Based Anonymous Feedback System for CCIS Students Using Sentiment Analysis and Algorithmic Concern Ranking
Project Team:	Team Mira
Methodology:	Agile Data Science 2.0

PHASES	TASKS (ACTIVITIES)	RESPONSIBLE	START DATE	END DATE	PROGRESS	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	DELIVERABLE
Preparation	Review literature on sentiment analysis and concern ranking	Team Lead (Viloria)	Aug 3	Aug 5	For confirmation														Literature review
	Study existing anonymous feedback systems	All Members	Aug 6	Aug 8	For confirmation														Comparative analysis
	Assess pretrained sentiment models and API options	Lead Dev (Queypo)	Aug 6	Aug 9	For confirmation														Technology assessment
	Research concern-ranking methods	Lead Dev (Queypo)	Aug 9	Aug 11	For confirmation														Ranking-method review
	Prepare data-gathering instruments	Doc Lead (Orenza)	Aug 8	Aug 10	For confirmation														Survey and interview instruments
	Conduct interviews and surveys	Team Lead (Viloria)	Aug 11	Aug 14	For confirmation														Survey and interview results
	Identify functional and non-functional requirements	Team Lead & Doc Lead	Aug 12	Aug 16	For confirmation														Software requirements
	Analyze problems and stakeholder needs	All Members	Aug 15	Aug 18	For confirmation														Problem statement
	Prioritize requirements and analytical deliverables	Team Lead & Developers	Aug 19	Aug 21	For confirmation														Prioritized development tasks
	Define iteration goals and evaluation criteria	Team Lead & Developers	Aug 21	Aug 22	For confirmation														Iteration goals and criteria
	Design system architecture	Lead Dev (Queypo)	Aug 20	Aug 23	For confirmation														Architecture diagram
	Create interface wireframes	UI/UX Dev (Queypo)	Aug 21	Aug 25	For confirmation														Interface wireframes
	Set up the development environment	Lead Dev (Queypo)	Aug 26	Aug 28	For confirmation														Development environment
	Develop the interface prototype	UI/UX Dev (Queypo)	Aug 26	Sep 4	For confirmation														Interface prototype
Records	Design the feedback and analysis database schema	Lead Dev (Queypo)	Aug 22	Aug 24	For confirmation														Database schema
	Set up the database and backend	Lead Dev (Queypo)	Aug 28	Sep 2	For confirmation														Database and backend
	Implement authentication and role permissions	Lead Dev (Queypo)	Sep 3	Sep 8	For confirmation														Authentication and access rules
	Develop feedback submission and input screening	Lead Dev (Queypo)	Sep 9	Sep 14	For confirmation														Validated feedback submission
	Implement anonymous feedback presentation	Lead Dev (Queypo)	Sep 12	Sep 17	For confirmation														Submitter identity protection
	Store feedback and analysis status records	Lead Dev (Queypo)	Sep 15	Sep 20	For confirmation														Feedback and analysis records
	Develop the feedback submission interface	UI/UX Dev (Queypo)	Sep 14	Sep 20	For confirmation														Feedback interface
	Configure Gemma through the Gemini API	Lead Dev (Queypo)	Sep 19	Sep 21	For confirmation														Model and API configuration
	Validate text and normalize duplicate checks	Lead Dev (Queypo)	Sep 22	Sep 25	For confirmation														Input and duplicate validation
	Integrate sentiment, topic and language analysis	Lead Dev (Queypo)	Sep 24	Sep 30	For confirmation														Validated analysis outputs
Charts	Display sentiment and analysis results	UI/UX Dev (Queypo)	Sep 30	Oct 2	For confirmation														Sentiment result views
	Build dashboard charts and verify filtered totals	UI/UX Dev (Queypo)	Oct 11	Oct 18	For confirmation														Sentiment, language and trend charts
Reports	Develop feedback reports and shared filters	Lead Dev (Queypo)	Oct 14	Oct 20	For confirmation														Filtered feedback reports
	Export reports to PDF and Excel	Lead Dev (Queypo)	Oct 21	Oct 24	For confirmation														PDF and XLSX exports
Recommendations	Design weighted concern-ranking rules	Lead Dev (Queypo)	Oct 1	Oct 5	For confirmation														Volume, sentiment, confidence and recency rules
	Implement topic grouping and weighted ranking	Lead Dev (Queypo)	Oct 4	Oct 10	For confirmation														Concern-ranking implementation
	Validate ranking weights and low-evidence indicators	Lead Dev (Queypo)	Oct 9	Oct 14	For confirmation														Controlled ranking test results
	Display ranked concerns with supporting examples	UI/UX Dev (Queypo)	Oct 11	Oct 14	For confirmation														Ranked concerns and evidence
Actions	Conduct user acceptance testing and stakeholder review	QA Tester (Viloria)	Oct 25	Oct 28	For confirmation														UAT results and review findings
	Refine data, prompts and interfaces from review findings	Lead Dev (Queypo)	Oct 26	Oct 30	For confirmation														Refined system
Testing and iteration	Review results and iterate across the five data-value stages	All Members	Aug 26	Oct 31	For confirmation														Iteration review log
	Evaluate initial sentiment outputs against human labels	QA Tester (Viloria)	Sep 28	Oct 2	For confirmation														Initial sentiment evaluation
	Run unit tests and fix defects	All Members	Oct 18	Oct 24	For confirmation														Unit test results and fixes
	Test the integrated feedback-to-report workflow	QA Tester (Viloria)	Oct 22	Oct 26	For confirmation														Integration and system test report
	Evaluate sentiment performance against human labels	QA Tester (Viloria)	Oct 23	Oct 26	For confirmation														Accuracy, precision, recall and F1 results
Documentation and delivery	Configure and verify local and online deployment	Lead Dev (Queypo)	Oct 18	Oct 22	For confirmation														Local and online deployment checks
	Prepare presentation materials	UI/UX Dev (Queypo)	Oct 25	Oct 28	For confirmation														Presentation deck
	Complete technical documentation and diagrams	Doc Lead (Orenza)	Oct 25	Oct 30	For confirmation														Technical documentation
	Conduct the internal presentation rehearsal	All Members	Oct 29	Oct 31	For confirmation														Presentation rehearsal

Figure 8 ResBack Gantt chart using Agile Data Science 2.0

Preparation Task details

PHASES	TASKS (ACTIVITIES)	RESPONSIBLE	START DATE	END DATE	PROGRESS
Preparation	Review literature on sentiment analysis and concern ranking	Team Lead (Viloria)	Aug 3	Aug 5	For confirmation
	Study existing anonymous feedback systems	All Members	Aug 6	Aug 8	For confirmation
	Assess pretrained sentiment models and API options	Lead Dev (Queypo)	Aug 6	Aug 9	For confirmation
	Research concern-ranking methods	Lead Dev (Queypo)	Aug 9	Aug 11	For confirmation
	Prepare data-gathering instruments	Doc Lead (Orenza)	Aug 8	Aug 10	For confirmation
	Conduct interviews and surveys	Team Lead (Viloria)	Aug 11	Aug 14	For confirmation
	Identify functional and non-functional requirements	Team Lead & Doc Lead	Aug 12	Aug 16	For confirmation
	Analyze problems and stakeholder needs	All Members	Aug 15	Aug 18	For confirmation
	Prioritize requirements and analytical deliverables	Team Lead & Developers	Aug 19	Aug 21	For confirmation
	Define iteration goals and evaluation criteria	Team Lead & Developers	Aug 21	Aug 22	For confirmation
	Design system architecture	Lead Dev (Queypo)	Aug 20	Aug 23	For confirmation
	Create interface wireframes	UI/UX Dev (Queypo)	Aug 21	Aug 25	For confirmation
	Set up the development environment	Lead Dev (Queypo)	Aug 26	Aug 28	For confirmation
	Develop the interface prototype	UI/UX Dev (Queypo)	Aug 26	Sep 4	For confirmation

Timeline and deliverables Same tasks and order

TASKS (ACTIVITIES)	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	DELIVERABLE
Review literature on sentiment analysis and concern ranking														Literature review
Study existing anonymous feedback systems														Comparative analysis
Assess pretrained sentiment models and API options														Technology assessment
Research concern-ranking methods														Ranking-method review
Prepare data-gathering instruments														Survey and interview instruments
Conduct interviews and surveys														Survey and interview results
Identify functional and non-functional requirements														Software requirements
Analyze problems and stakeholder needs														Problem statement
Prioritize requirements and analytical deliverables														Prioritized development tasks
Define iteration goals and evaluation criteria														Iteration goals and criteria
Design system architecture														Architecture diagram
Create interface wireframes														Interface wireframes
Set up the development environment														Development environment
Develop the interface prototype														Interface prototype

Figure 8 ResBack Gantt chart enlarged continuation panel 1 of 3

Records, Charts, Reports and Recommendations Task details

PHASES	TASKS (ACTIVITIES)	RESPONSIBLE	START DATE	END DATE	PROGRESS
Records	Design the feedback and analysis database schema	Lead Dev (Queypo)	Aug 22	Aug 24	For confirmation
	Set up the database and backend	Lead Dev (Queypo)	Aug 28	Sep 2	For confirmation
	Implement authentication and role permissions	Lead Dev (Queypo)	Sep 3	Sep 8	For confirmation
	Develop feedback submission and input screening	Lead Dev (Queypo)	Sep 9	Sep 14	For confirmation
	Implement anonymous feedback presentation	Lead Dev (Queypo)	Sep 12	Sep 17	For confirmation
	Store feedback and analysis status records	Lead Dev (Queypo)	Sep 15	Sep 20	For confirmation
	Develop the feedback submission interface	UI/UX Dev (Queypo)	Sep 14	Sep 20	For confirmation
	Configure Gemma through the Gemini API	Lead Dev (Queypo)	Sep 19	Sep 21	For confirmation
	Validate text and normalize duplicate checks	Lead Dev (Queypo)	Sep 22	Sep 25	For confirmation
	Integrate sentiment, topic and language analysis	Lead Dev (Queypo)	Sep 24	Sep 30	For confirmation
Charts	Display sentiment and analysis results	UI/UX Dev (Queypo)	Sep 30	Oct 2	For confirmation
	Build dashboard charts and verify filtered totals	UI/UX Dev (Queypo)	Oct 11	Oct 18	For confirmation
Reports	Develop feedback reports and shared filters	Lead Dev (Queypo)	Oct 14	Oct 20	For confirmation
	Export reports to PDF and Excel	Lead Dev (Queypo)	Oct 21	Oct 24	For confirmation
Recommendations	Design weighted concern-ranking rules	Lead Dev (Queypo)	Oct 1	Oct 5	For confirmation
	Implement topic grouping and weighted ranking	Lead Dev (Queypo)	Oct 4	Oct 10	For confirmation
	Validate ranking weights and low-evidence indicators	Lead Dev (Queypo)	Oct 9	Oct 14	For confirmation
	Display ranked concerns with supporting examples	UI/UX Dev (Queypo)	Oct 11	Oct 14	For confirmation

Timeline and deliverables Same tasks and order

TASKS (ACTIVITIES)	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	DELIVERABLE
Design the feedback and analysis database schema														Database schema
Set up the database and backend														Database and backend
Implement authentication and role permissions														Authentication and access rules
Develop feedback submission and input screening														Validated feedback submission
Implement anonymous feedback presentation														Submitter identity protection
Store feedback and analysis status records														Feedback and analysis records
Develop the feedback submission interface														Feedback interface
Configure Gemma through the Gemini API														Model and API configuration
Validate text and normalize duplicate checks														Input and duplicate validation
Integrate sentiment, topic and language analysis														Validated analysis outputs
Display sentiment and analysis results														Sentiment result views
Build dashboard charts and verify filtered totals														Sentiment, language and trend charts
Develop feedback reports and shared filters														Filtered feedback reports
Export reports to PDF and Excel														PDF and XLSX exports
Design weighted concern-ranking rules														Volume, sentiment, confidence and recency rules
Implement topic grouping and weighted ranking														Concern-ranking implementation
Validate ranking weights and low-evidence indicators														Controlled ranking test results
Display ranked concerns with supporting examples														Ranked concerns and evidence

Figure 8 ResBack Gantt chart enlarged continuation panel 2 of 3

Actions, Testing and iteration, Documentation and delivery Task details

PHASES	TASKS (ACTIVITIES)	RESPONSIBLE	START DATE	END DATE	PROGRESS
Actions	Conduct user acceptance testing and stakeholder review	QA Tester (Viloria)	Oct 25	Oct 28	For confirmation
	Refine data, prompts and interfaces from review findings	Lead Dev (Queypo)	Oct 26	Oct 30	For confirmation
Testing and iteration	Review results and iterate across the five data-value stages	All Members	Aug 26	Oct 31	For confirmation
	Evaluate initial sentiment outputs against human labels	QA Tester (Viloria)	Sep 28	Oct 2	For confirmation
	Run unit tests and fix defects	All Members	Oct 18	Oct 24	For confirmation
	Test the integrated feedback-to-report workflow	QA Tester (Viloria)	Oct 22	Oct 26	For confirmation
	Evaluate sentiment performance against human labels	QA Tester (Viloria)	Oct 23	Oct 26	For confirmation
Documentation and delivery	Configure and verify local and online deployment	Lead Dev (Queypo)	Oct 18	Oct 22	For confirmation
	Prepare presentation materials	UI/UX Dev (Queypo)	Oct 25	Oct 28	For confirmation
	Complete technical documentation and diagrams	Doc Lead (Orenza)	Oct 25	Oct 30	For confirmation
	Conduct the internal presentation rehearsal	All Members	Oct 29	Oct 31	For confirmation

Timeline and deliverables Same tasks and order

TASKS (ACTIVITIES)	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	DELIVERABLE
Conduct user acceptance testing and stakeholder review														UAT results and review findings
Refine data, prompts and interfaces from review findings														Refined system
Review results and iterate across the five data-value stages														Iteration review log
Evaluate initial sentiment outputs against human labels														Initial sentiment evaluation
Run unit tests and fix defects														Unit test results and fixes
Test the integrated feedback-to-report workflow														Integration and system test report
Evaluate sentiment performance against human labels														Accuracy, precision, recall and F1 results
Configure and verify local and online deployment														Local and online deployment checks
Prepare presentation materials														Presentation deck
Complete technical documentation and diagrams														Technical documentation
Conduct the internal presentation rehearsal														Presentation rehearsal

Figure 8 ResBack Gantt chart enlarged continuation panel 3 of 3

APPENDIX C USE CASE DIAGRAM

Figure 9 shows the 17 ResBack use cases and their actor associations, including Google Gemini API as the external supporting actor for accepted-feedback analysis. Student, Faculty, Admin, and Super Admin permissions are represented within the system boundary. Super Admin inherits administrative capabilities, while feedback screening and accepted-feedback analysis are associated with submission.

Appendix D provides the corresponding use case descriptions. Figure numbers identify each diagram's position in this paper; labels within the source diagram sheets remain unchanged.

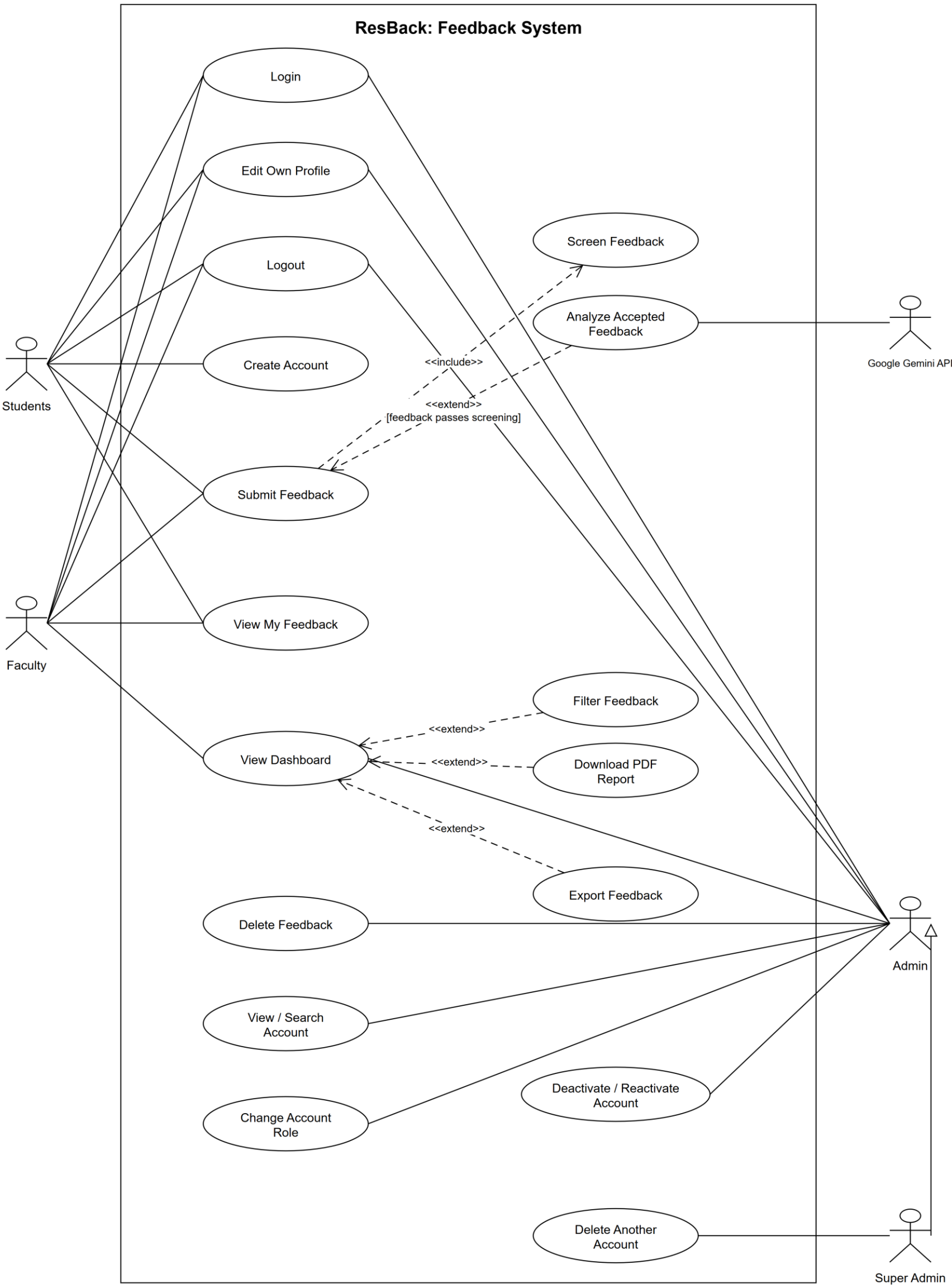


Figure 9 ResBack UML use case diagram

APPENDIX D USE CASE DESCRIPTIONS

The following 17 descriptions correspond to the use cases in Figure 9. Each table identifies the actors, preconditions, trigger, main steps, alternate and exception flows, postconditions, and non-functional requirements. The tables retain the previously prepared Lab 5 description format.

UC-01 Create Account

UC-02 Login

UC-03 Submit Feedback

UC-04 Screen Feedback

UC-05 Edit Own Profile

UC-06 Analyze Accepted Feedback

UC-07 View My Feedback

UC-08 View Dashboard

UC-09 Filter Feedback

UC-10 Download PDF Report

UC-11 Export Feedback

UC-12 Delete Feedback

UC-13 View / Search Account

UC-14 Delete Another Account

UC-15 Change Account Role

UC-16 Deactivate / Reactivate Account

UC-17 Logout

Table 5 UC-01 Create Account

Use Case Name	Create Account	
Use Case ID	UC-01	
Use Case Description	Allows a new Student to register an account and access the feedback submission form.	
Primary Actor	Student registering a new account	
Secondary Actor	None	
Precondition	The Student is not signed in and can access the registration page.	
Postcondition	A Student account is saved and an authenticated session opens the feedback form.	
Normal Flow (Main Flow)	User Action	System Response
	1. Selects Create Account.	2. Displays the registration form.
	3. Enters a nickname, email, password and password confirmation; submits the form.	4. Validates the required fields, uniqueness and password confirmation.
	5. None	6. Saves the account with the Student role and a hashed password.
	7. None	8. Starts an authenticated session and displays the feedback form.

Use Case Name	Create Account
Alternate Flow	1a. A Student who already has an account selects Login (UC-02) instead of registering again.
Exception Flow	4a. Invalid or duplicate details prevent account creation. The system identifies the fields to correct. A storage failure prevents successful registration.
Non-Functional Requirements	Passwords must be hashed. Validation must not reveal stored passwords. The authenticated session must be regenerated.

Table 6 UC-02 Login

Use Case Name	Login	
Use Case ID	UC-02	
Use Case Description	Allows an existing user to access ResBack according to the account role.	
Primary Actor	Student, Faculty, Admin or Super Admin	
Secondary Actor	None	
Precondition	The user has an active account and is not currently signed in.	
Postcondition	The user is authenticated and reaches the feedback form or dashboard allowed for the role.	
Normal Flow (Main Flow)	User Action	System Response
	1. Opens Login.	2. Displays the email and password fields.
	3. Enters the registered email and password; selects Login.	4. Validates the input, verifies the credentials and checks that the account is active.
	5. None	6. Starts an authenticated session and regenerates the session identifier.
	7. None	8. Opens the Student feedback form or the Faculty, Admin or Super Admin dashboard according to the account role.

Use Case Name	Login
Alternate Flow	1a. A Student without an account selects Create Account (UC-01). Successful registration signs the Student in automatically.
Exception Flow	4a. Incorrect credentials or an inactive account prevents login. Missing or invalid fields return validation errors.
Non-Functional Requirements	Password entry must be masked. Authentication errors must not expose stored credentials. Sessions must be protected from fixation.

Table 7 UC-03 Submit Feedback

Use Case Name	Submit Feedback	
Use Case ID	UC-03	
Use Case Description	Allows a Student or Faculty member to submit confidential CCIS feedback for screening and, when accepted, automatic analysis.	
Primary Actor	Student or Faculty	
Secondary Actor	None	
Precondition	The submitter is signed in with an active Student or Faculty account and can access the active CCIS feedback category. Admin and Super Admin cannot submit feedback.	
Postcondition	The feedback is saved with its screening and analysis result. The submitter receives the submission result.	
Normal Flow (Main Flow)	User Action	System Response
	1. Opens Submit Feedback.	2. Displays the CCIS feedback form.
	3. Enters feedback and selects Submit.	4. Validates the content, category, duplicate submission and submission limit.
	5. None	6. Performs Screen Feedback (UC-04) and saves accepted feedback as pending.
	7. None	8. Performs Analyze Accepted Feedback (UC-06) and

Use Case Name	Submit Feedback	
		displays the stored status and available analysis.
Alternate Flow	6a. If screening detects prohibited content, the system saves the feedback as rejected, displays the rejection result and skips UC-06.	
Exception Flow	4a. Invalid content/category, duplicate feedback or exceeding the submission limit prevents saving. 8a. Analysis failure leaves the saved feedback marked failed.	
Non-Functional Requirements	Feedback history must remain private to its owner. The result must distinguish an unsaved submission from a saved record whose analysis failed.	

Table 8 UC-04 Screen Feedback

Use Case Name	Screen Feedback	
Use Case ID	UC-04	
Use Case Description	Checks submitted feedback against configured prohibited-content rules before analysis.	
Primary Actor	Student or Faculty through Submit Feedback (UC-03)	
Secondary Actor	None	
Precondition	Feedback has passed submission validation and duplicate and submission-limit checks.	
Postcondition	The screening decision is returned to Submit Feedback; rejected content is excluded from AI analysis.	
Normal Flow (Main Flow)	User Action	System Response
	1. None	2. Receives the validated feedback text from Submit Feedback (UC-03).
	3. None	4. Normalizes the text and configured blocked terms and their variants.
	5. None	6. Checks the normalized text for prohibited matches.
	7. None	8. Returns an accepted decision to UC-03 when no prohibited match is found; UC-

Use Case Name	Screen Feedback	
		03 saves the feedback as pending and continues to UC-06.
Alternate Flow	6a. A prohibited match returns a rejected decision to UC-03. The system saves the record as rejected, displays the rejection result and skips UC-06.	
Exception Flow	4a/6a. A screening error prevents successful completion; the system must not report the submission as successfully analyzed.	
Non-Functional Requirements	Screening must apply the configured rules consistently. It must preserve the original feedback text and keep rejected content out of dashboard reports.	

Table 9 UC-05 Edit Own Profile

Use Case Name	Edit Own Profile	
Use Case ID	UC-05	
Use Case Description	Allows a signed-in user to update the user's own nickname and optional profile photo.	
Primary Actor	Student, Faculty, Admin or Super Admin	
Secondary Actor	None	
Precondition	The user is signed in with an active account.	
Postcondition	Valid profile changes are saved and a confirmation is displayed.	
Normal Flow (Main Flow)	User Action	System Response
	1. Opens the profile page.	2. Displays the current nickname and profile photo.
	3. Changes the nickname or selects a new photo; selects Save.	4. Validates nickname uniqueness and the permitted photo type and size.
	5. None	6. Stores or removes the profile photo as requested and updates only the signed-in user's profile.
	7. None	8. Displays the updated profile and a confirmation message.
Alternate Flow	3a. The user chooses to remove the existing photo before selecting Save. The user may leave the page without saving;	

Use Case Name	Edit Own Profile
	existing details remain.
Exception Flow	4a. A duplicate/invalid nickname or an invalid photo prevents the update. 6a. A photo-storage failure prevents successful completion.
Non-Functional Requirements	Users must only edit their own profiles. Nicknames protect real-name privacy. Photo uploads must be validated before storage.

Table 10 UC-06 Analyze Accepted Feedback

Use Case Name	Analyze Accepted Feedback	
Use Case ID	UC-06	
Use Case Description	Classifies accepted feedback by sentiment, language and concern topics using Gemma through the Gemini API.	
Primary Actor	Student or Faculty through Submit Feedback (UC-03)	
Secondary Actor	Google Gemini API (hosted Gemma inference service)	
Precondition	Screen Feedback (UC-04) accepted the content; the feedback is saved as pending and the analysis service is configured.	
Postcondition	A valid analysis result is saved, the feedback is marked analyzed and its category concern rankings are rebuilt.	
Normal Flow (Main Flow)	User Action	System Response
	1. None	2. Starts analysis for the accepted feedback record saved as pending by UC-03.
	3. None	4. Sends the feedback to the Gemini API with the required classification instructions.
	5. None	6. Receives and validates the returned sentiment, confidence, keywords, topics and language data.
	7. None	8. Saves the result, marks the

Use Case Name	Analyze Accepted Feedback	
		feedback analyzed and rebuilds the concern rankings; returns control to UC-03 to display the result.
Alternate Flow	4a. A temporary request failure may trigger a retry before the final result is determined.	
Exception Flow	4a/6a. Missing configuration, connection failure, API error or unusable output marks the saved feedback failed; it does not create a successful analysis result.	
Non-Functional Requirements	API credentials must remain confidential and transport must use TLS. Returned data must be validated. A service failure must preserve the submitted record.	

Table 11 UC-07 View My Feedback

Use Case Name	View My Feedback	
Use Case ID	UC-07	
Use Case Description	Allows a submitter to review only the feedback records belonging to the signed-in account.	
Primary Actor	Student or Faculty	
Secondary Actor	None	
Precondition	The Student or Faculty member is signed in with an active account. Feedback history is restricted to the account owner.	
Postcondition	The user sees the personal feedback history with processing status and available analysis.	
Normal Flow (Main Flow)	User Action	System Response
	1. Selects View My Feedback.	2. Checks that the signed-in user is permitted to access personal feedback history.
	3. None	4. Retrieves only the user's feedback records with their categories and available analysis, newest first.
	5. None	6. Displays the personal feedback history, processing statuses and available analysis in a paginated list.

Use Case Name	View My Feedback	
	7. Selects the next or previous history page when available.	8. Displays the requested page of the same user's records.
Alternate Flow	4a. If the user has no submissions, the system displays an empty history. 6a. Records without analysis show their current status.	
Exception Flow	1a. An expired session requires Login (UC-02). A retrieval failure prevents the history from being displayed.	
Non-Functional Requirements	History queries must enforce ownership. Pagination must not expose another user's records. Status labels must distinguish rejected, pending, failed and analyzed feedback.	

Table 12 UC-08 View Dashboard

Use Case Name	View Dashboard	
Use Case ID	UC-08	
Use Case Description	Allows authorized users to review CCIS feedback summaries, sentiment and language distributions, trends and ranked concerns.	
Primary Actor	Faculty, Admin or Super Admin	
Secondary Actor	None	
Precondition	The user is signed in with an active account and a role permitted to access the dashboard.	
Postcondition	The dashboard displays the available reportable feedback and analytics for the current scope.	
Normal Flow (Main Flow)	User Action	System Response
	1. Opens View Dashboard.	2. Checks dashboard access and retrieves reportable CCIS feedback.
	3. None	4. Calculates feedback counts, sentiment and language distributions and trends.
	5. None	6. Builds the leading concern rankings and retrieves the paginated feedback list.
	7. None	8. Displays the dashboard

Use Case Name	View Dashboard	
		summaries, charts, concerns and feedback, with Filter Feedback (UC-09), Download PDF Report (UC-10) and Export Feedback (UC-11).
Alternate Flow	2a. With no reportable records, the dashboard shows zero totals and an empty list. After step 8, the user may activate UC-09, UC-10 or UC-11.	
Exception Flow	1a. A Student or other unauthorized user is denied dashboard access. Invalid scope values or a retrieval failure prevent the requested view.	
Non-Functional Requirements	Access must be role controlled. Rejected feedback must be excluded from reports. Dashboard records must not disclose submitters' email addresses or real names.	

Table 13 UC-09 Filter Feedback

Use Case Name	Filter Feedback	
Use Case ID	UC-09	
Use Case Description	Narrows dashboard feedback and analytics by supported language and date-range criteria.	
Primary Actor	Faculty, Admin or Super Admin	
Secondary Actor	None	
Precondition	View Dashboard (UC-08) is open and the user has dashboard permission.	
Postcondition	Dashboard records and analytics reflect the selected valid filter criteria.	
Normal Flow (Main Flow)	User Action	System Response
	1. Opens the dashboard filter controls.	2. Displays supported language and date-range options for CCIS.
	3. Selects the desired criteria and applies the filters.	4. Validates the selected criteria.
	5. None	6. Retrieves matching reportable feedback and recalculates analytics for the selected scope.
	7. None	8. Displays the filtered feedback and analytics; retains

Use Case Name	Filter Feedback	
		the scope for PDF and spreadsheet downloads.
Alternate Flow	3a. The user clears optional filters to restore the default CCIS view. 6a. No matching records produces an empty result.	
Exception Flow	4a. Unsupported language/category values or invalid date ranges prevent applying the requested filter. A retrieval failure prevents refreshed results.	
Non-Functional Requirements	The same criteria must be applied consistently to records, analytics and downloads. Filter use must not bypass access restrictions.	

Table 14 UC-10 Download PDF Report

Use Case Name	Download PDF Report	
Use Case ID	UC-10	
Use Case Description	Produces a downloadable PDF containing CCIS feedback summaries, concern rankings and reportable feedback for the selected scope.	
Primary Actor	Faculty, Admin or Super Admin	
Secondary Actor	None	
Precondition	The user has dashboard permission and the requested language and date filters are valid.	
Postcondition	A PDF report is generated and returned for download.	
Normal Flow (Main Flow)	User Action	System Response
	1. Selects Download PDF Report for the current dashboard scope.	2. Checks access and validates the requested language and date filters.
	3. None	4. Retrieves matching reportable CCIS feedback and calculates summaries and concern rankings.
	5. None	6. Renders the PDF report with the selected scope and page numbering.
	7. None	8. Returns the PDF for

Use Case Name	Download PDF Report	
		download with a descriptive filename.
Alternate Flow	1a. With no optional filters, the report uses the default CCIS scope. 4a. With no matching feedback, the report contains the available zero/empty summaries.	
Exception Flow	2a. Invalid filters or unauthorized access prevents download. 4a/6a. A retrieval or PDF-generation failure prevents a completed report.	
Non-Functional Requirements	The PDF must be readable with consistent pagination. It must reflect the selected scope, exclude rejected feedback and avoid submitters' identifying account data.	

Table 15 UC-11 Export Feedback

Use Case Name	Export Feedback	
Use Case ID	UC-11	
Use Case Description	Exports matching reportable CCIS feedback and available analysis to an Excel spreadsheet.	
Primary Actor	Faculty, Admin or Super Admin	
Secondary Actor	None	
Precondition	The user has dashboard permission and the requested filters are valid.	
Postcondition	An XLSX spreadsheet is generated and returned for download.	
Normal Flow (Main Flow)	User Action	System Response
	1. Selects Export Feedback for the current dashboard scope.	2. Checks access and validates the requested language and date filters.
	3. None	4. Retrieves matching reportable CCIS feedback and available analysis.
	5. None	6. Writes feedback, dates, statuses and analysis into an XLSX spreadsheet with column headings and formatted values.
	7. None	8. Returns the XLSX file for

Use Case Name	Export Feedback	
		download with a descriptive filename.
Alternate Flow	1a. With no optional filters, the export uses the default CCIS scope. 4a. No matching feedback produces a spreadsheet containing the column headings.	
Exception Flow	2a. Invalid filters or unauthorized access prevents export. 4a/6a. A retrieval or spreadsheet-generation failure prevents a completed download.	
Non-Functional Requirements	The export must preserve the selected scope and exclude rejected feedback and identifying account fields. Feedback text must be written as text rather than executable spreadsheet formulas.	

Table 16 UC-12 Delete Feedback

Use Case Name	Delete Feedback	
Use Case ID	UC-12	
Use Case Description	Allows an administrator to permanently remove a selected feedback record and its associated analysis.	
Primary Actor	Admin or Super Admin	
Secondary Actor	None	
Precondition	The administrator is signed in with an active account and the selected feedback exists.	
Postcondition	The feedback and associated analysis are removed; the dashboard displays a deletion confirmation.	
Normal Flow (Main Flow)	User Action	System Response
	1. Locates a feedback record on the dashboard.	2. Displays the record and the permitted delete control.
	3. Selects Delete Feedback.	4. Presents the deletion confirmation.
	5. Confirms deletion.	6. Checks administrator permission and permanently deletes the record and associated analysis.
	7. None	8. Displays confirmation and the remaining feedback records.

Use Case Name	Delete Feedback
Alternate Flow	5a. The administrator cancels confirmation; the feedback remains unchanged.
Exception Flow	6a. A Faculty or Student user cannot delete feedback. A missing record or database failure prevents successful deletion.
Non-Functional Requirements	Deletion must enforce administrator authorization and request protection. Confirmation must make the permanent effect clear. Associated analysis must not remain orphaned.

Table 17 UC-13 View / Search Account

Use Case Name	View / Search Account	
Use Case ID	UC-13	
Use Case Description	Allows an administrator to view registered accounts and search them by nickname.	
Primary Actor	Admin or Super Admin	
Secondary Actor	None	
Precondition	The administrator is signed in with an active account and has account-management access.	
Postcondition	The account list or nickname search results are displayed.	
Normal Flow (Main Flow)	User Action	System Response
	1. Opens View / Search Account.	2. Displays accounts with nickname, role, status and registration date.
	3. Enters a nickname or part of a nickname and submits the search.	4. Validates the search text.
	5. None	6. Retrieves accounts matching the nickname search.
	7. None	8. Displays the paginated results while retaining the search criteria.
Alternate Flow	3a. An empty search displays the full paginated list. 6a. No	

Use Case Name	View / Search Account
	matching nickname displays an empty result. After step 8, the administrator may select another result page or an authorized account action (UC-14, UC-15 or UC-16).
Exception Flow	1a. An unauthorized user is denied access. Invalid search input or a retrieval failure prevents the requested result.
Non-Functional Requirements	Account lists must use nicknames and omit email addresses and real names. Search must enforce administrator access and validate input.

Table 18 UC-14 Delete Another Account

Use Case Name	Delete Another Account	
Use Case ID	UC-14	
Use Case Description	Allows a Super Admin to permanently delete an account other than the currently signed-in account.	
Primary Actor	Super Admin	
Secondary Actor	None	
Precondition	The Super Admin is signed in with an active account; the target account exists and belongs to another user.	
Postcondition	The target account is deleted and the updated account list shows a confirmation.	
Normal Flow (Main Flow)	User Action	System Response
	1. Finds the target using View / Search Account (UC-13).	2. Displays the account and its permitted delete control.
	3. Selects Delete Another Account.	4. Presents confirmation identifying the target by nickname.
	5. Confirms the deletion.	6. Verifies Super Admin permission and checks that the target is not the signed-in account.
	7. None	8. Deletes the target account and displays the updated

Use Case Name	Delete Another Account	
		account list with a success message.
Alternate Flow	5a. The Super Admin cancels confirmation; the target account remains unchanged.	
Exception Flow	6a. A regular Admin or any other role cannot delete accounts. Self-deletion is blocked. A missing target or database failure prevents successful completion.	
Non-Functional Requirements	Super Admin authorization and request protection must be enforced. Confirmation must make permanent deletion clear without exposing email addresses or real names.	

Table 19 UC-15 Change Account Role

Use Case Name	Change Account Role	
Use Case ID	UC-15	
Use Case Description	Allows an administrator to assign a permitted role to another registered account.	
Primary Actor	Admin or Super Admin	
Secondary Actor	None	
Precondition	The administrator is signed in with an active account and has opened View / Search Account (UC-13). The target account exists and is not the signed-in account.	
Postcondition	The permitted role is saved for the target account and a confirmation is displayed.	
Normal Flow (Main Flow)	User Action	System Response
	1. Opens View / Search Account (UC-13).	2. Displays account nicknames, current roles, statuses and permitted role controls.
	3. Selects another account, chooses a permitted role and selects Save.	4. Checks administrator authority and blocks self-role changes. Admin may manage only Student/Faculty targets; Super Admin may also manage administrator targets.
	5. None	6. Validates the selected role.

Use Case Name	Change Account Role	
		Admin may assign Student or Faculty; Super Admin may also assign Admin or Super Admin.
	7. None	8. Updates the target account role and displays confirmation.
Alternate Flow	3a. The administrator leaves the page without saving; the current role remains unchanged.	
Exception Flow	4a. Self-role changes or unauthorized management of an administrator account are blocked. 6a. An invalid or disallowed role prevents the update. A missing target or database failure prevents successful completion.	
Non-Functional Requirements	Authorization must be enforced on the server. Requests must be protected against forgery. Account controls must identify users by nickname and protect the signed-in account.	

Table 20 UC-16 Deactivate / Reactivate Account

Use Case Name	Deactivate / Reactivate Account	
Use Case ID	UC-16	
Use Case Description	Allows an administrator to deactivate or reactivate another account within the administrator's authority.	
Primary Actor	Admin or Super Admin	
Secondary Actor	None	
Precondition	The administrator is signed in with an active account and has opened View / Search Account (UC-13). The target account exists and is not the signed-in account.	
Postcondition	The target account status is changed and a confirmation is displayed. Deactivated accounts cannot log in or access operations that require an active account.	
Normal Flow (Main Flow)	User Action	System Response
	1. Opens View / Search Account (UC-13).	2. Displays account nicknames, roles, statuses and permitted status controls.
	3. Selects Deactivate for an active target or Reactivate for an inactive target.	4. Checks administrator authority and blocks changes to the signed-in account. Admin may manage only Student/Faculty targets; Super Admin may also manage administrator targets.

Use Case Name	Deactivate / Reactivate Account	
	5. None	6. Changes the target's active status to the opposite state and saves it.
	7. None	8. Displays the updated status and a confirmation message.
Alternate Flow	3a. For an inactive account, Reactivate restores active status. For an active account, Deactivate removes active status.	
Exception Flow	4a. Changing the signed-in account's status or unauthorized management of an administrator account is blocked. A missing target or database failure prevents successful completion.	
Non-Functional Requirements	Status updates must enforce role restrictions and request protection. Active-account checks must prevent a deactivated user from accessing protected operations.	

Table 21 UC-17 Logout

Use Case Name	Logout	
Use Case ID	UC-17	
Use Case Description	Allows a signed-in actor to end the current authenticated session.	
Primary Actor	Student, Faculty, Admin or Super Admin	
Secondary Actor	None	
Precondition	The actor has an authenticated session.	
Postcondition	The actor is signed out, the session is invalidated and the Login page is displayed.	
Normal Flow (Main Flow)	User Action	System Response
	1. Selects Logout.	2. Ends authentication for the current session.
	3. None	4. Invalidates the session and clears its stored data.
	5. None	6. Regenerates the request-protection token.
	7. None	8. Redirects the actor to Login (UC-02).
Alternate Flow	1a. If the session has already expired, access to the authenticated logout route redirects the actor to Login.	
Exception Flow	2a. An invalid request-protection token prevents the logout request from being processed; the actor must reload the page	

Use Case Name	Logout
	and submit a valid request.
Non-Functional Requirements	Logout must use a protected POST request. Invalidated sessions must not authorize further access to authenticated pages.

APPENDIX E ACTIVITY DIAGRAMS

Figures 10–21 show the twelve ResBack activities. Swimlanes separate user and system responsibilities. Every activity has one final endpoint; additional exits in the corrected diagrams loop back to an action. Larger workflows include continuation panels for readability.

Figure 10. Registration validates account details, handles duplicate values, creates an active Student account, and starts an authenticated session.

Figure 11. Login checks credentials and account status before routing the user according to role. Extra endings return to an action through a loop.

Figure 12. Submission covers confirmation, validation, duplicate and rate checks, screening, storage, and accepted or rejected results.

Figure 13. Personal feedback history displays the signed-in user’s submissions, processing status, and pagination.

Figure 14. Dashboard review validates date and language filters, retrieves matching feedback, summarizes sentiment, and ranks concerns.

Figure 15. Account management covers search, protected-account checks, role and status changes, and permitted account deletion. Disallowed or canceled actions return to the relevant action.

Figure 16. Analysis performs synchronous Gemma inference, validates the response, normalizes language and concern topics, stores results, and rebuilds rankings. Failed analysis remains visible as a failed status.

Figure 17. Excel export applies the selected filters and downloads feedback without submitter identities or rejected submissions.

Figure 18. PDF reporting uses the same filters and presents summaries and supporting feedback, including an empty state when appropriate.

Figure 19. Feedback deletion checks permission and the selected record. Cancellation preserves the record and returns to the previous action.

Figure 20. Profile editing validates the nickname and photo, handles photo replacement or removal, and saves permitted changes.

Figure 21. Logout invalidates the authenticated session, refreshes the CSRF token, and redirects the user.

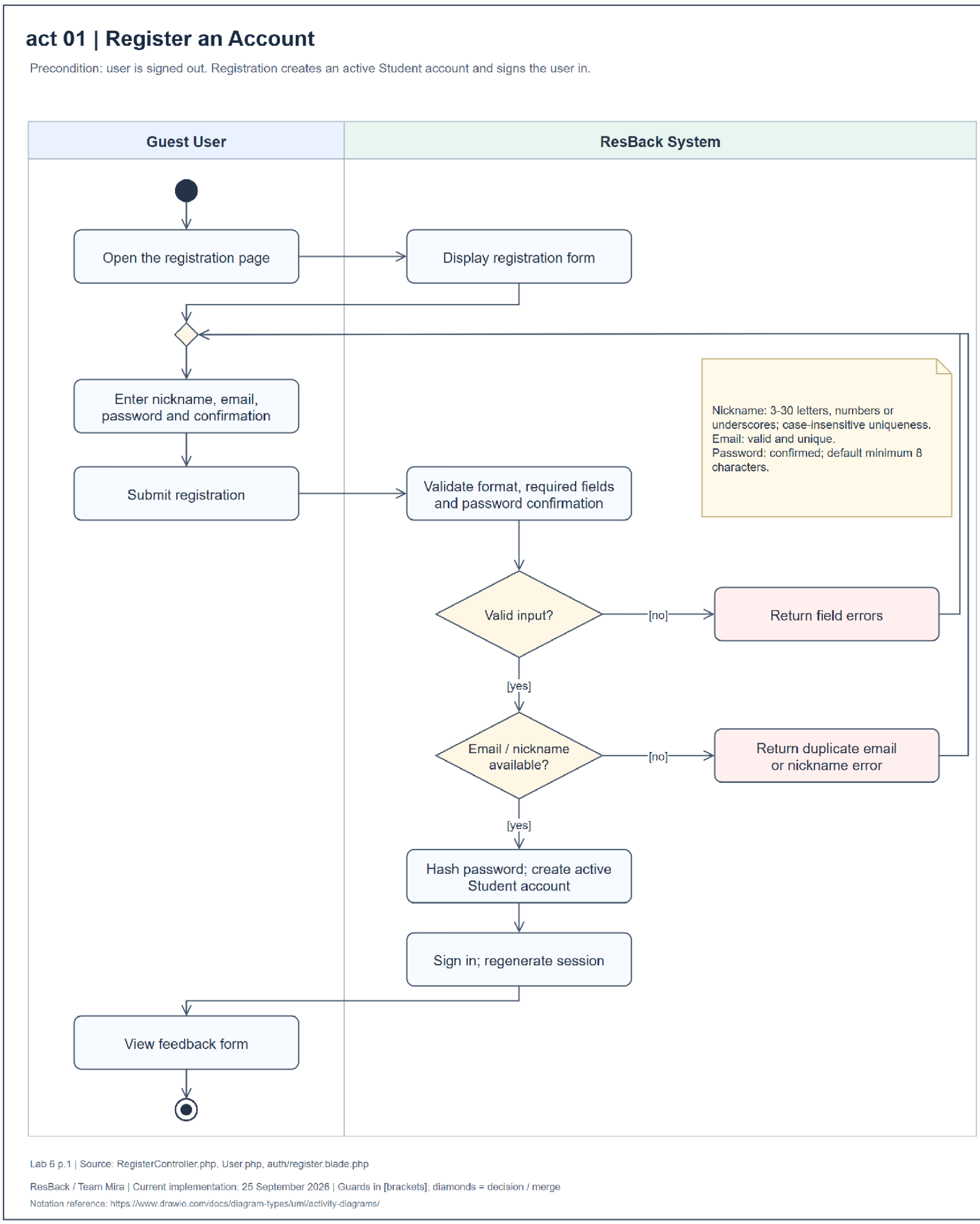


Figure 10 Activity diagram for register an account

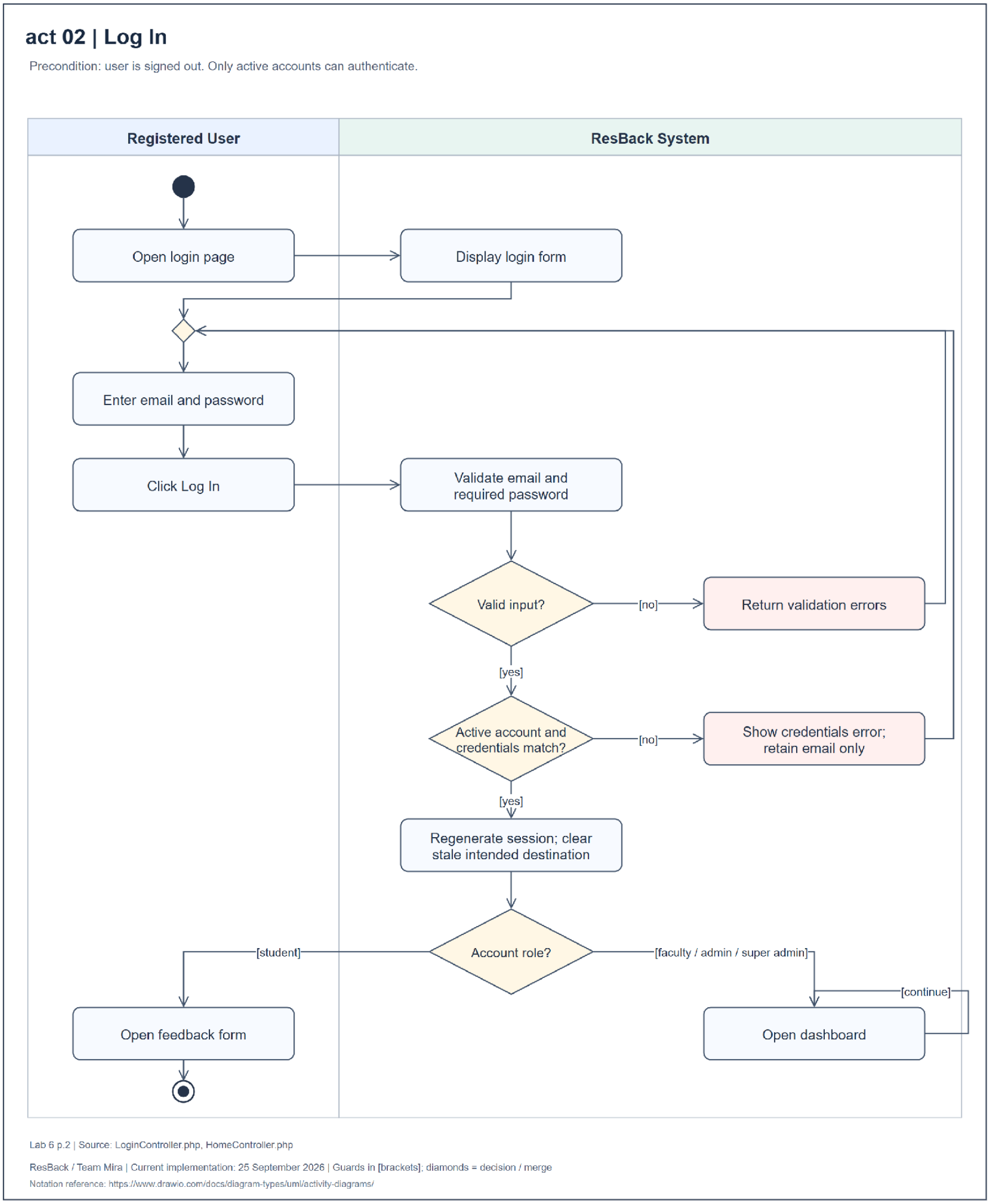


Figure 11 Activity diagram for log in

ACT 03 Submit Feedback | Complete flow | Panel 1 of 3

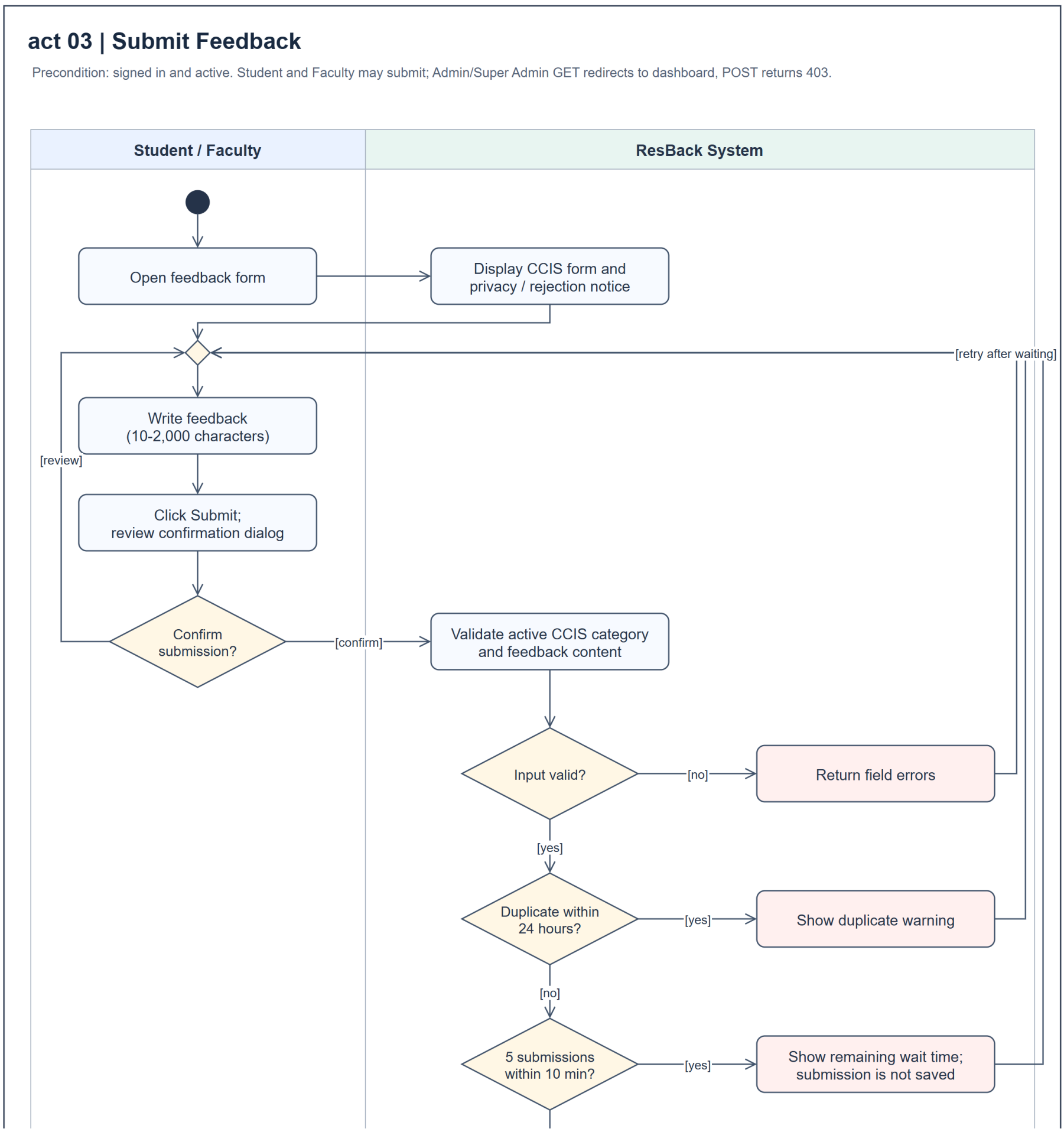


Figure 12 Activity diagram for submit feedback complete flow panel 1 of 3

ACT 03 Submit Feedback | Complete flow | Panel 2 of 3

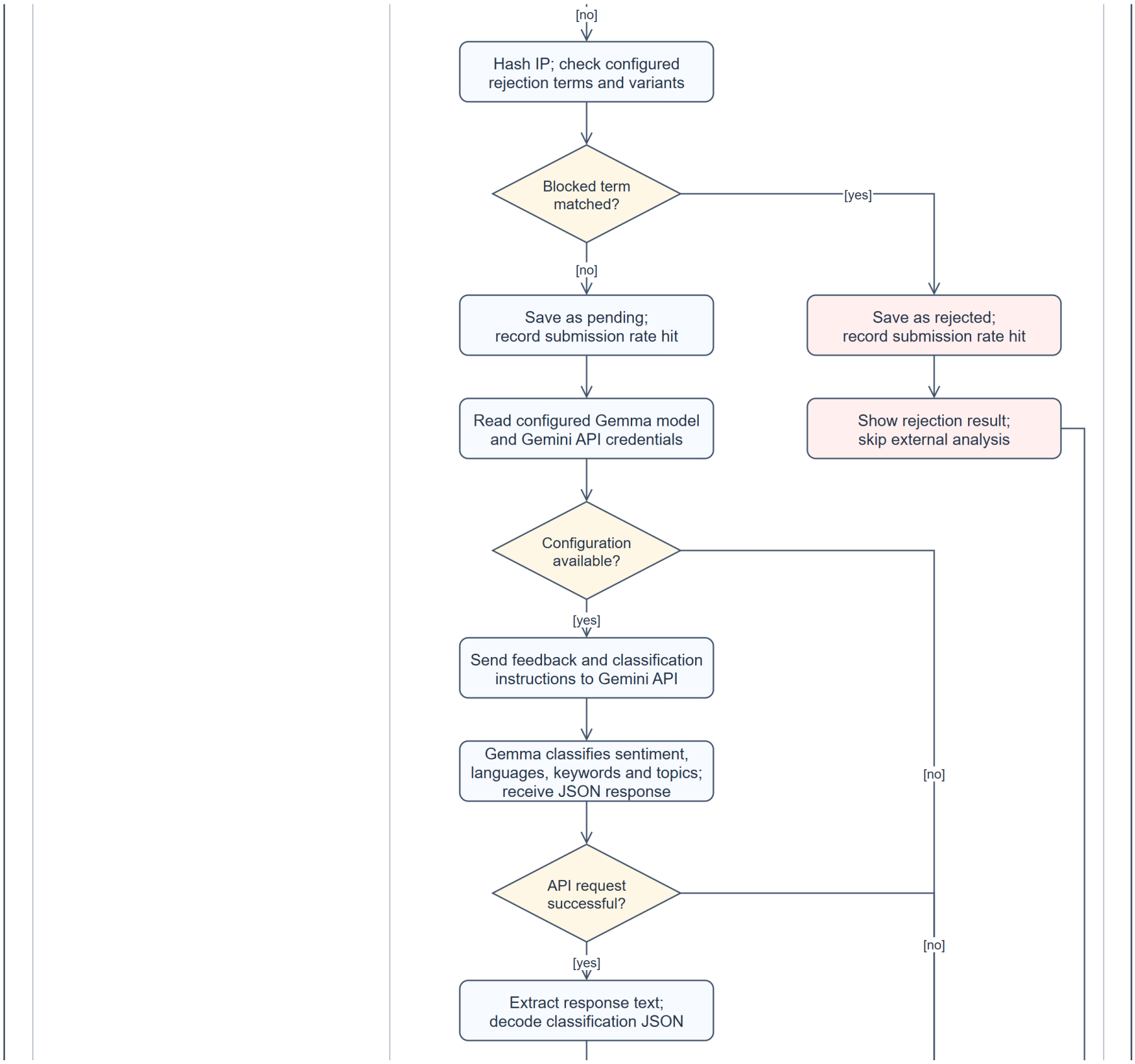


Figure 12 Activity diagram for submit feedback complete flow panel 2 of 3

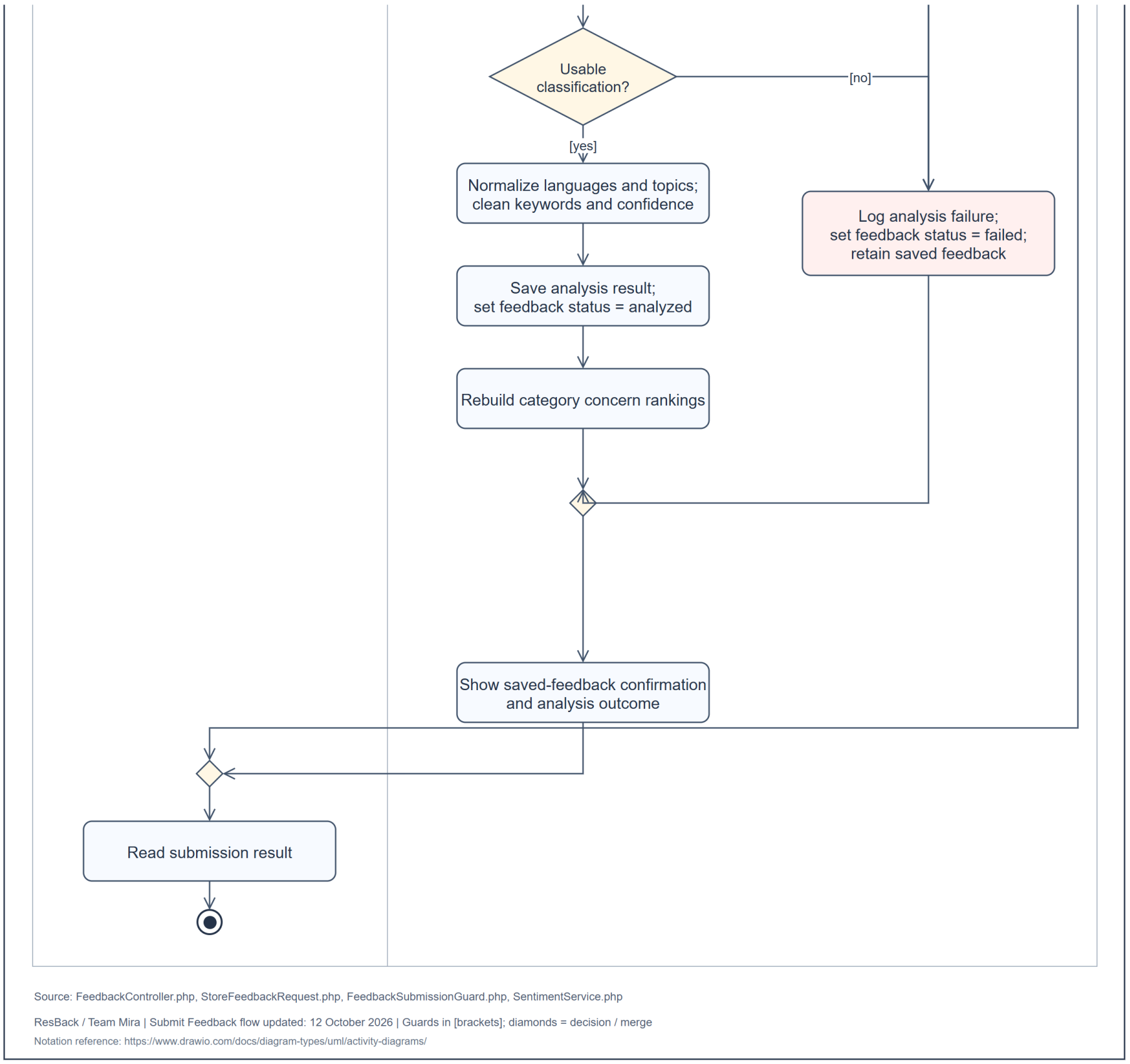


Figure 12 Activity diagram for submit feedback complete flow panel 3 of 3

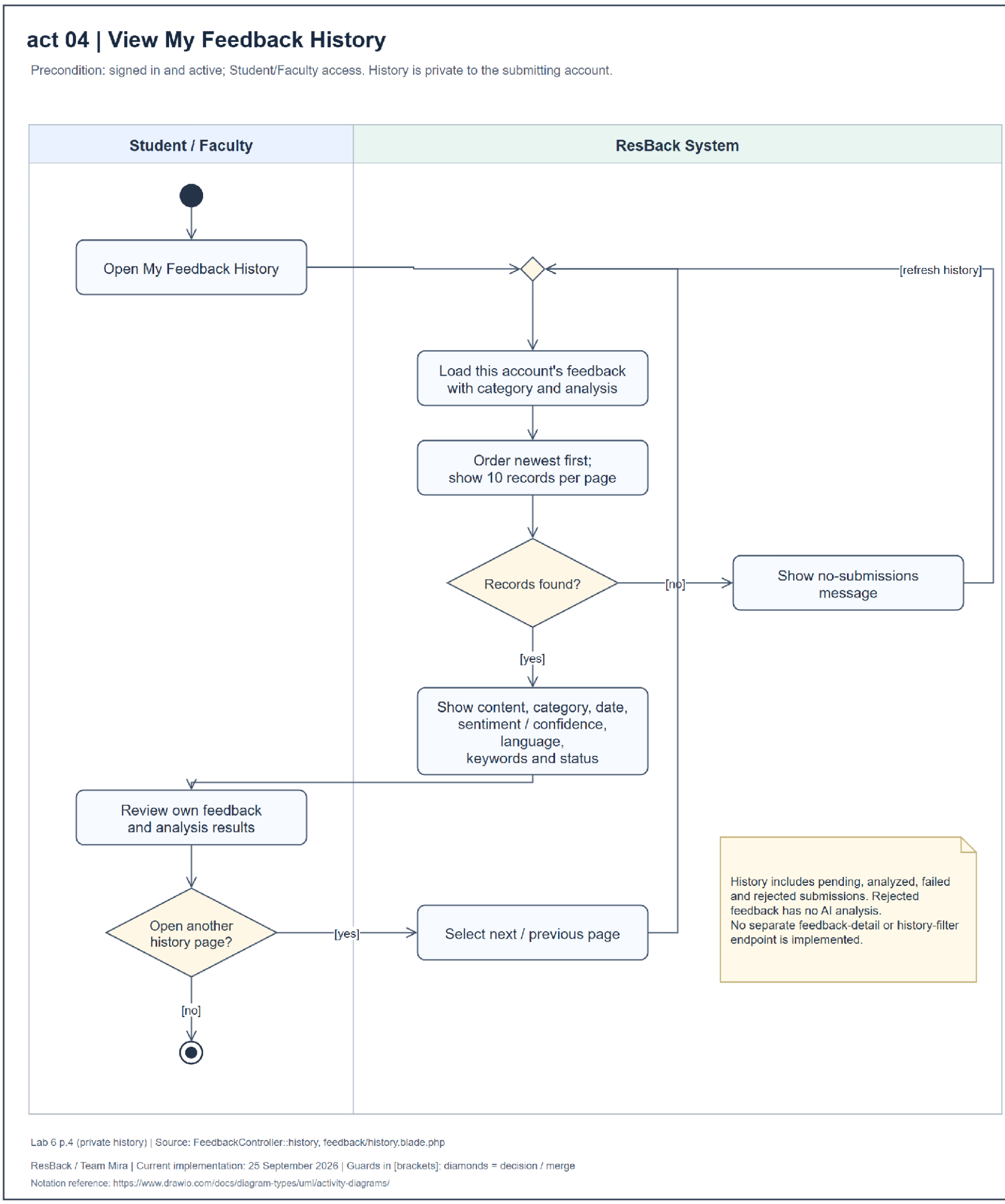


Figure 13 Activity diagram for view my feedback history

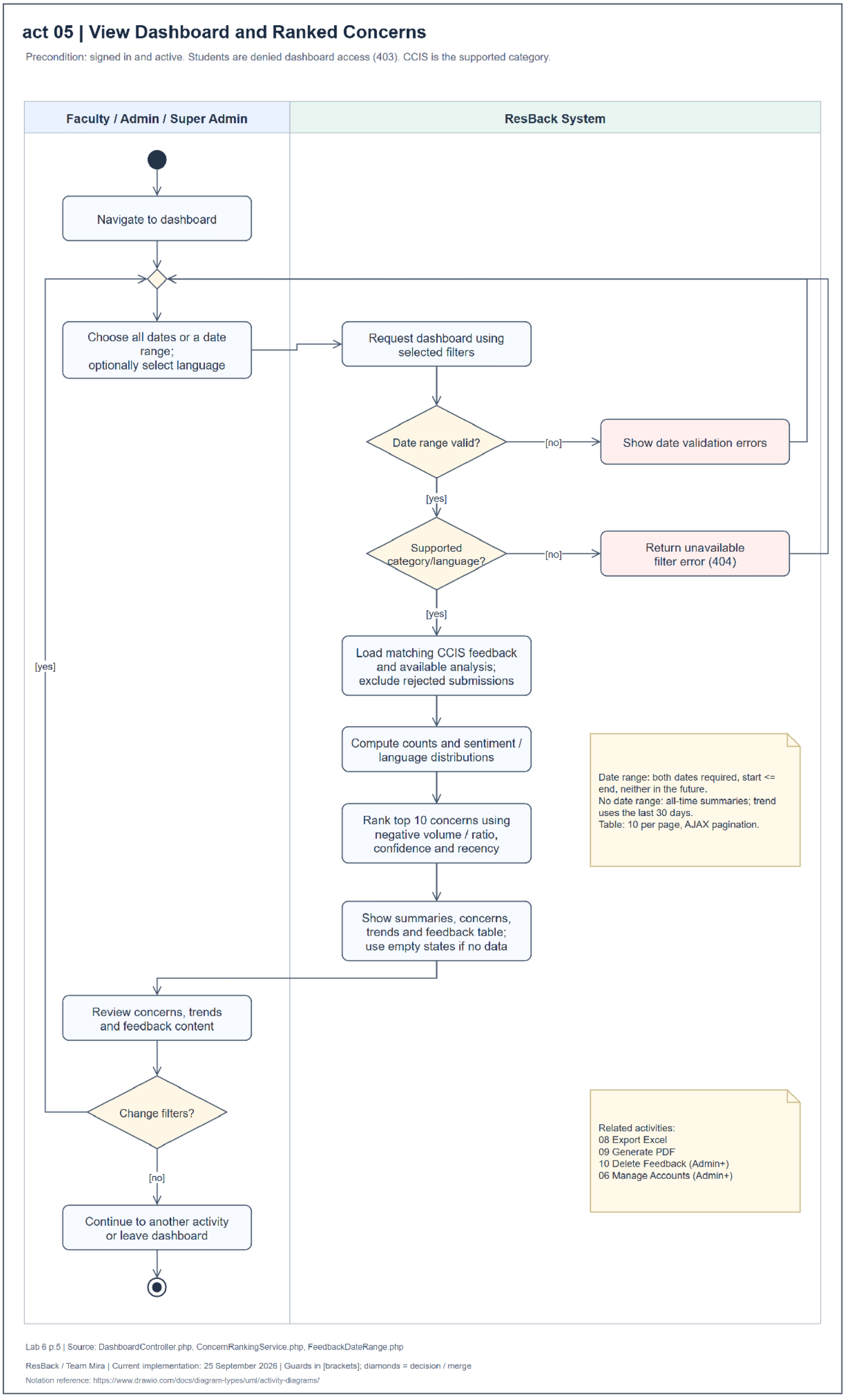


Figure 14 Activity diagram for view dashboard and ranked concerns

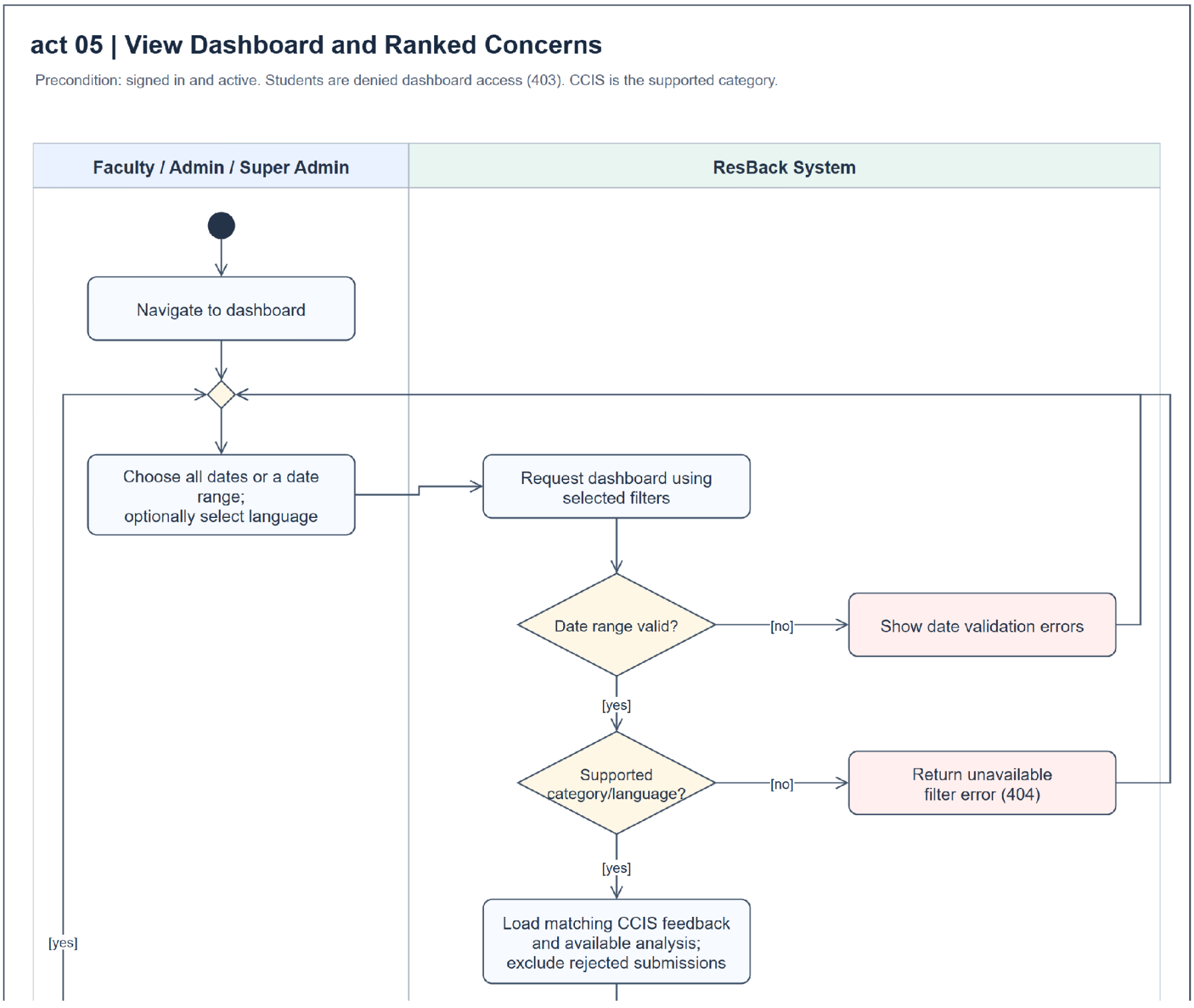


Figure 14 Activity diagram for view dashboard and ranked concerns enlarged continuation panel 1 of 2

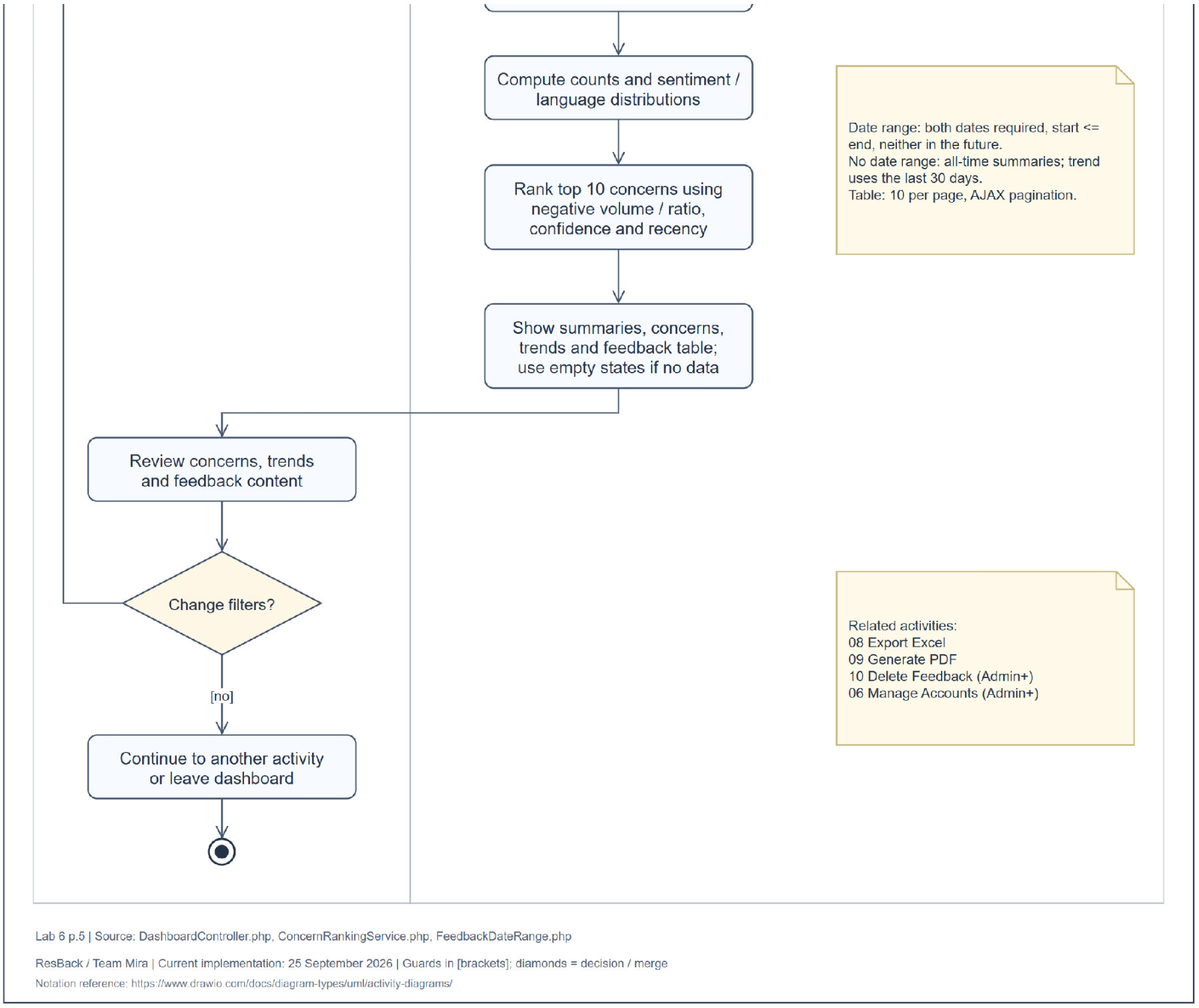


Figure 14 Activity diagram for view dashboard and ranked concerns enlarged continuation panel 2 of 2

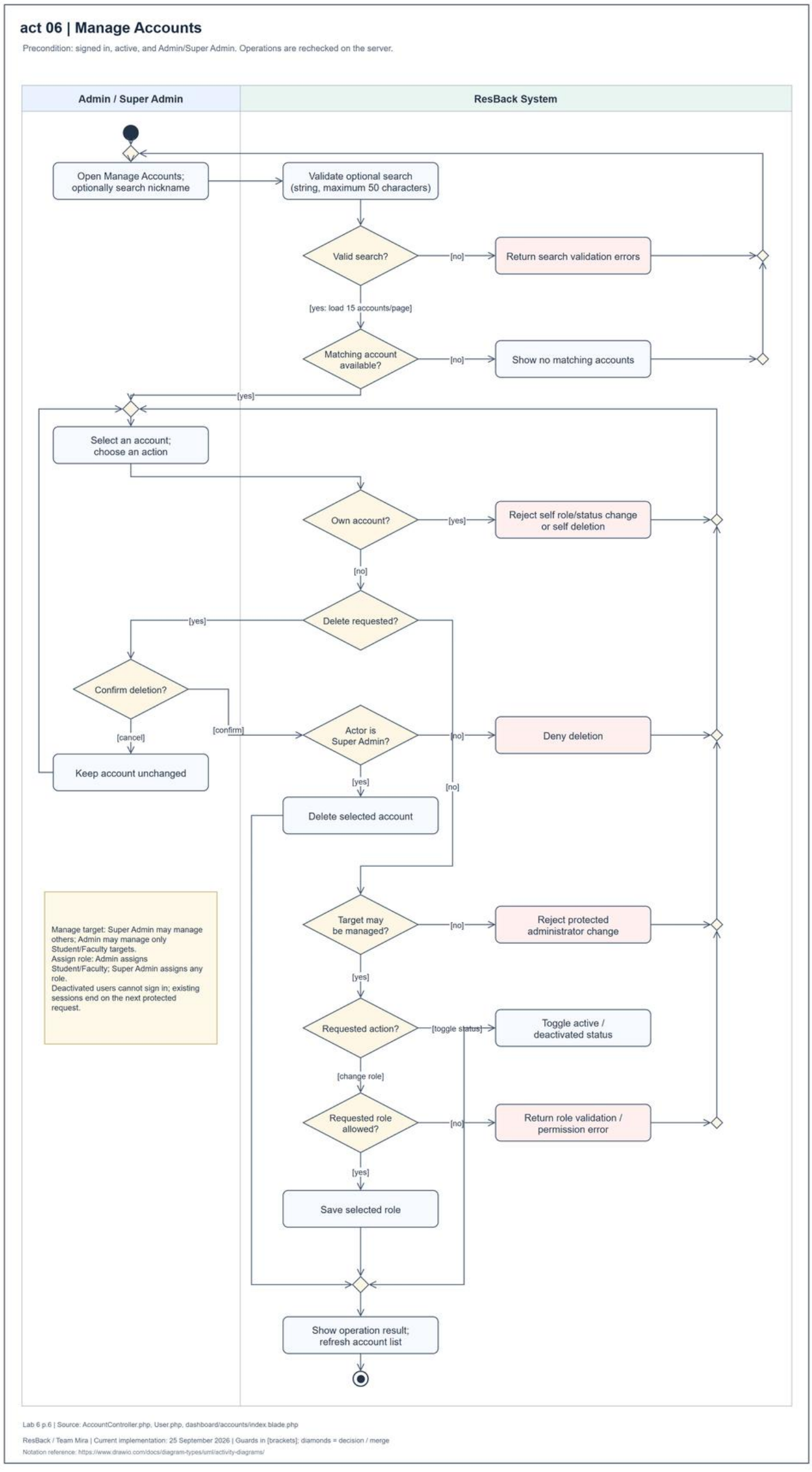


Figure 15 Activity diagram for manage accounts

ACT 06 Manage Accounts | Panel 1 of 2

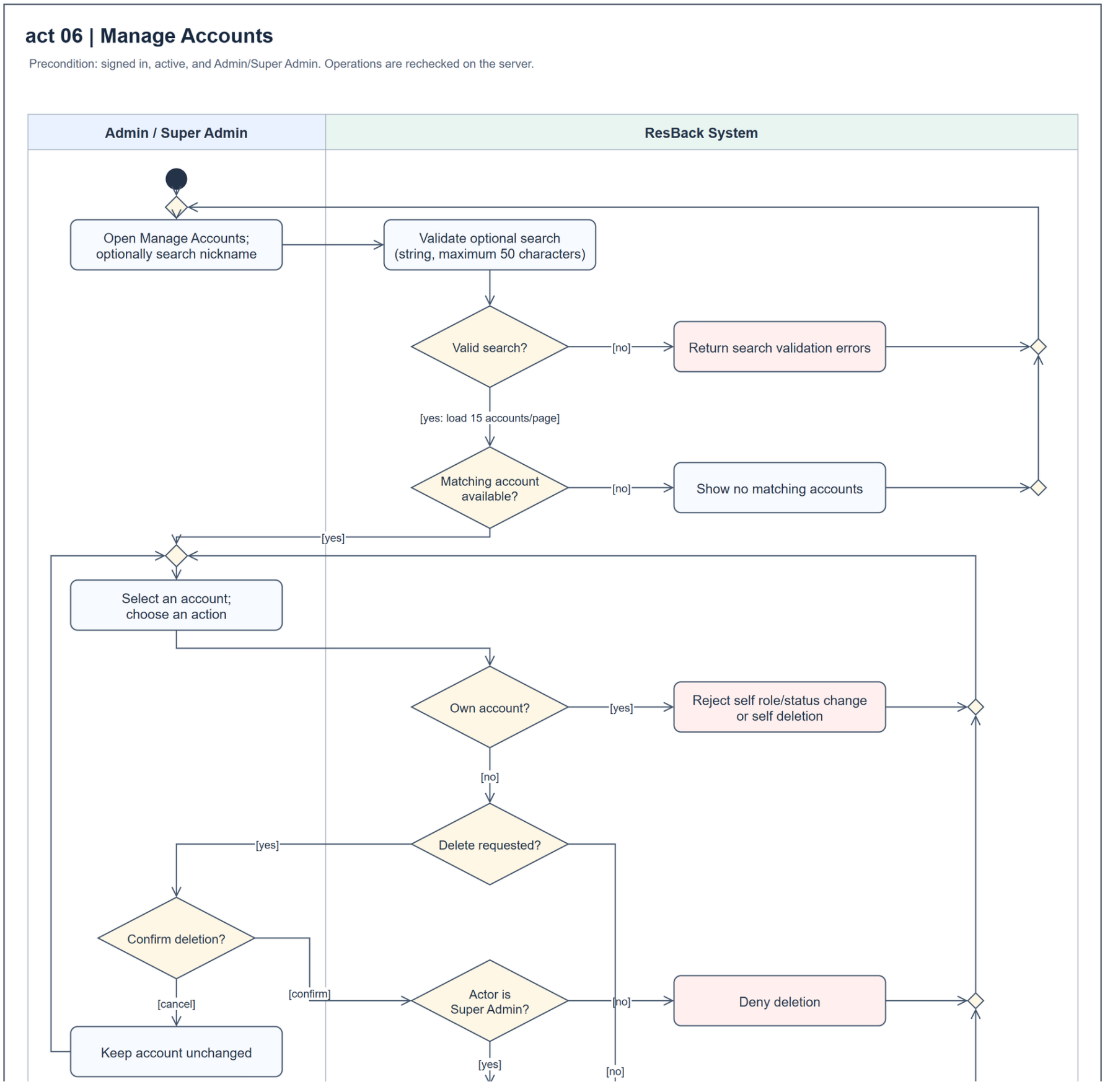


Figure 15 Activity diagram for manage accounts enlarged continuation panel 1 of 2

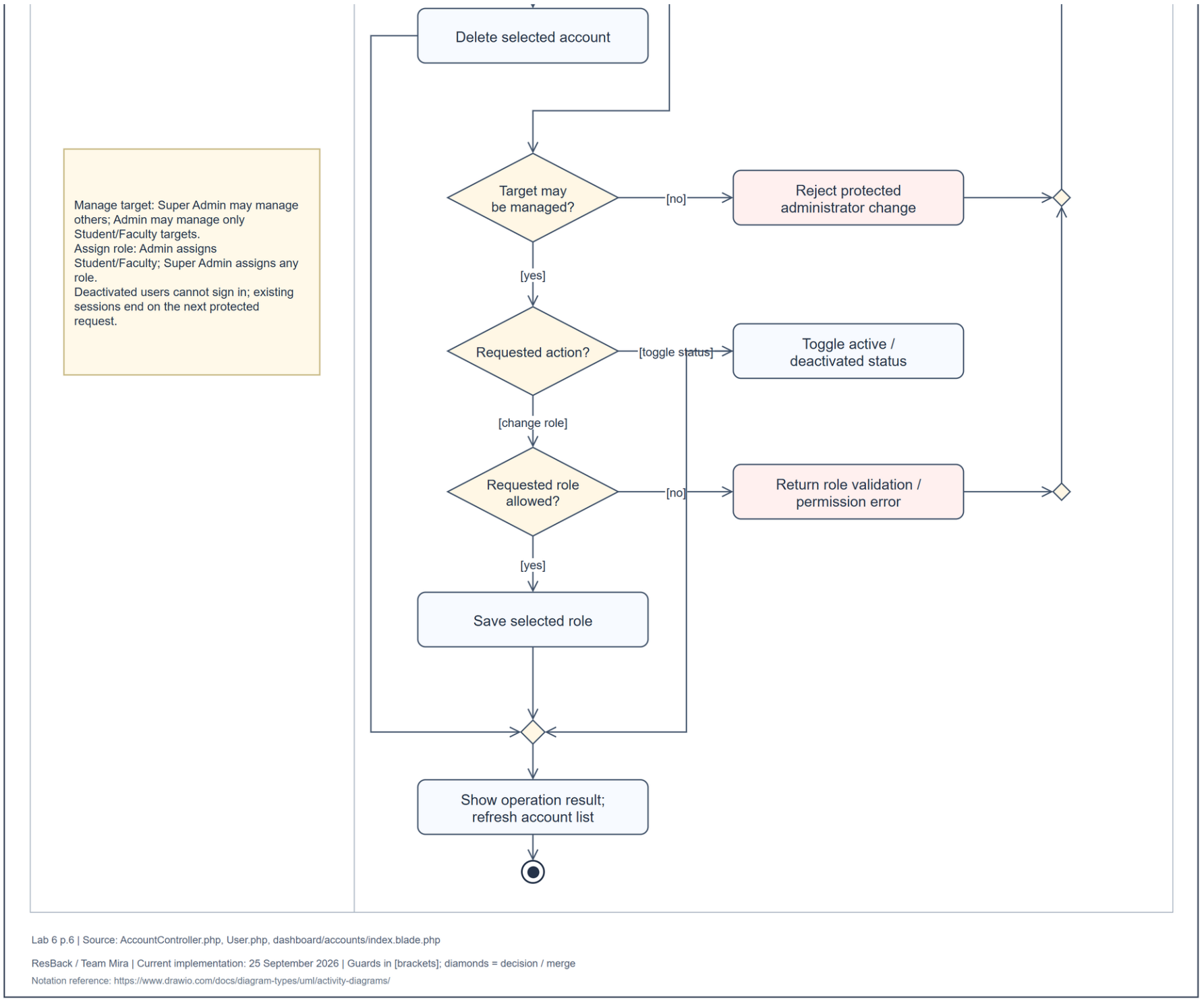


Figure 15 Activity diagram for manage accounts enlarged continuation panel 2 of 2

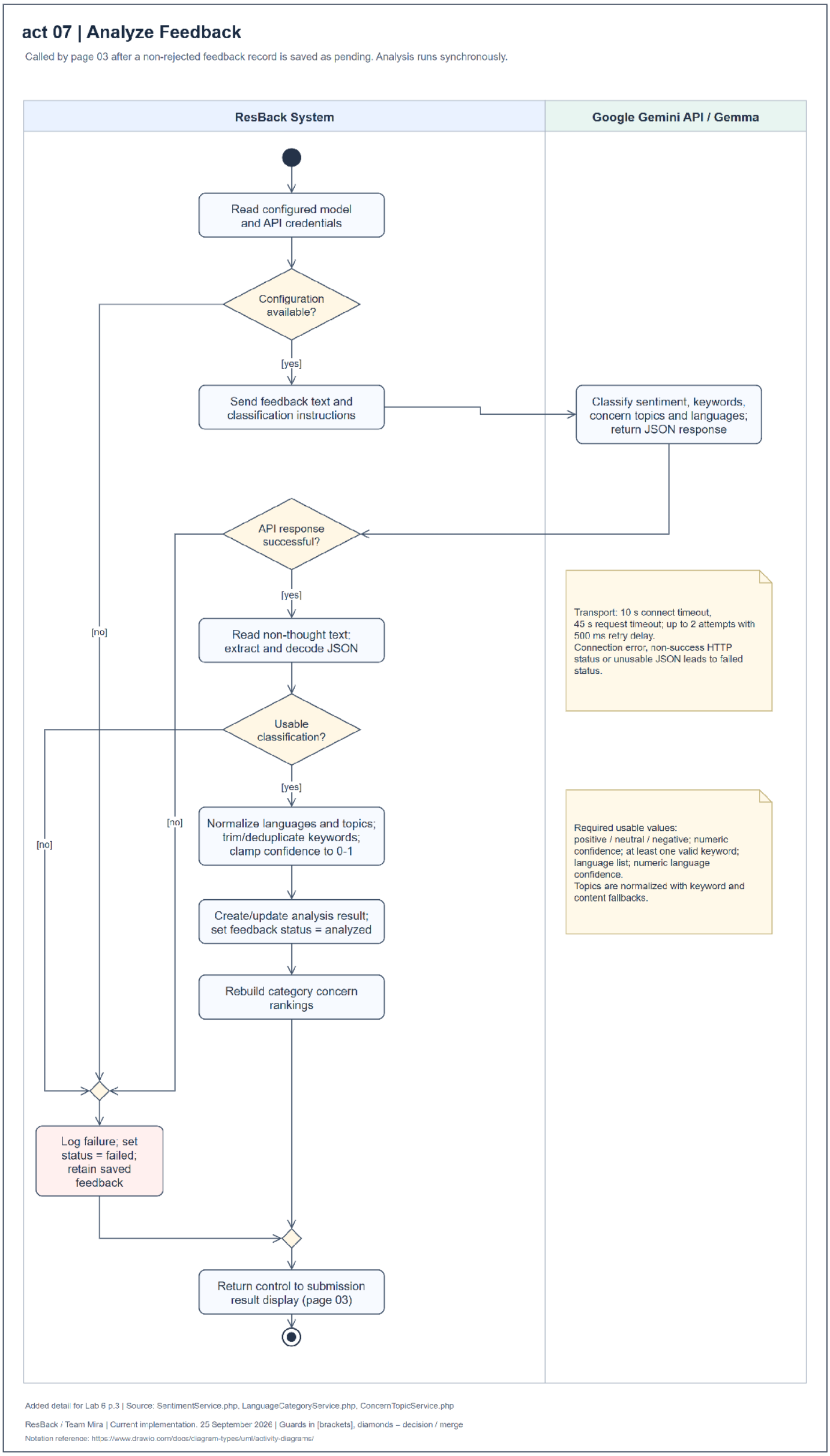


Figure 16 Activity diagram for analyze feedback

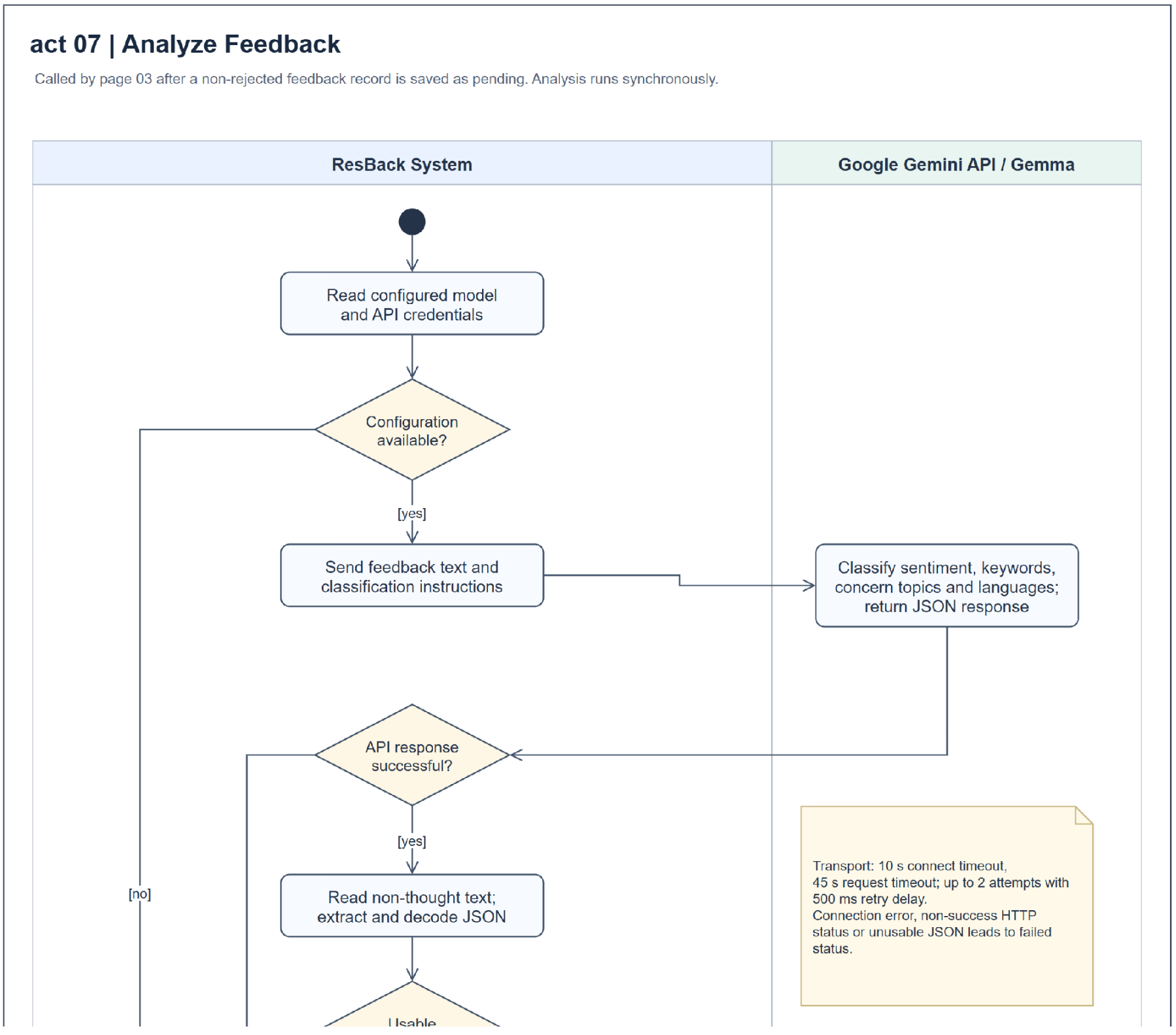


Figure 16 Activity diagram for analyze feedback enlarged continuation panel 1 of 2

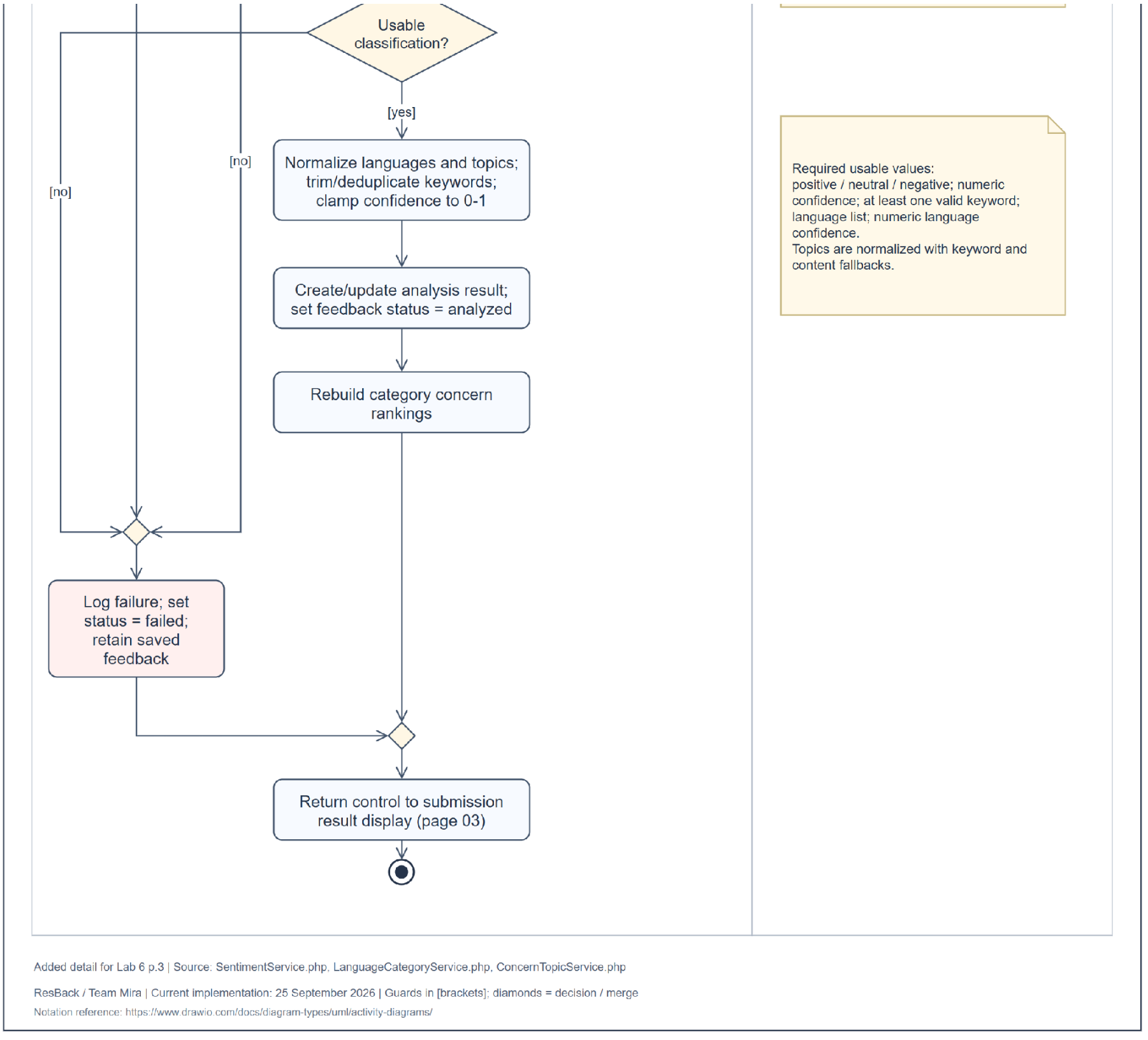


Figure 16 Activity diagram for analyze feedback enlarged continuation panel 2 of 2

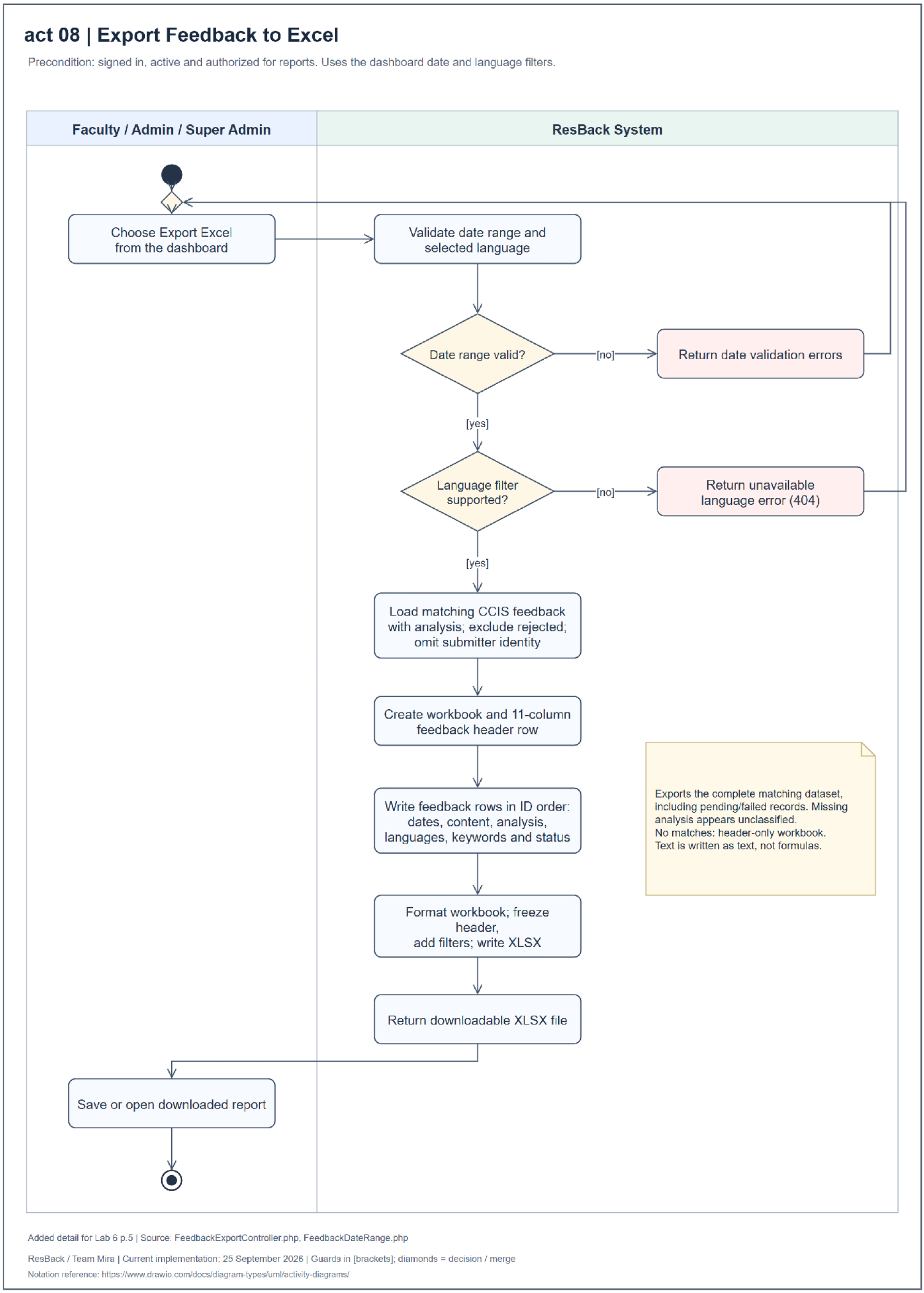


Figure 17 Activity diagram for export feedback to Excel

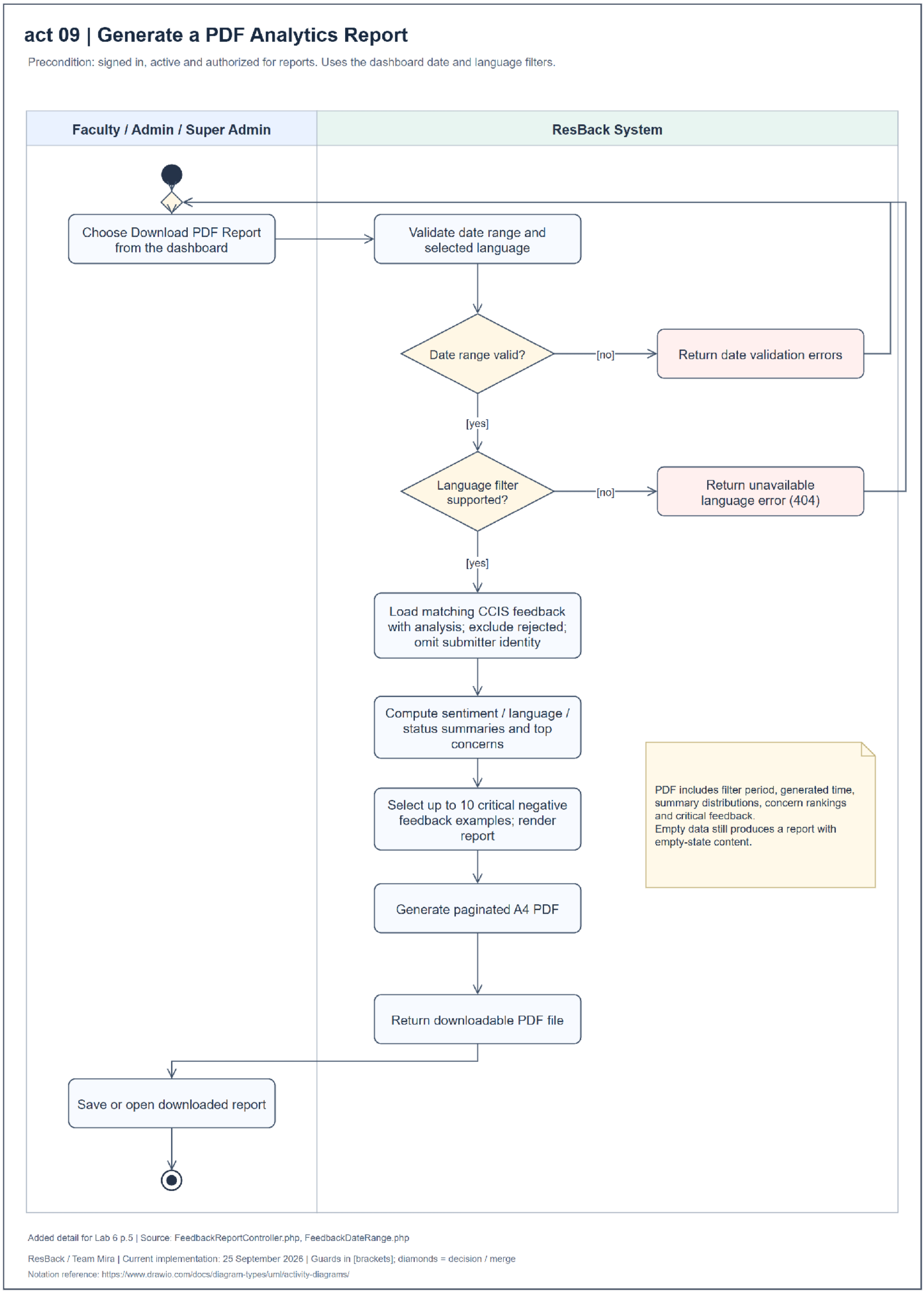


Figure 18 Activity diagram for generate a PDF analytics report

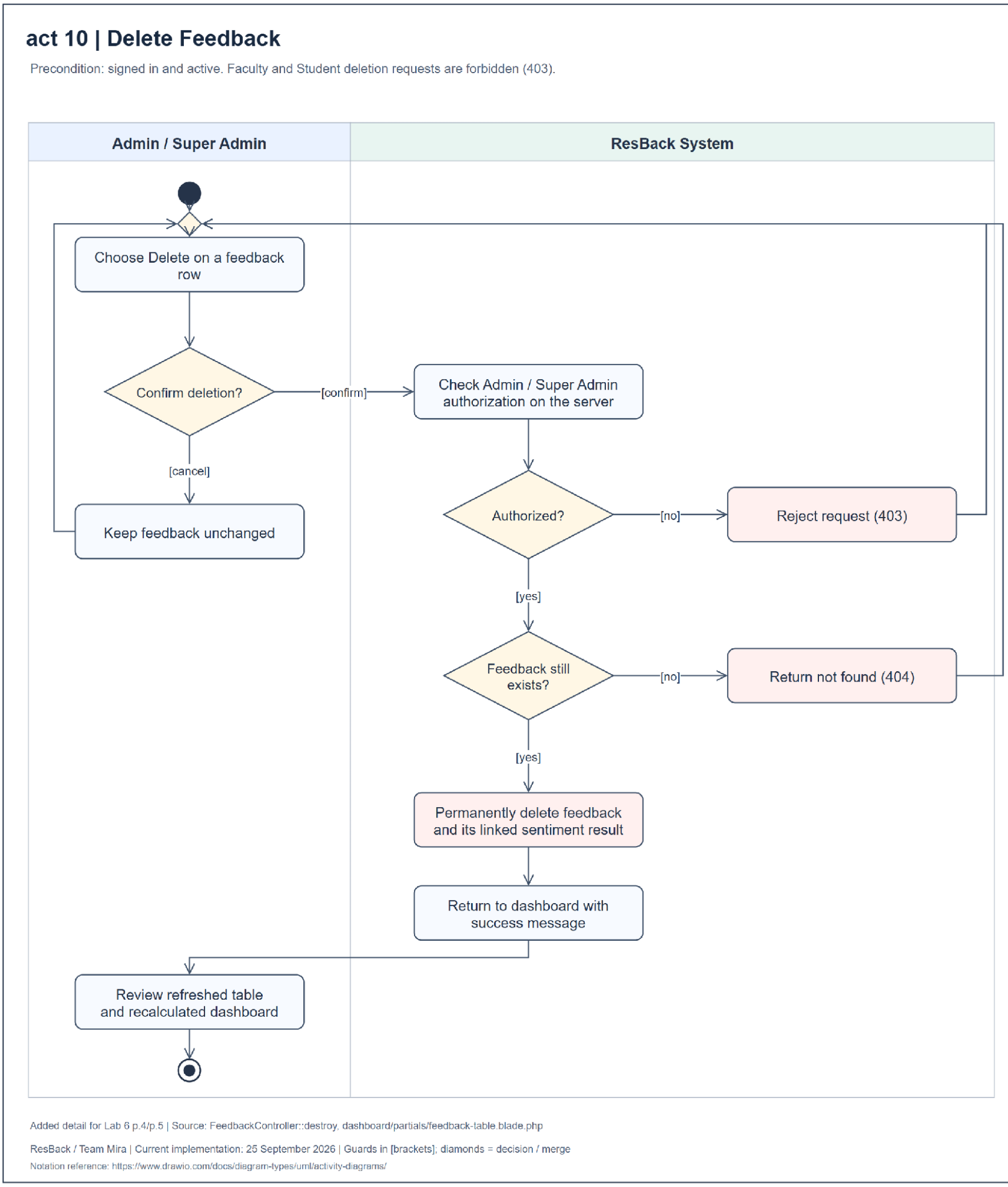


Figure 19 Activity diagram for delete feedback

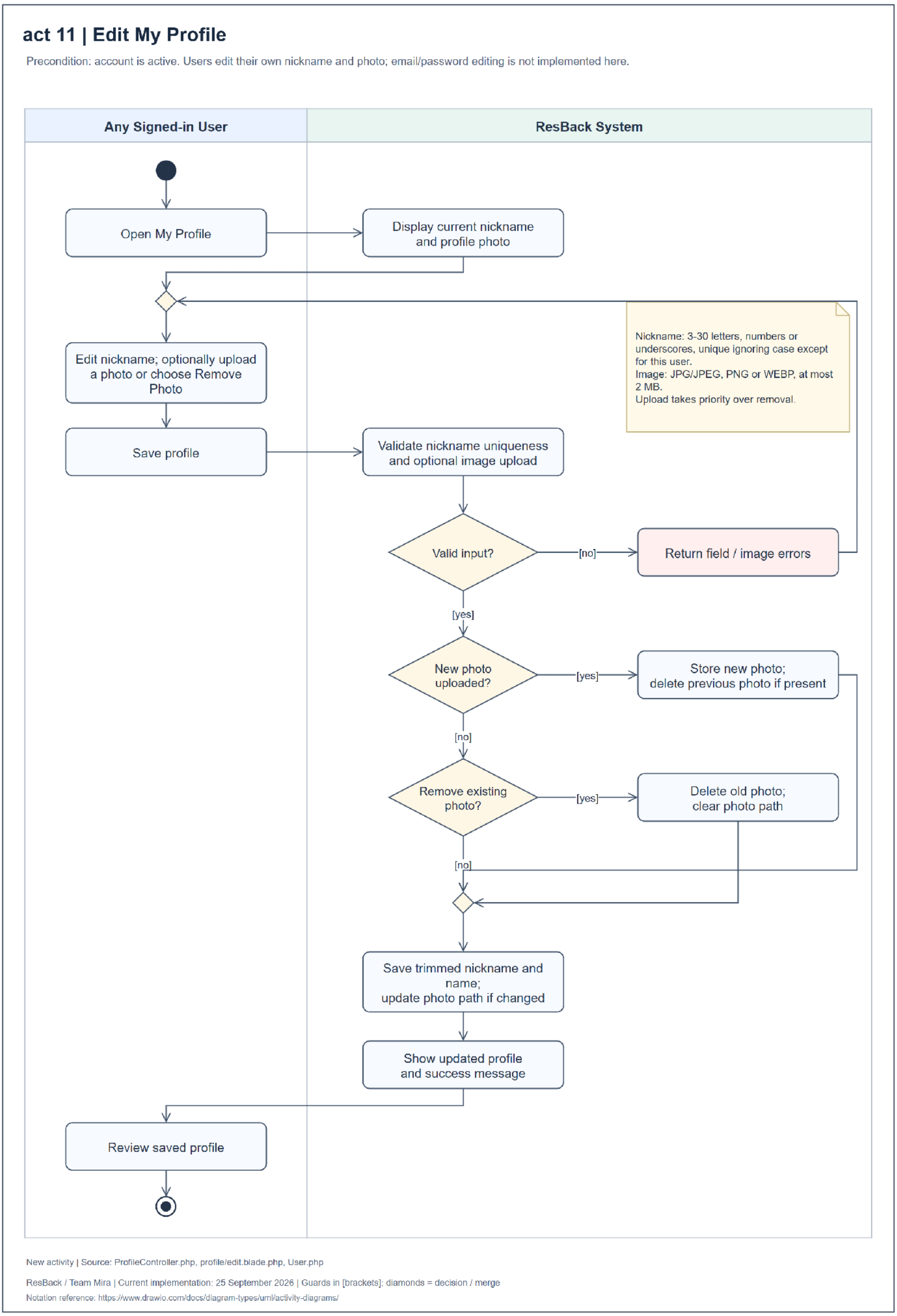


Figure 20 Activity diagram for edit my profile

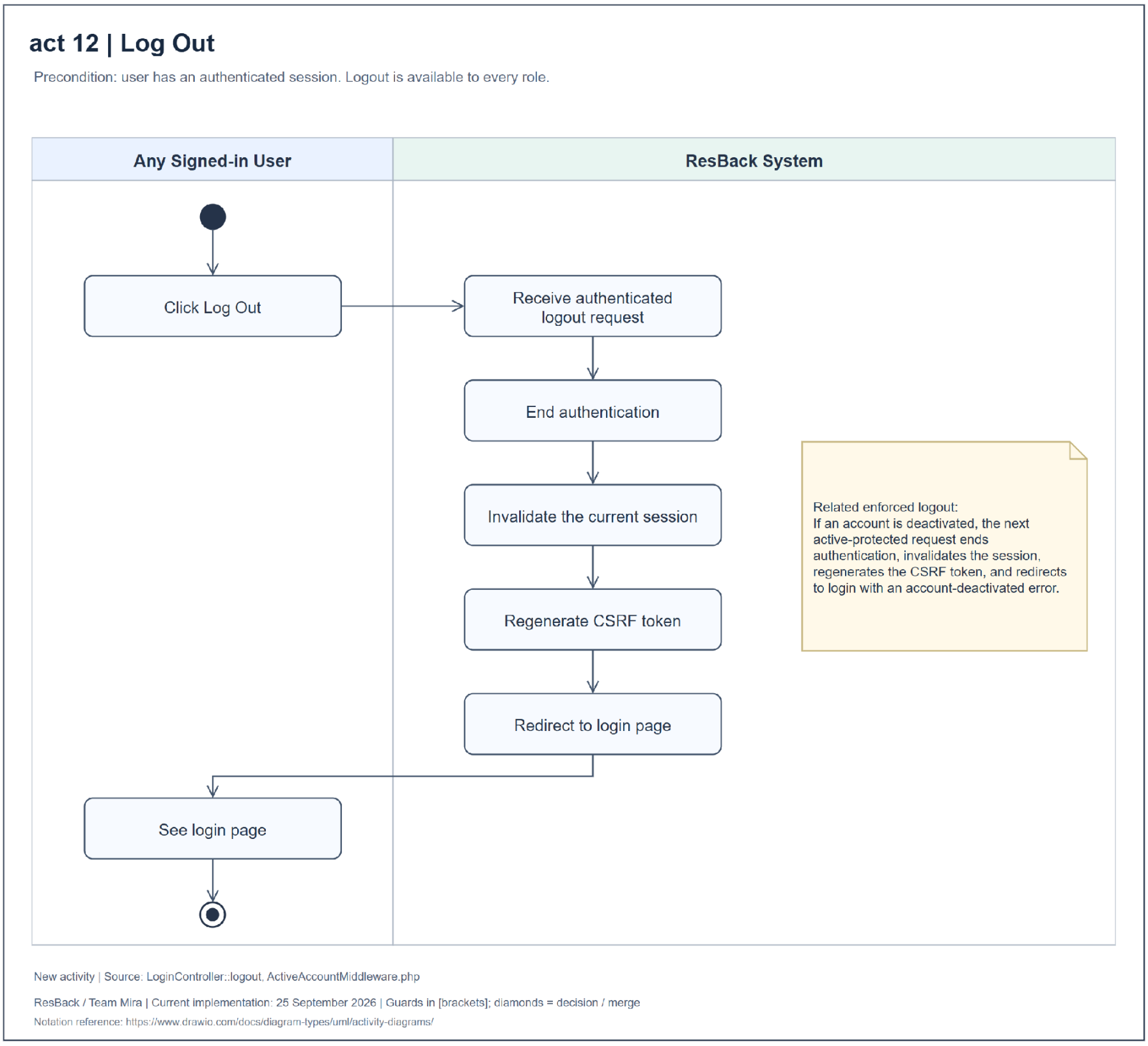


Figure 21 Activity diagram for log out

APPENDIX F CLASS DIAGRAMS

Figures 22–27 present six class views. Plus, hash, and minus symbols identify public, protected, and private members. The source sheets retain their original 01–06 labels.

Figure 22. The core domain model shows record attributes, operations, ownership, analysis relationships, and deletion behavior. User roles are stored values rather than subclasses.

Figure 23. Authentication, accounts, and profiles connect registration, login, routing, account administration, and profile editing to the User model.

Figure 24. Feedback submission and protection connect validation, rate limiting, duplicate detection, screening, and analysis. Screening occurs before external inference.

Figure 25. Sentiment analysis and concern ranking connect model-output normalization, language and topic categories, optional date boundaries, and ranking reconstruction.

Figure 26. Dashboard and report generation share filters, ranked concerns, models, and export utilities. Review outputs omit submitter identification and rejected feedback.

Figure 27. Account-access middleware checks authentication, active status, and roles. Controller authorization adds account-management restrictions.

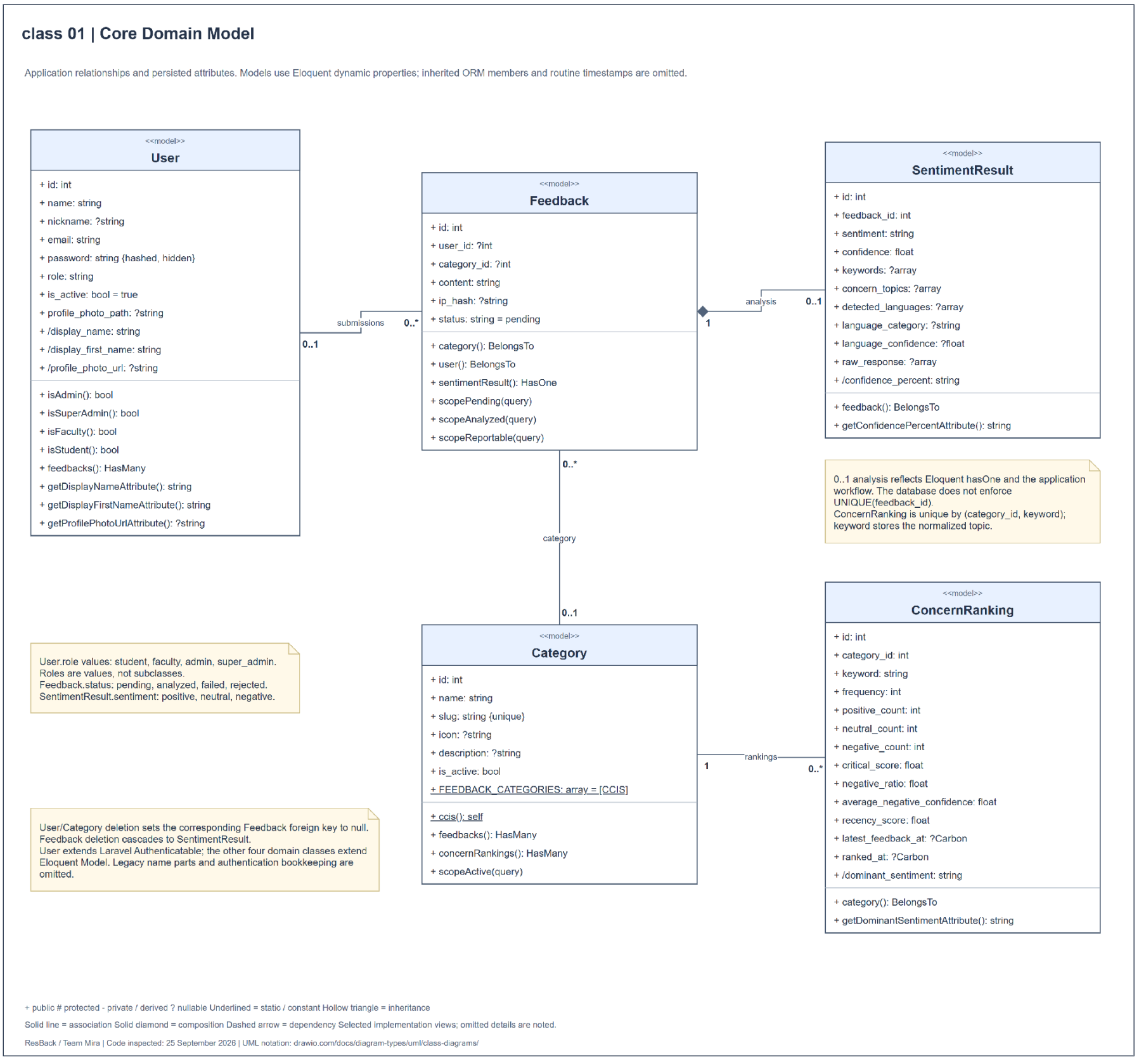


Figure 22 Class diagram for core domain model

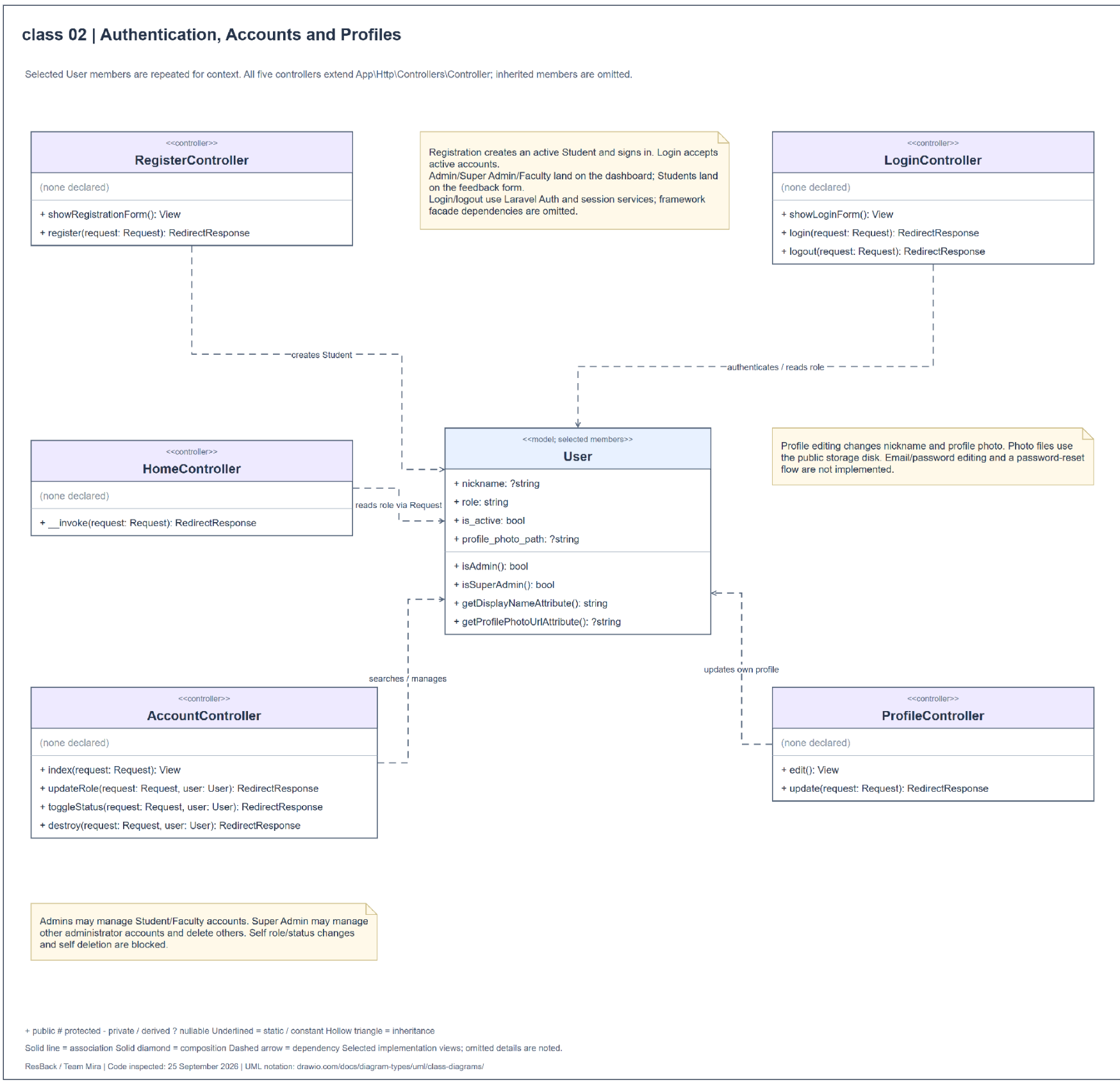


Figure 23 Class diagram for authentication accounts and profiles

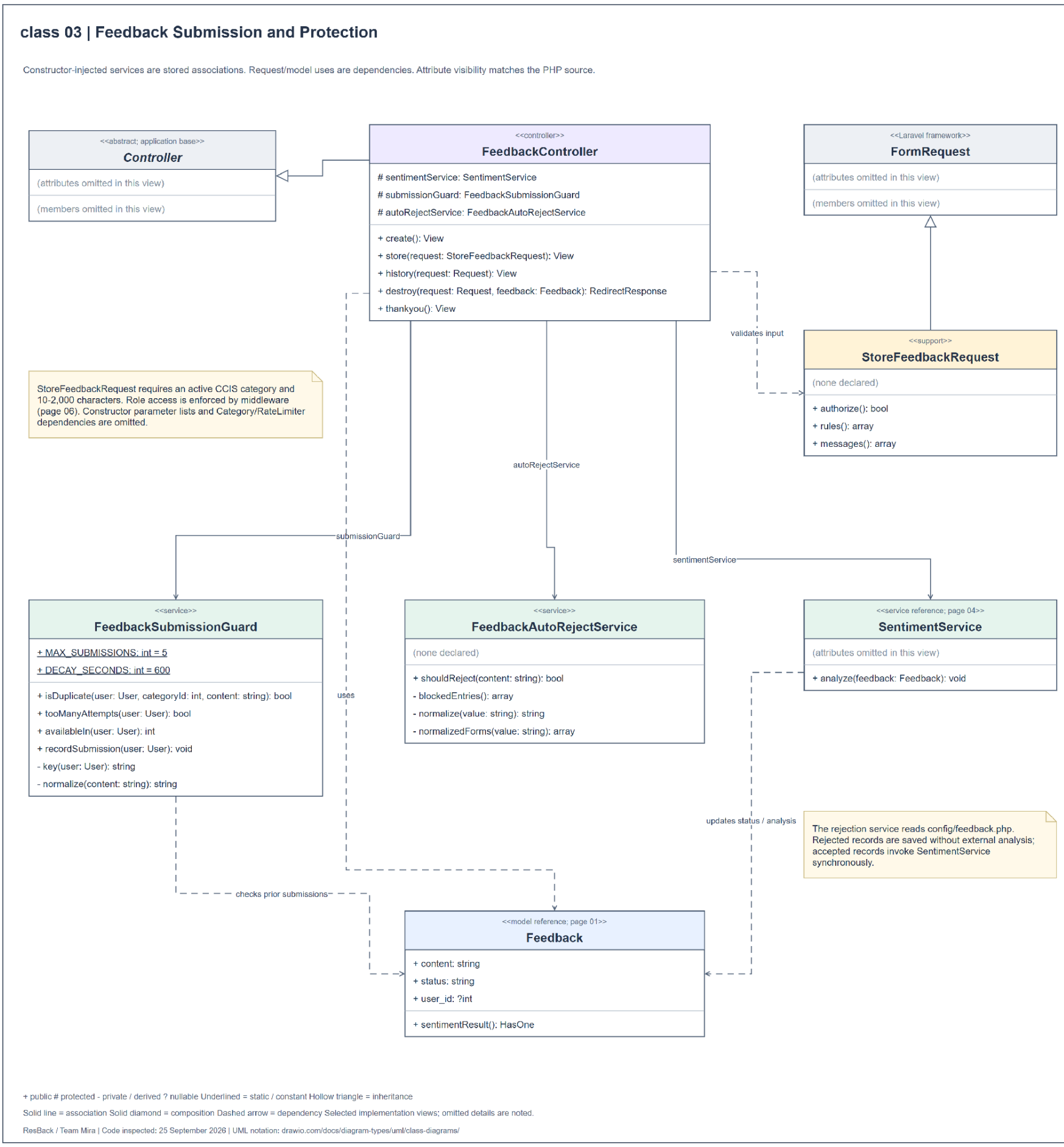


Figure 24 Class diagram for feedback submission and protection

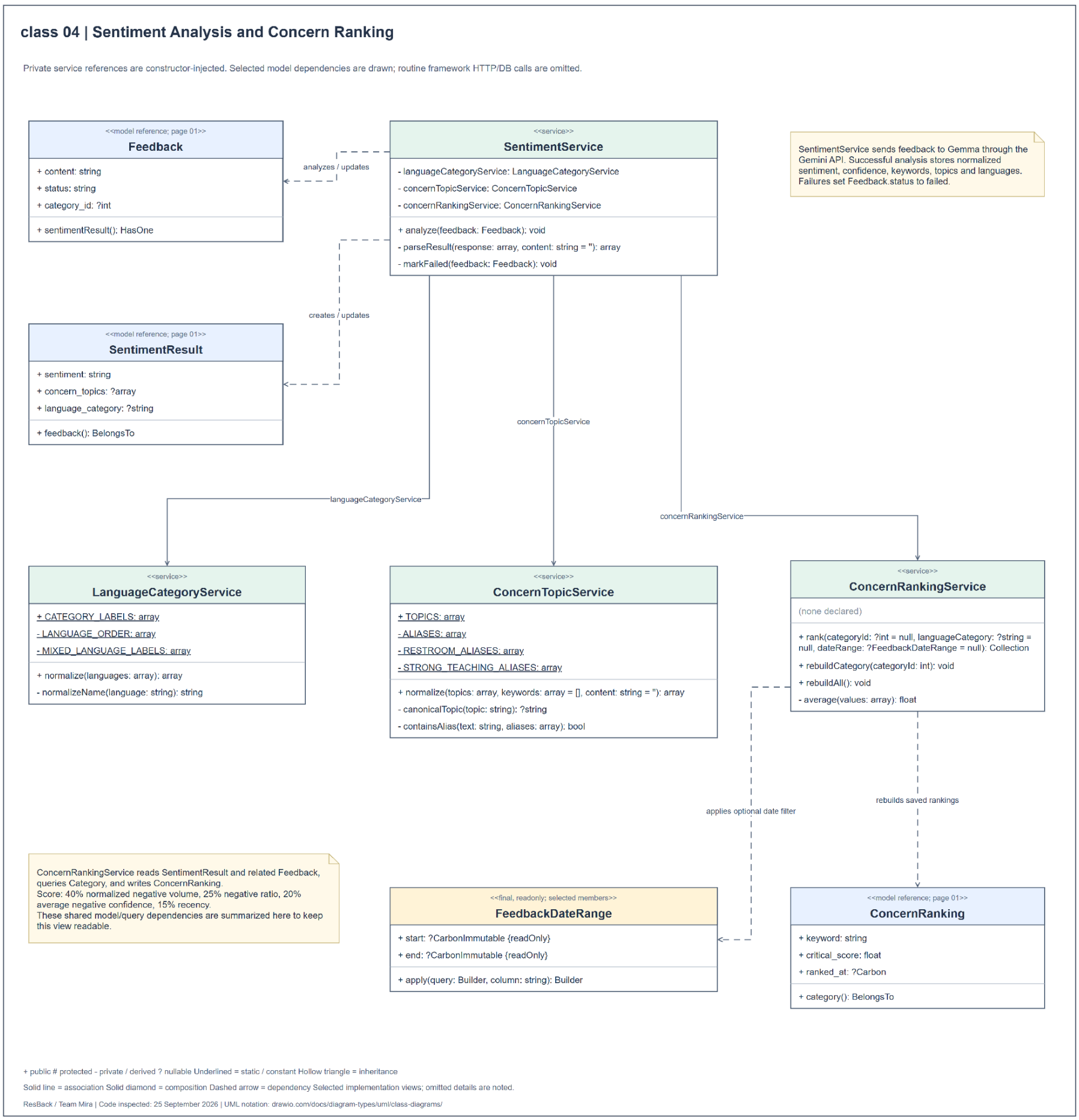


Figure 25 Class diagram for sentiment analysis and concern ranking

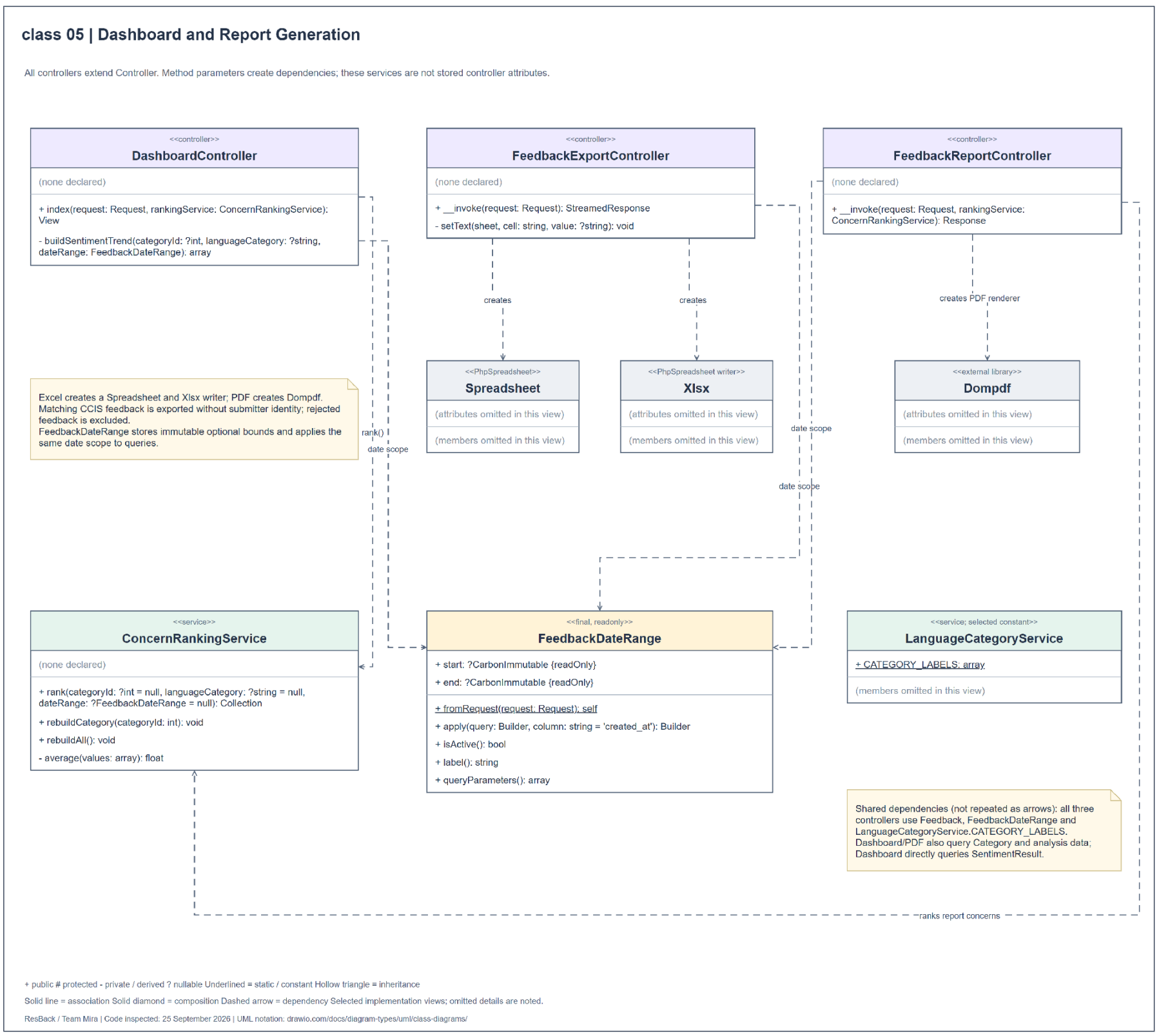


Figure 26 Class diagram for dashboard and report generation

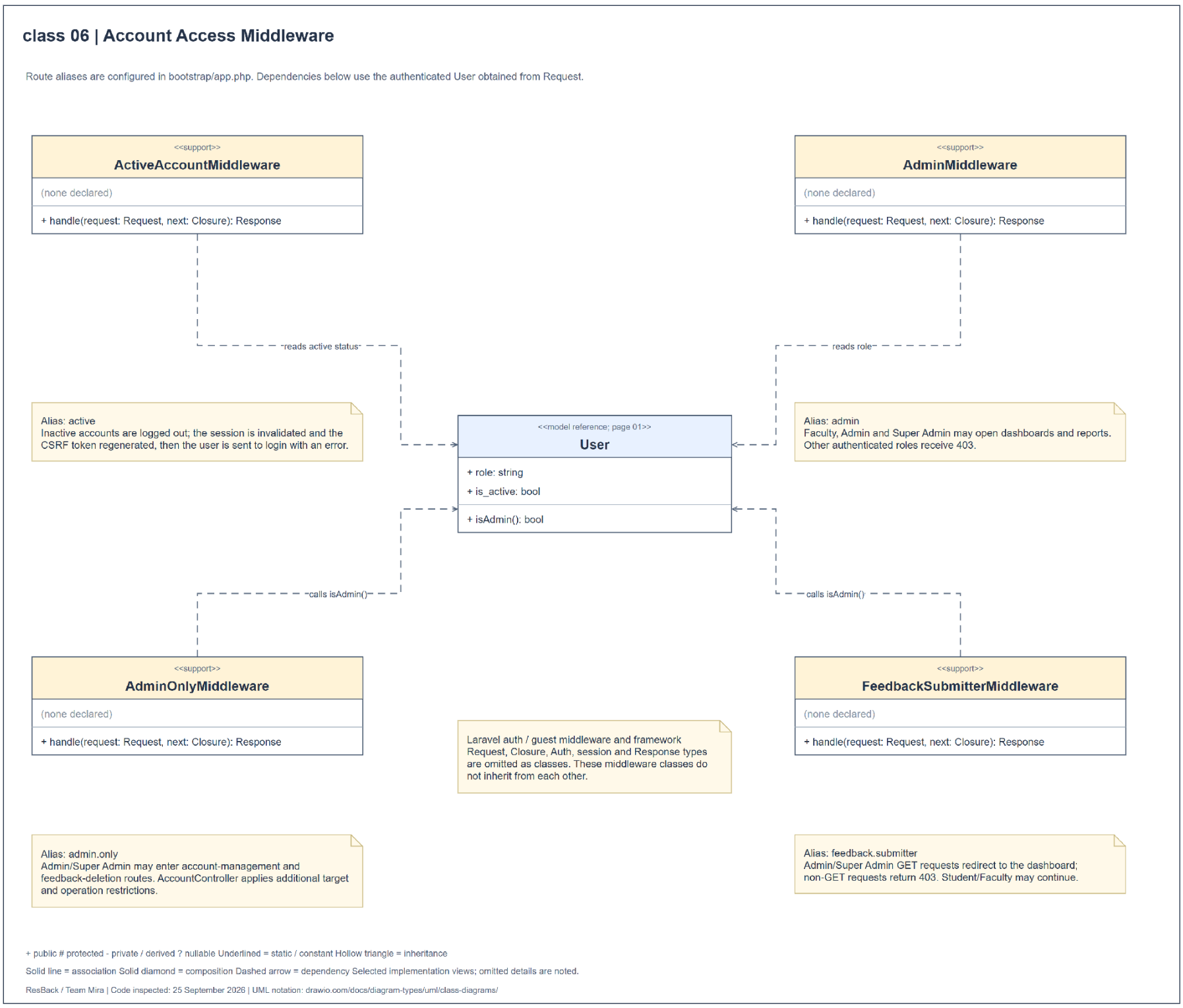


Figure 27 Class diagram for account access middleware

APPENDIX G COMPONENT DIAGRAM

Figure 28 shows the logical modules inside the Laravel application: Web Presentation, Identity and Accounts, Feedback Management, Sentiment Analysis, Concern Ranking, and Dashboard and Reporting. These are logical components of the existing application rather than separately deployed services.

Provided and required interfaces identify service contracts. Delegation connectors link application boundaries to internal components. The diagram also includes external inference, relational storage, profile-photo storage, and report libraries.

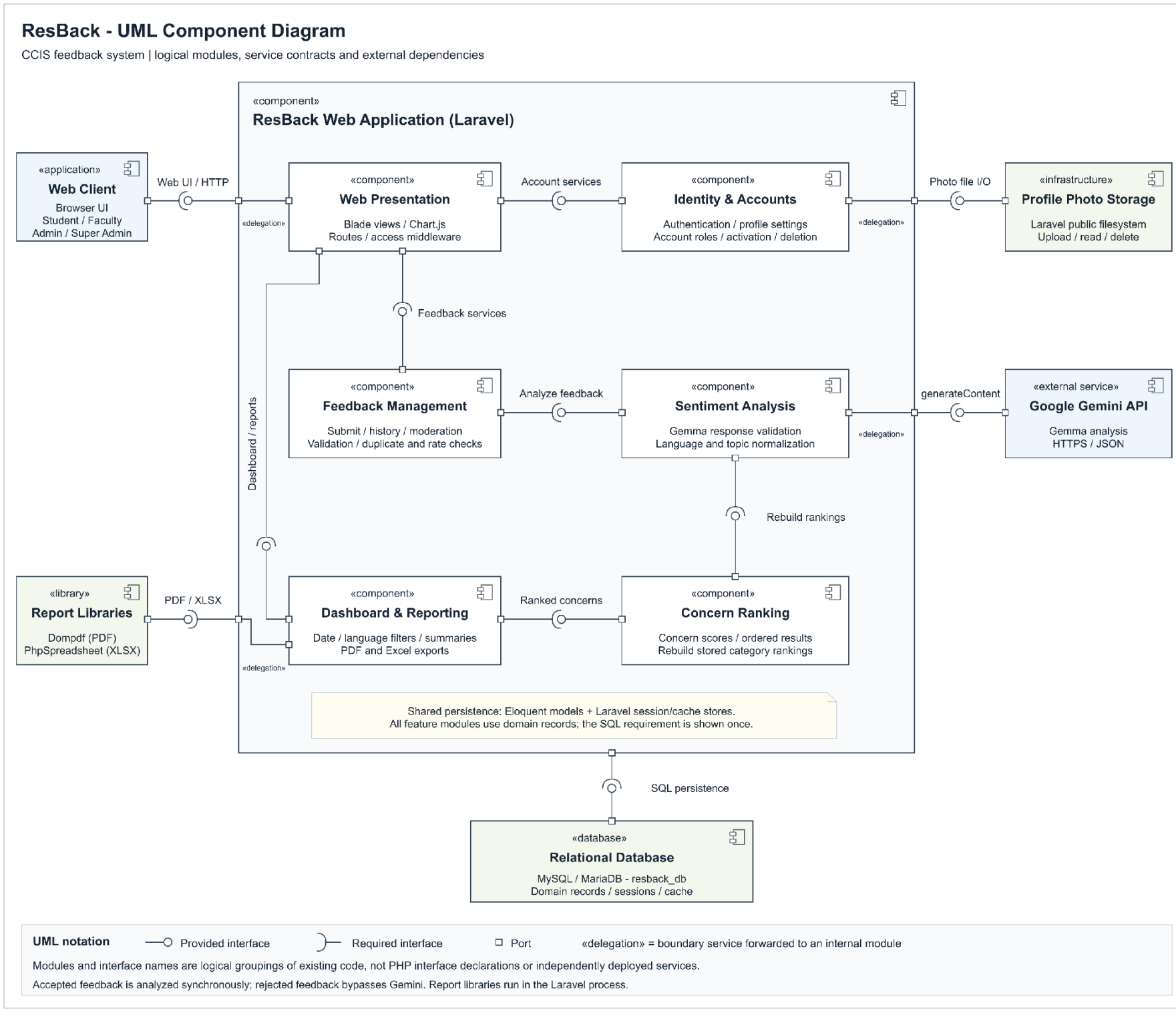


Figure 28 ResBack UML component diagram

APPENDIX H PACKAGE DIAGRAM

Figure 29 groups the application into packages for presentation, account access, feedback handling, analysis, reporting, and persistence. Dashed dependency arrows point toward the package being used. The Google API remains an external dependency.

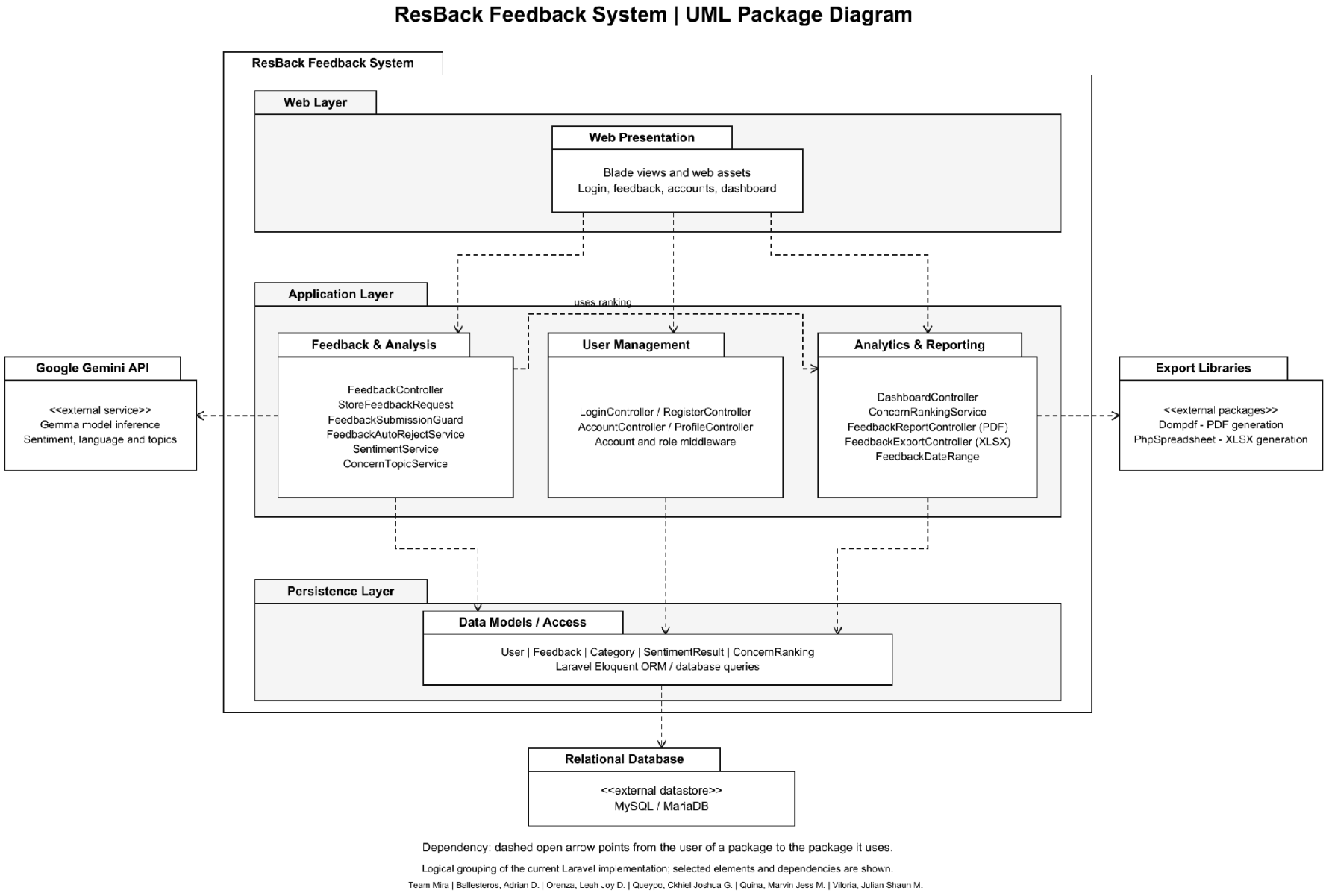


Figure 29 ResBack UML package diagram

APPENDIX I DEPLOYMENT DIAGRAMS LOCAL AND ONLINE

Figure 30 shows the local development deployment: the web browser, PHP development server, XAMPP MySQL or MariaDB, application files, and external Google Gemini API. Web and database services run locally, while Gemma inference requires internet access.

Figure 31 shows the proposed online deployment on Render with Supabase PostgreSQL and profile-photo storage. This remains a target architecture and requires database compatibility, persistent-storage integration, deployment configuration, and operational verification.

The local and online diagrams are separate views. Solid communication paths identify the protocols between execution environments and services.

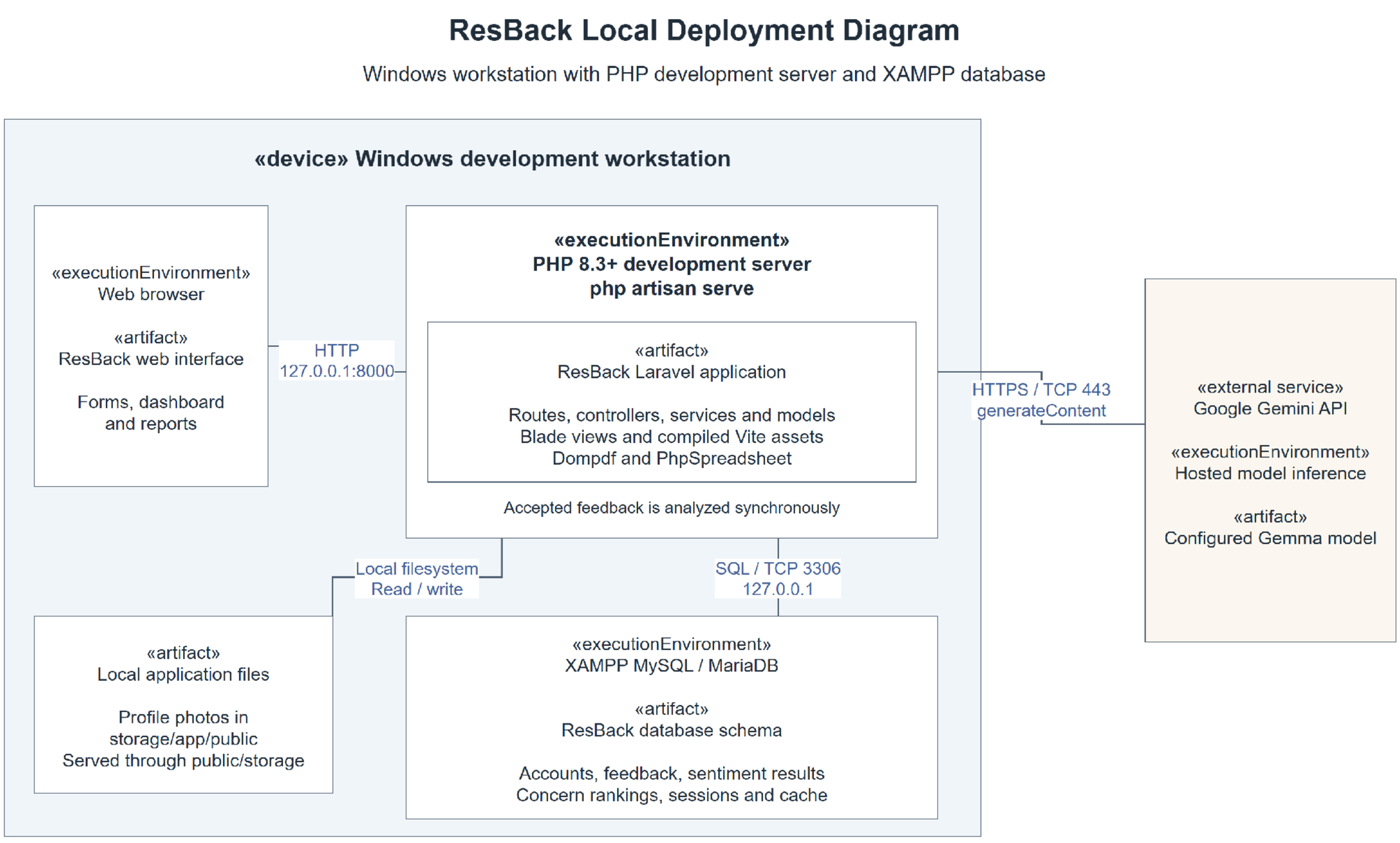


Figure 30 ResBack UML local deployment diagram

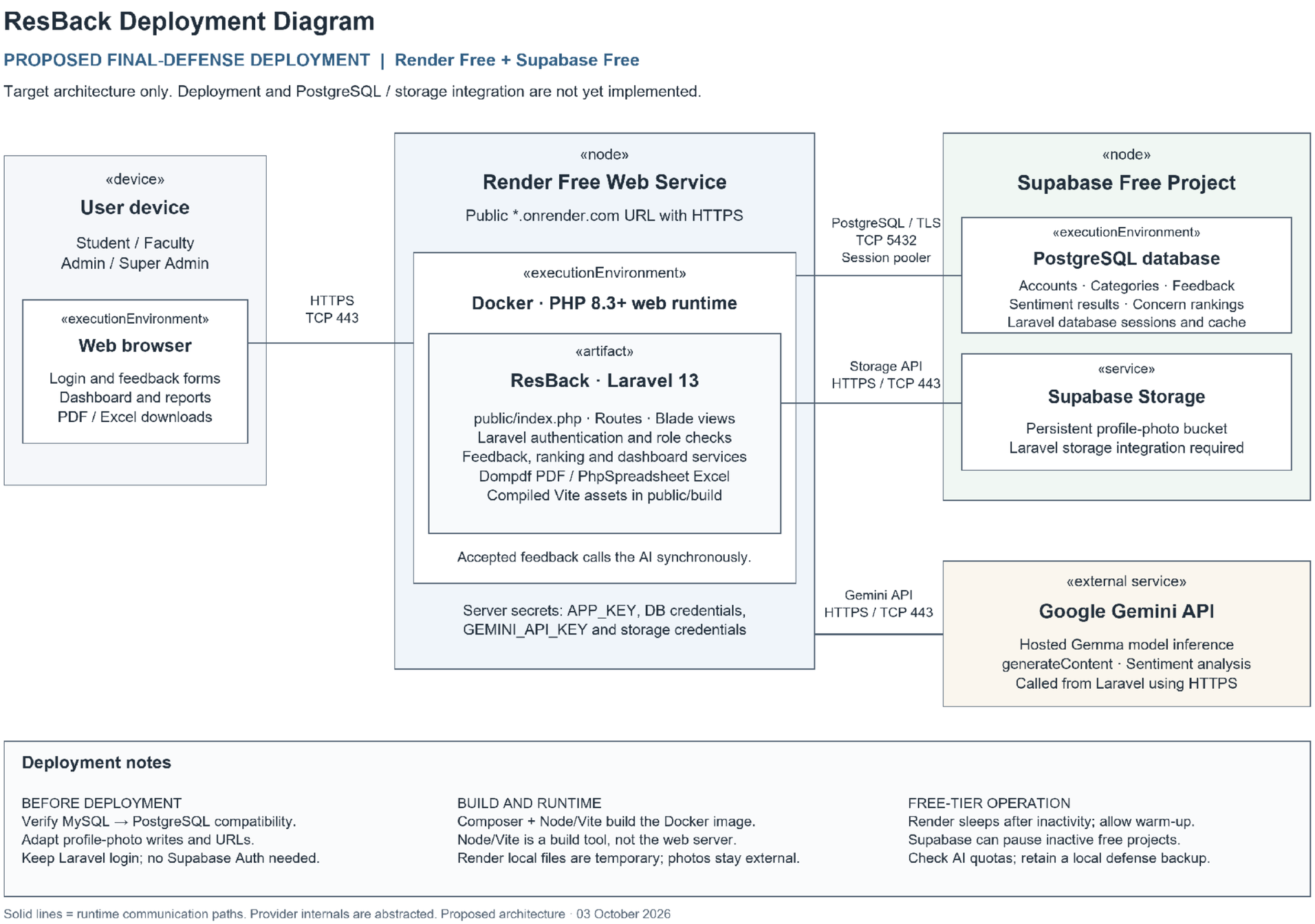


Figure 31 Proposed online deployment diagram